700.

SERVICE BULLETIN NO. 14

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700 SERIES CALCULATORS

This Service Bulletin outlines the main differences between models in the 700/720 A/B/C series of Wang Calculators. Sections are included on *New Features*, *I/O Operations*, and *Conversions*.

700/720 C - NEW FEATURES

The 700/720 C's have all the features of the 700/720 B, plus those described below:

1. BLOCK TRANSFER COMMANDS.

These commands provide for the transfer of program steps, in either direction, between the 700/720 C and a *high speed device*. The 708 (external memory), the 709 (dual tape cassette) and the 710 (disc) are high speed devices. *Program steps* may be transferred singly or in blocks of 8, 16, 32, 64, 128 or 256 either from the 700/720 C to the high speed device, or from the high speed device to the 700/720 C. This is accomplished as follows:

(a) Up to sixteen high speed devices may be connected to a 700/720 C. Each of these units is assigned an unique address from 0000_2 to 1111_2 by setting the address switches to the correct configuration. These switches are marked 1, 2, 4, 8, and when down = 0. Note that the 709 dual cassette is considered as *two* devices, and thus requires two address assignments.

(b) A high speed device is selected by a GROUP 2, 00 XX command, where XX = the device address as determined by the switches. After a device has been addressed in this manner, it will stay selected until a GROUP 2, 00 YY command is issued, where YY ≠ XX; the device whose address is YY is now active, and device XX is not.

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700 SERIES CALCULATORS
DIFFERENCES BETWEEN MODELS

	700 A	700 B	700 C	720 A	720 B	720 C
FEATURES	120 Registers - 960 Steps			248 Registers - 1948 Steps		
	Two Level Subroutine	Five Level Subroutine				
	No Block Transfer		Block Transfer 708/709/710	No Block Transfer		No Block Transfer 708/709/710
	No Binary Decisions		Binary Decisions	No Binary Decisions		Binary Decisions
	No Learn & Run Mode		Learn & Run Mode	No Learn & Run Mode		Learn & Run Mode
	No Plotter	702 Plotter		No Plotter	702 Plotter	
CARDS	5923 5927 5928		5930 5975 5976	5923 5927 5928		5930 5975 5976
ROM	P3A P4A	700 B1	700 D4	720-2A Replace with 720 B1	720 B1	720 D3

TABLE I

(c) Once selected, a high speed device will respond to the *Transfer Commands*. These are two sets of seven commands which control the exchange of data between the 700/720 and the external device. This, allows the user to transfer 1, 8, 16, 32, 64, 128 or 256 program steps at one time.

Commands needed to transfer program steps to the 700/720 C:

08 00	1 program step	08 04	64 program steps
08 01	8 program steps	08 05	128 program steps
08 02	16 program steps	08 06	256 program steps
08 03	32 program steps		

Commands needed to transfer program steps from the 700/720 C:

08 08	1 program step	08 12	64 program steps
08 09	8 program steps	08 13	128 program steps
08 10	16 program steps	08 14	256 program steps
08 11	32 program steps		

Data is transferred between the program step in the 700/720 C indicated by the contents of the X Register, and the location in the high speed device determined by the contents of the Y Register; these addresses must be in the X and Y Registers prior to the transfer command.

(d) Any 700/720 C instruction, including non-high speed device peripheral instructions (701, 702, 703, 704, 705, 706/707, 711 and 714), may be executed after a GROUP 2, OO XX command has been issued to select a high speed device: they will be executed, the high speed device will remain selected and will respond to the transfer commands.

EXAMPLE: To transfer eight program steps (that have been read in from a 714 Mark Sense Card Reader) from step 4 of the 700 C to step 9 of the 708 whose address is 12:

STEP	KEY	COMMENTS
000 001	GROUP 2 0012	Device #12 is selected and will now respond to transfer commands.
002 003 004 005 006 007 008 009 010 011	GROUP 2 MARK ---- (eight steps read in from 714) ---- ----	Any program may be run or commands executed from Keyboard: i.e., read 8 steps from 714 Mark Sense Card Reader into steps 004 through 011 of 700 C.
012	9	708 address to Y Register.
013	↑	
014	4	700 C address to X Register.
015	0809	Transfer eight steps from 700 C (004 through 011) to 708 (9 through 16).

Note that GROUP 2 MARK (04 08) does not release the 708: an 00 XX would be required.

2. BINARY DECISIONS.

The three 700/720 A/B numeric *Skips* ($Y \geq X$, 0507; $Y = X$, 0509; and $Y < X$, 0508) are only *decimal* comparisons. They assume that both the X and Y Registers contain decimal data in the 700/720 format: 0000_2 or 0001_2 in the two sign positions, and the remaining digits $0000_2 \leq X \leq 1001_2$. If the X and Y Registers contain digits other than the above, the results of the compare are unpredictable, and the instructions should not be used. Two new decisions are incorporated into the 700/720 C; they compare the contents of X and Y in a *binary* manner, and all digits may contain $0000_2 \leq X \leq 1111_2$. These instructions are:

0807 : Skip if $Y \equiv X$ (binary)

0815 : Skip if $Y \not\equiv X$ (binary)

3. LEARN AND RUN MODE.

This mode is entered by depressing both the LEARN and the RUN switches on the 700/720 C. When this is done, the following commands can be executed:

(a) *BACKSTEP*, which allows the user to decrement the Program Counter. The display will appear as in the LEARN mode. Depressing *VERIFY PROGRAM* will cause the Program Counter to be decremented by 1 (the opposite of *STEP* in LEARN mode).

(b) *INSERT*, which puts a GO command at the location at which the Program Counter is set, and moves all subsequent program codes in core (up to and including the first END PROGRAM) to the next step; a new code may then be entered to replace the GO. If a second program is in core, its first step will be replaced by the END PROGRAM of the first program. To *INSERT*:

1. In LEARN or RUN mode, SET PC to the location at which the new command is to be inserted;
2. Depress LEARN & RUN;
3. Depress SET PC;
4. Key in desired new command.

EXAMPLE: To insert code 0702 (2) into step 005:

KEY:	I. SET PC 005	II. LEARN & RUN SET PC	III. 2
	After executing I, II or III, core contains:		
STEP			
003	0700	0700	0700
004	0701	0701	0701
005	0703	0514 GO	0702
006	0704	0703	0703
007	0512 END PRG.	0704	0704
008	0705	0512 END PRG.	0512
009	0706	0706	0706
010	0707	0707	0707

(c) *DELETE*, which deletes a program step and moves all subsequent program codes in core (up to and including the first END PROGRAM) to the step previous to the one in which they were located. This is in effect the opposite of *INSERT*. To *DELETE*:

1. In LEARN or RUN mode, SET PC to the step which is to be deleted;
2. Depress LEARN and RUN;
3. Depress RECORD PROGRAM.

EXAMPLE: If step 005 is deleted, the code in step 006 is moved to step 005, the code in step 007 is moved to step 006, etc.

(d) *SEARCH, MARK & STOP*, which sets the Program Counter to the step immediately after the mark that is searched:

1. Depress RUN and LEARN switches simultaneously;
2. Depress PRIME;
3. Select the appropriate mark.

The display is now at the step immediately following the mark.

4. REWIND TAPE (709 only).

This instruction assumes that the Cassette Drive has been previously selected by a GROUP 2, 00 XX command. At any time after selection, a GROUP 2, 01 00 will rewind the tape.

GENERAL 700 INPUT/OUTPUT CONNECTOR PIN DIAGRAM

NAME	PIN NO.	REMARKS
		INPUTS FROM EXTERNAL DEVICE
$\overline{GKA_0}$	1	BCD 1
$\overline{GKA_1}$	2	2
$\overline{GKA_2}$	3	4
$\overline{GKA_3}$	4	8 — High Order Code
$\overline{GKB_0}$	5	BCD 1
$\overline{GKB_1}$	6	2
$\overline{GKB_2}$	7	4
$\overline{GKB_3}$	8	8 — Low Order Code
$\overline{GISN}$	9	Input Strobe from External Device
$\overline{PRMS}$	10	Prime Output
$\overline{GLRN}$	11	Remote Learn Control
$\overline{N_0B}$	12	BCD 1
$\overline{N_1B}$	13	2
$\overline{N_2B}$	14	4
$\overline{N_3B}$	15	8 — 700 Display Digit Position of "X" Register
$\overline{Rb_0B}$	16	BCD 1
$\overline{Rb_1B}$	17	2
$\overline{Rb_2B}$	18	4
$\overline{Rb_3B}$	19	8 — Output From 700 Display "X" Register

NAME	PIN NO.	REMARKS
		GENERAL OUTPUTS
$\overline{GI_oB_oF_b}$	20	BCD 1
$\overline{GI_oB_1F_b}$	21	2
$\overline{GI_oB_2F_b}$	22	4
$\overline{GI_oB_3F_b}$	23	8 — Low Order Code
$\overline{GI_oA_oF_b}$	24	BCD 1
$\overline{GI_oA_1F_b}$	25	2
$\overline{GI_oA_2F_b}$	26	4
$\overline{GI_oA_3F_b}$	27	8 — High Order Code
$\overline{I_oB_1}$	28	I/O Bus & Group I & II Code
$\overline{I_oB_2}$	29	
$\overline{I_oB_3}$	30	
$\overline{GISO}$	31	Output Data Strobe
$\overline{GKBD}$	32	Keyboard Condition Indicator
±OV	33	Common
±OV	34	Common
±OV	35	Common
Chassis Ground	36	Chassis Ground

'1' = Ready
('1' = +4V)

700/720 I/O OPERATION

The Flow Charts show I/O operations for the 700 Series, excluding Typewriter systems. The first chart refers to all 700/720 Models, A, B and C. The second chart describes the operation of the 700/720 C with High-Speed devices (708, 709, 710-1), using the GROUP 2, 00XX command. Note that the signals on the charts are shown in the *true* form, but that they are actually exchanged between units in their *negated* aspect.

1. *ALL MODELS.* A GROUP command starts all I/O operations and sets the IOB as shown on the Flow Chart. The second part of the command goes out on the General I/O lines: GI_0A_xF contains the high order, and GI_0B_xF contains the low order of this code. A strobe, GISO accompanies this data, which is decoded in the device that is being addressed. This device, once selected, takes control of the operations from the calculator.

The 700/720 informs the sending device that it can accept data by a GKBD = 0 signal. The device then puts the data for the calculator on the GKA_x and GKB_x lines and issues a GISN strobe. When all the data has been sent, a GO code (0514) accompanied by GISN informs the 700/720 of the fact, and the calculator resumes control of the operations.

There are at present no Wang devices that receive data from the I/O jack, as shown on the right side of the Flow Chart, but the data is available.

700/720 - A/B I/O OPERATION

1. GROUP COMMAND (1st part)

	IOB_3	IOB_2	IOB_1
GROUP 1			
from KB	1	0	0
from Program	1	1	0
GROUP 2			
from KB	1	0	1
from Program	1	1	1

2. GROUP COMMAND (2nd part)

GI_0A_3F through GI_0B_0F

XX YY

3. OUTPUT STROBE

Generated by 700

GISO = ⊓ (20 - 30μs)

Device is now selected and takes control over from 700/720.

DATA INPUT INTO 700/720

GLRN = 1, RUN mode
= 0, LEARN mode

GKA_3 through GKB_0 = data to be entered

GKBD = 0 (700 Ready)

NO

YES

Generated by device
GISN = ⊓
Strobes data into 700/720

Is data entered a GO code? $GKA_3 \rightarrow GKB_0$ = 0514

NO

YES

Control returns to 700

DATA OUTPUT FROM 700/720

RUN or LEARN	RUN only
Typewriter Jack	I/O Jack

WRITE or
WRITE α

701,702,
707,711

GKBD = 0

NO

YES

DATA from X Register in $\overline{Rb_0B}$ through Rb_3B

POSITION of digit in display in N_0B through N_3B

SIGN of number at $N_xB = 0000_2$

SIGN of exponent at $N_xB = 1101_2$

+ = 0000_2
− = 0001_2

All data rec'd.?

YES

NO

Issue GO code $GKA_3 \rightarrow GKB_0$ = 0514

NOTE: GKBD = 1 when the 700/720 is BUSY.

2. *700/720 C.* In addition to the I/O operation described in paragraph 1, the C models can address High-Speed devices such as the 708, 709 and 710 (External Memory, Dual Cassettes and Disc). Sixteen of these may be used with a 700/720 C (each cassette drive in the 709 is a separate device); four switches on each unit determine its address. A device is selected by a GROUP 2, 00XX command, (A on Flow Chart) where XX = 0000_2 through 1111_2 and equals the switch settings. It then sends a GO to the 700/720 C; at this time:

(a) The calculator resumes control.

(b) The High-Speed device, for all practical purposes, becomes a part of the 700/720 C and may be addressed by a Transfer command. (0800 - 0807, 0808 - 0814).

(c) The calculator can execute any instruction, including standard peripheral commands (701, 702, 703, 704, 705, 706/707, 711, 714), and the High-Speed unit will remain selected.

If a GROUP 2, 00XX is issued, and if XX ≠ the switch settings, the device is released.

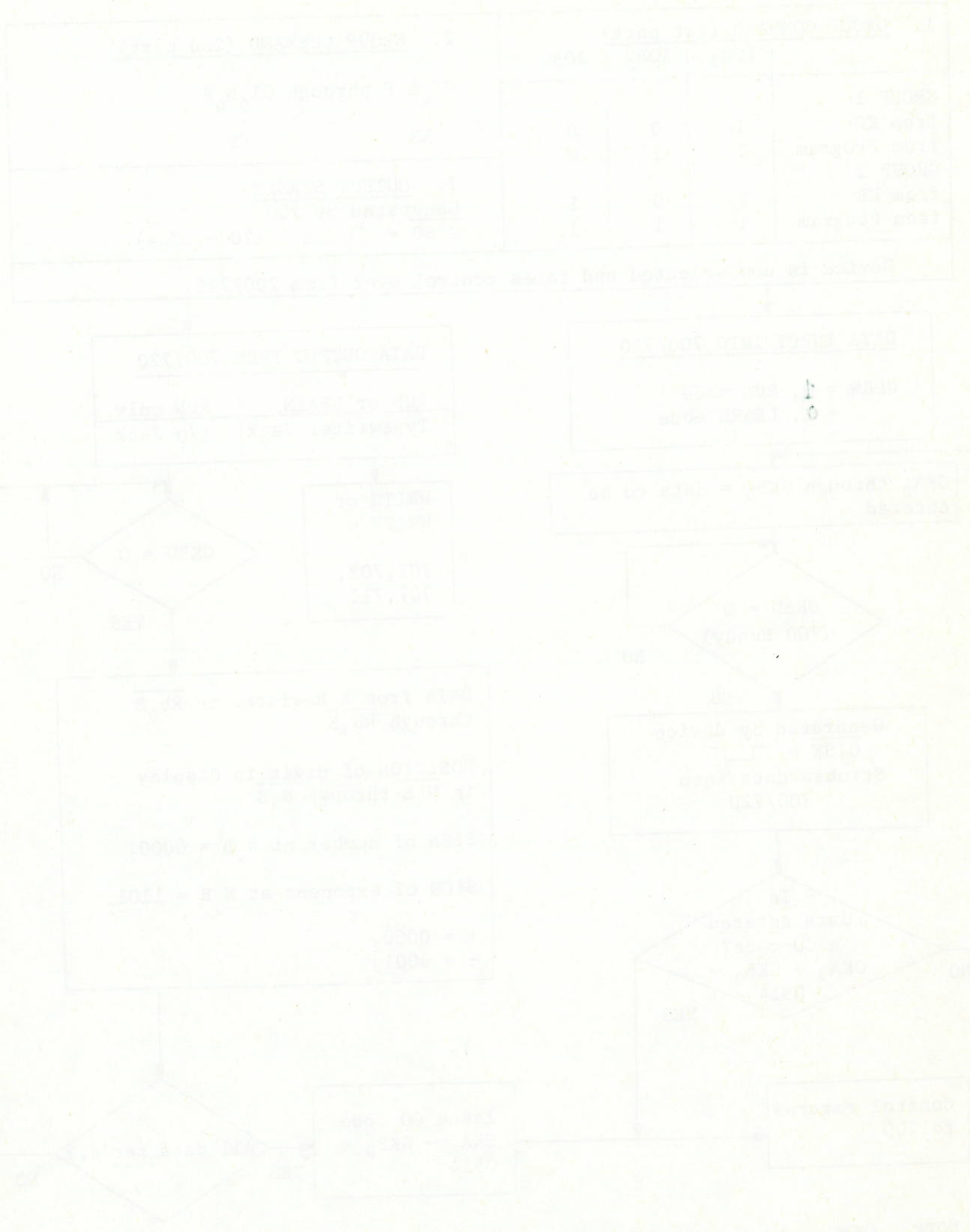
The Transfer Commands consist of four parts:

(a) The Address of the data to be transferred is sent to the device, which stores it. It is taken by the 700 from the Y Register (where it is in decimal form), converted to binary, and sent in three groups of eight bits, high order first, to the device (B on Flow Chart).

(b) The transfer parameters are sent out next. This information consists of one byte (= step = 8 bits) and informs the device of both the direction of transfer and of the number of bytes to be transferred (C on Flow Chart).

(c) The actual data transfer then takes place, as directed by the 700/720 C. The flow can be either from the calculator to the device, or vice-versa (E on Flow Chart).

(d) After all the data has been transferred, the device may send an error signal, which will turn the Error light on in the 700/720 C. The 709 and the 710 both use this feature, while the 708 does not.



PART B — Loading the information onto the disk

Now, assuming the Disk Drive is ON, PRIME and SET have been keyed on the Disk Drive, and GROUP 2, 00 00 have been keyed on the calculator, we proceed as follows:

700/720 Calculator Keystroke Operations

7.) Key 12288
8.) Key ↑
9.) Key 928 (key 1952 if you have a Model 720).
10.) With the "80" Toggle Switch in the "UP" position, depress the [10] Special Function key. This will generate the (08 10) code.

Result: The contents of storage registers [02] and [03] are now on the disk in addresses 12288 through 12303.

PART C — Recalling the data back into the calculator.

Since we are recalling data from the disk into a different location in the calculator — storage registers [04] and [05] — we now need to know the location of the first program step corresponding to storage register [04] . Referring to the calculator manual again, we find that in the Model 700 Calculator, the step in question is Step No. 912 (no. 1936 on the Model 720).

We now proceed as follows: (Note that since the Y-Display still contains the disk address 12288, we do not have to re-enter it.)

700/720 Calculator Keystroke Operations

11.) Key 912 (key 1936 if you have a Model 720).
12.) With the "80" Toggle Switch in the "UP" position, key the [02] Special Function key. This will generate the (08 02) code.

Result: The number 100 will now be in storage registers [02], [03], [04], and [05] and our example is complete.

USING THE DISK UNDER PROGRAM CONTROL

Example 5:

Write a set of subroutines that will allow you to easily store and recall data from the disk by specifying the data register number of the Disk Drive (some number between 0 and 32,767) in the Y-Display, and the number to be stored in the X-Display (if a recall command is given, the number should appear in the X-Display).

From the above, it is evident that we will need a subroutine for storing a number, and a separate subroutine for recalling a number. Since both of these subroutines will have to calculate the address associated with the data register specified in the Y-Display, we can save ourselves time and effort by writing a third routine, which, when called by the other two, will calculate the addresses necessary to store and recall the data.

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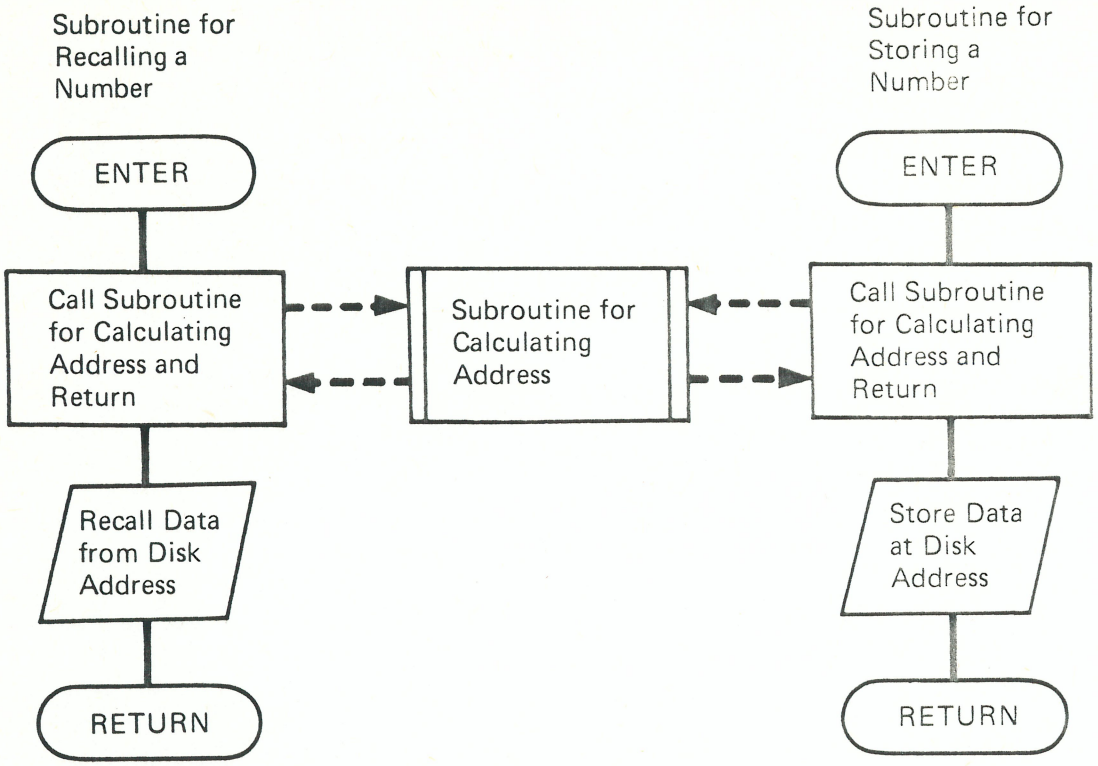

One more point should be considered before going further. As you know, the core of the calculator is arranged in such a way that 2 data registers can be created out of every 16 program steps as follows:

Registers	015	014	013	012	011	010	009	008	007	006	005	004	003	002	001	000	
246												05	05	06	06	07	High-Order Codes
247												12	15	00	04	02	Low-Order Codes

Program Step Numbers

That is, in order to use one data register, we still need to specify 16 program steps. Conversely, by specifying one data register of storage area, we are, at the same time, specifying a second register as well.

This fact will have important implications when we attempt to store or recall a specific data register since we will always be storing or recalling two registers at a time. When we recall a data register from the disk, since we will be recalling 2 registers, we will have to specify which one of the two we wish to use once both are in a work area in the calculator (a buffer).

Likewise, when we store a given data register on the disk, we will actually be storing two registers. This means that while storing one piece of data in one register, we will be writing over and destroying the data in another register. We can prevent this loss of data-already-stored, however. All we have to do is recall the 2 registers in question from the disk into a "buffer", update the one register of these two that we want to hold our new data value, and then store both of the registers back onto the disk.

In other words, the subroutine to recall data will use the address calculating subroutine, while the subroutine to store data in a register will use both the address calculating routine, and the register recall subroutine.

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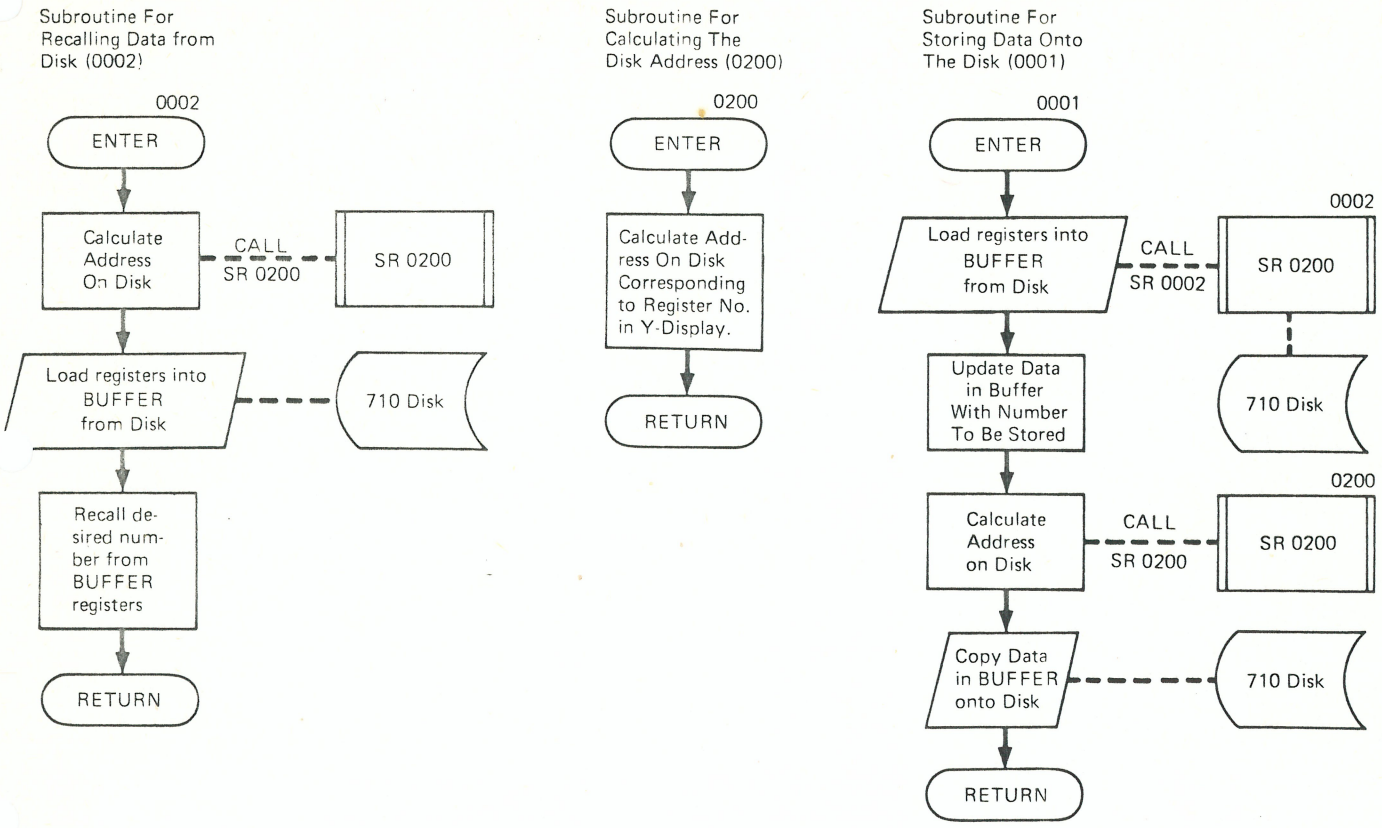
The basic flow diagrams for the three subroutines follow:



The following listing represents the three subroutines just described. (With the 700 in RUN Mode, and the 710-1 Disk connected and ON, key PRIME and SET on the Disk control panel.)

Operating Instructions:

To store a number in Register No. n on the disk:

1.) Key "n" on the 700 keyboard ("n" can be any integer from 0 to 32767).
2.) Key ↑
3.) Key the number you want to store.
4.) With all the toggle switches in the "DOWN" position, key the [01] Special Function key.

Result: The number in the X-Display is now stored on the disk at the position corresponding to Register No. "n" which was keyed into the Y-Display.

To recall a number from Register No. n on the disk into the X-Display:

1.) Key "n" on the 700 keyboard ("n" can be any integer from 0 to 32767).
2.) Key ↑
3.) With all the toggle switches in the "DOWN" position, key the [02] Special Function key.

Result: The number appearing in the X-Display was recalled from the data register on the disk corresponding to the register number in the Y-Display.

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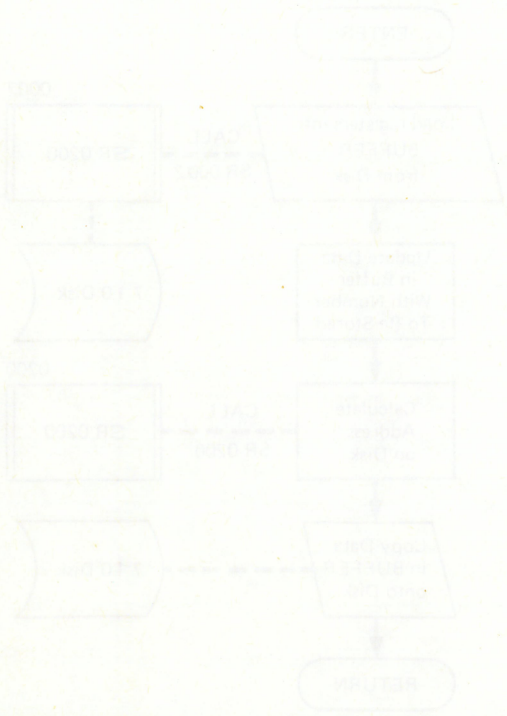
700 PROGRAM TITLE: VER. NO: 656

STEP	CODE	KEY	COMMENT
0000	04 08	MARK	
0001	02 00	0200	Calc. Address
0002	04 14	ST Y	
0003	00 01	REG 001	
0004	04 04	ST DIR	Save X + Y
0005	00 00	REG 000	
0006	07 02	2	Calculate
0007	06 03	/	Address On
0008	07 01	1	Disk Corres-
0009	07 06	6	ponding To
0010	06 06	EX 69	Desired
0011	06 08	INT X	Register
0012	06 02	X	
0013	07 01	1	
0014	07 09	9	Address of
0015	07 05	5	Buffer
0016	07 02	2	
0017	05 11	RETURN	
0018	04 08	MARK	
0019	00 02	0002	Recall A #
0020	02 00	0200	Load
0021	08 02	READ 16	Buffer
0022	04 15	RE Y	Calc. Indic.
0023	00 01	REG 001	0 ⇒ Reg. 3
0024	07 02	2	1 ⇒ Reg. 2
0025	06 03	/	
0026	06 05	DN	
0027	06 08	INT X	
0028	06 01	-	
0029	04 12	WRITE A	skip if Y=0
0030	04 11	WRITE	Recall De-
0031	04 05	RE DIR	sired #
0032	00 02	REG 002	from buffer
0033	04 12	WRITE A	skip if Y≠0
0034	05 11	RETURN	
0035	04 05	RE DIR	
0036	00 03	REG 003	
0037	04 15	RE Y	
0038	00 01	REG 001	
0039	05 11	RETURN	
0040	04 08	MARK	
0041	00 01	0001	Store A #
0042	00 02	0002	Load Buffer
0043	04 15	RE Y	
0044	00 01	REG 001	Cal. Ind.
0045	07 02	2	0 ⇒ Reg. 3
0046	06 03	/	1 ⇒ Reg. 2
0047	06 05	DN	
0048	06 08	INT X	
0049	06 01	-	
0050	04 05	RE DIR	
0051	00 00	REG 000	
0052	04 12	WRITE A	skip if Y=0
0053	04 11	WRITE	Update Buffer
0054	04 04	ST DIR	With # to
0055	00 02	REG 002	be Stored
0056	04 12	WRITE A	skip if Y≠0
0057	05 11	RETURN	
0058	04 04	ST DIR	
0059	00 03	REG 003	
0060	04 15	RE Y	Calculate
0061	00 01	REG 001	Address
0062	02 00	0200	on Disk
0063	08 10	WRITE 16	Copy Buffer Onto D
0064	04 15	RE Y	
0065	00 01	REG 001	
0066	04 05	RE DIR	Restore X and Y
0067	00 00	REG 000	
0068	05 11	RETURN	
0069	05 12	END PROG	

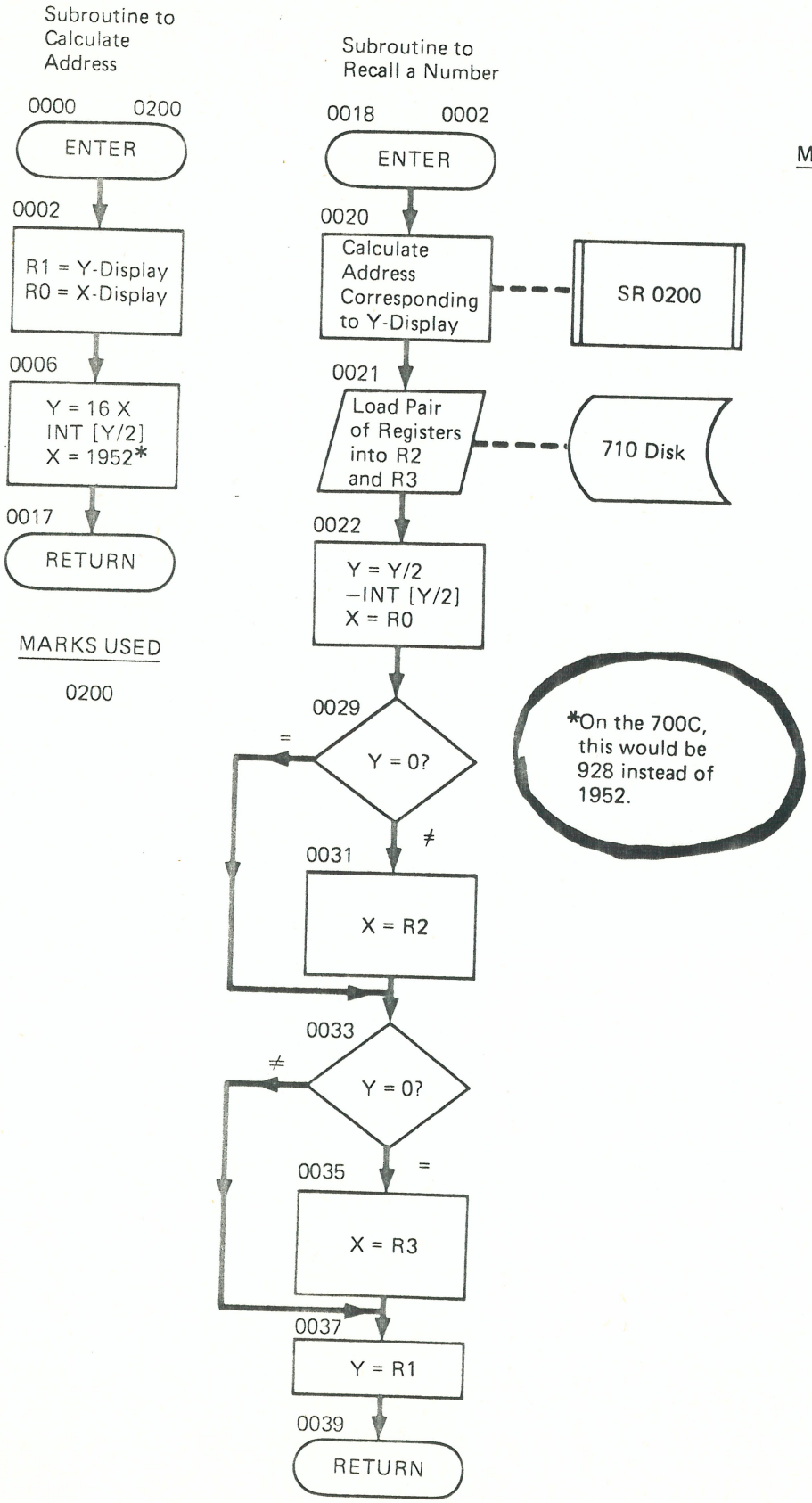
As you can see, the contents of the X-Display are saved in Register 00, while the contents of the Y-Display are saved in Register 01. Registers 02 and 03 are used as the buffer area for the recalling and storing procedures.

Note:

If you have a Model 700C instead of a 720C, the above program can be used, if Steps 13 through 16 are changed from "1952" to "928".



Full flow charts for the three subroutines follow.



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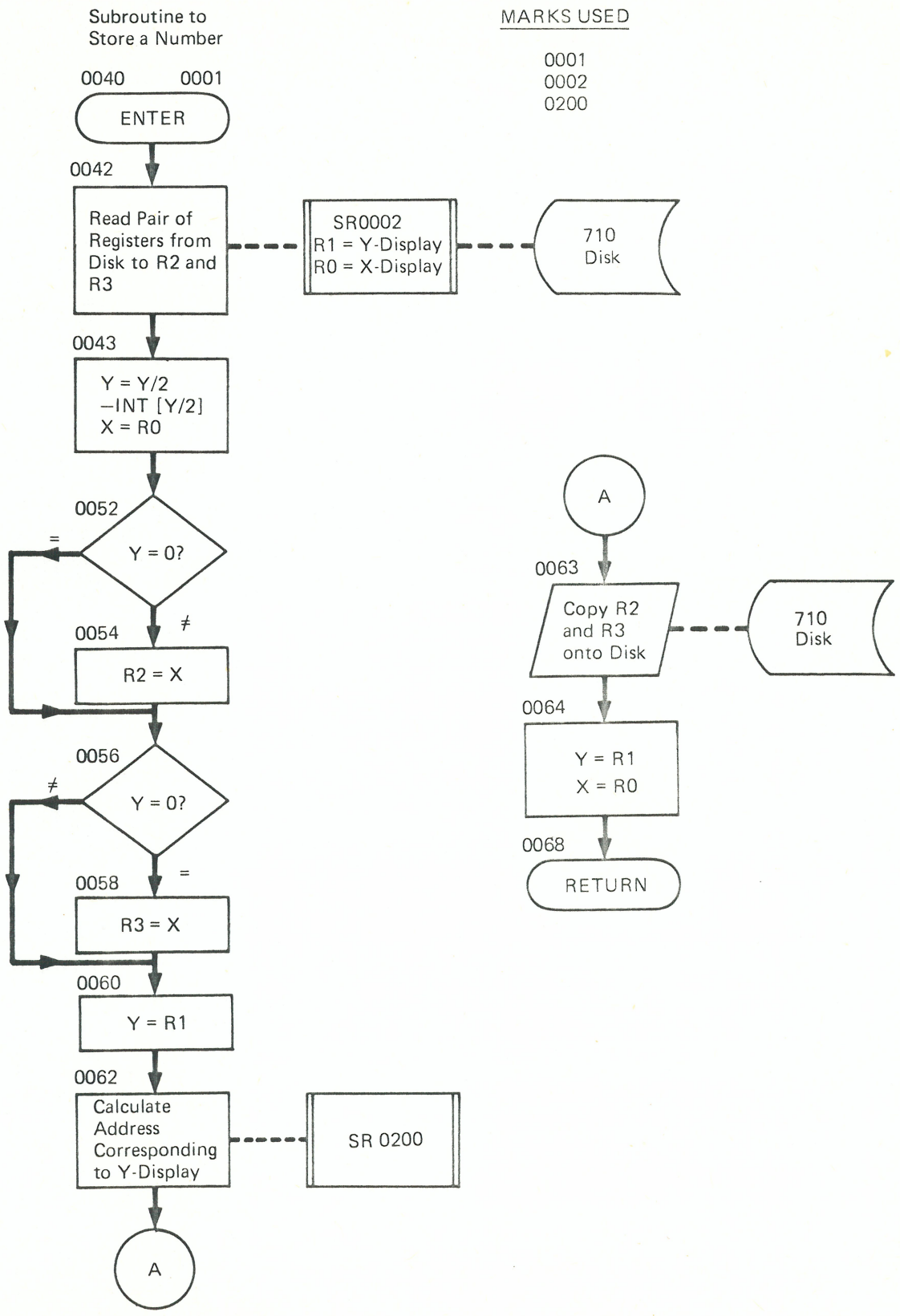

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TYPEWRITER COMMANDS

All Write Alpha Commands

FUNCTIONS	TOGGLE SWITCH SETTING	SPECIAL FUNCTION KEY	CODE
→ Space	00	02	0002
Back Space	00	03	0003
Index (LF)	10	10	0110
→ Shift Up	10	03	0103
→ Shift Down	10	02	0102
→ CR/LF	10	08	0108
Power On	120	00	1200
Power Off	120	01	1201
Tab (not available on 702)	00	08	0008
Clear Tab (not available on 702)	00	11	0011
Set Tab (not available on 702)	00	10	0010
Reverse Index (702 only)	01	11	0111

The above commands must be preceded by a **WRITE ALPHA** and end with an **END ALPHA.**

MANUAL ENTRY CODES FOR PRINTING

CHARACTER: (U/L Case)	TOGGLE SWITCH SETTING	SPECIAL FUNCTION KEY	CODES :
A/a	10	12	0112
B/b	20	00	0200
C/c	20	12	0212
D/d	20	13	0213
E/e	20	05	0205
F/f	00	14	0014
G/g	00	15	0015
H/h	20	01	0201
I/i	10	04	0104
J/j	00	07	0007
K/k	20	04	0204
L/l /1	20	09	0209
M/m	10	15	0115
N/n	20	06	0206
O/o	10	09	0109
P/p	00	05	0005
Q/q	00	04	0004
R/r	10	13	0113
S/s	10	01	0101
T/t	20	07	0207
U/u	20	14	0214
V/v	10	14	0114
W/w	10	00	0100
X/x	20	15	0215
Y/y	00	01	0001
Z/z	30	07	0307
[/] or 1/± L/	30	15	0315
@/2	30	06	0306
#/3	30	14	0314
$/4	30	09	0309
%/5	30	05	0305
¢/6	30	04	0304
&/7	30	13	0313
*/8	30	12	0312
(/9	30	00	0300
)/0	30	01	0301
-/-	00	00	0000
+/=	00	06	0006
¼ ½ or °/! !/!	10	07	0107
:/;	00	13	0013
"/'	10	05	0105
,/,	00	12	0012
./.	10	06	0106
?/ /	00	09	0009

THE ABOVE COMMANDS NEED TO BE PRECEDED BY A WRITE ALPHA AND END WITH AN END ALPHA.

3.0 Printed Circuit Board Placement

No. 5938R1 ROM (Under-Chassis Mount)

SPARE
14
SPARE
13
W.P.N. 2105928 Input/Output Buffer Board 5928 (A+B)
12
No. 5927 PCB – KB Control, LRN, RUN, BLK Display
11 W.P.N. 210 5927
No. 5922 PCB – ROM Decoder
10 W.P.N. 210 5922
No. 5923 PCB – JH, JL, JAD, O.F.
9 W.P.N. 210 5923
No, 5918 PCB – ALU, ADDER
8 W.P.N. 210 5918
No. 5920 PCB – BUS No. 2
7 W.P.N. 210 5920
No. 5919 PCB – BUS No. 1
6 W.P.N. 210 5919
No. 5921 PCB – SENSE AMPS, OSC, BIT DRIVE
5 W.P.N. 210 5921
No. 5887 PCB – CORE CARD
4 W.P.N. 210 5887
No. 5917 PCB – CORE DRIVER
3 W.P.N. 210 5917
No. 5926 PCB – KBD, CORE ADDRESS, "D" BUS
2 W.P.N. 210 5926
No. 5924 PCB – TAPE IN-OUT NIXIE DECODER
1 W.P.N. 210 5924
(Without tubes W.P.N. 210 5936 – With tubes W.P.N. 210 59361)

5936 NIXIE DISPLAY BOARD (2)

LINE VOLTAGE FAILURE PROTECT

W.P.N. 210 0335 +5V +12V Reg 335
W.P.N. 210 0336 +5V +12V Reg 336
CAP
CAP
POWER TRANSFORMER Wang Part No. 410 0059

"Y" NIXIE DRIVER W.P.N. 210 0505 L505
"X" NIXIE DRIVER W.P.N. 210 0505 L505

EMPTY CONNECTOR
TAPE DRIVE SENSORY CONNECTOR
TAPE DRIVE CONNECTOR A.C.

TAPE DRIVE
Wang Part No. 278 5230
50 cps Model
Wang Part No. 278 5231

MODEL 700 CHASSIS LAYOUT

A

1. GROUP COMMAND (1st part)

	IOB_3	IOB_2	IOB_1
from KB	1	0	1
from Program	1	1	1

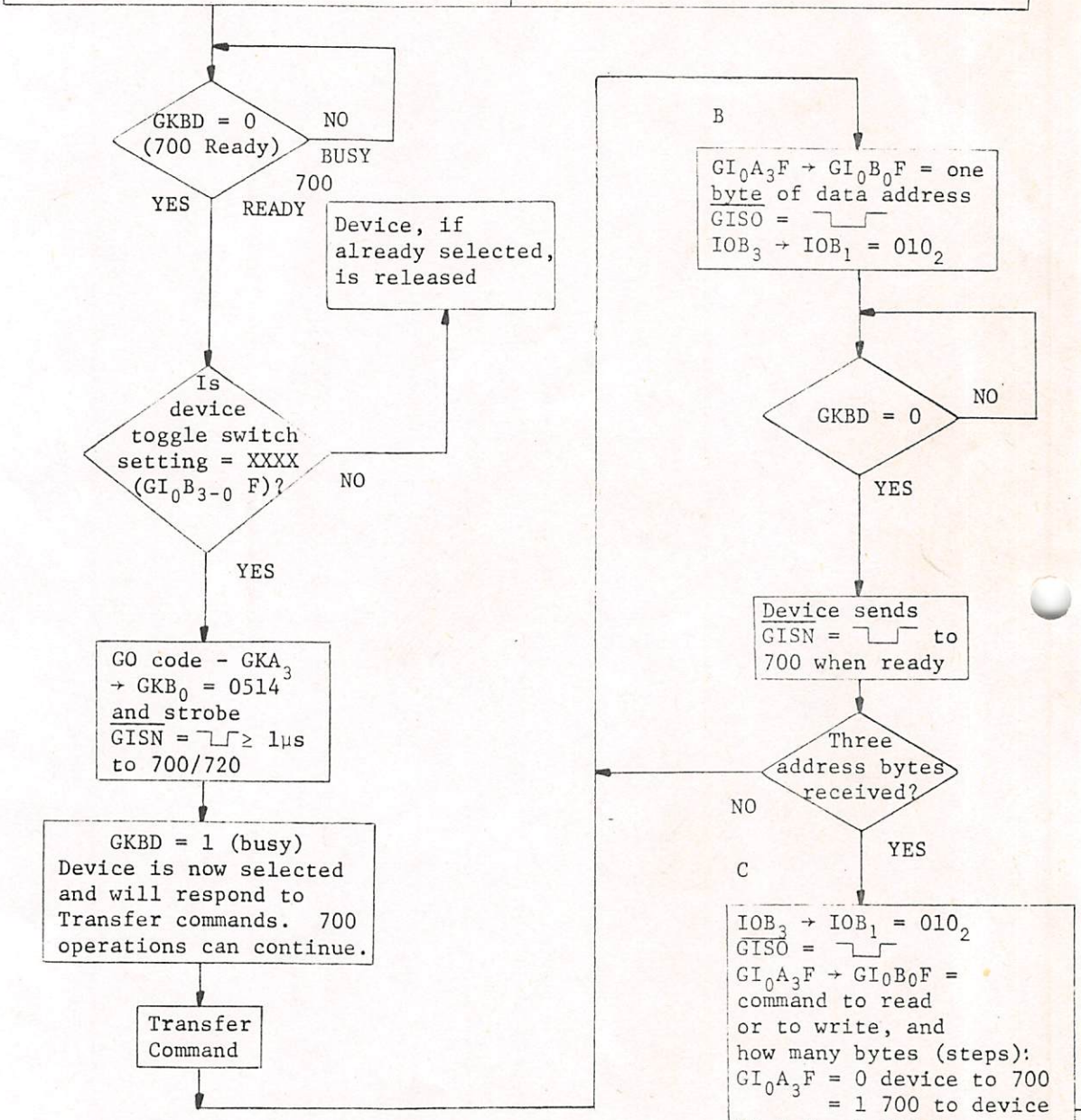
2. GROUP COMMAND (2nd part)

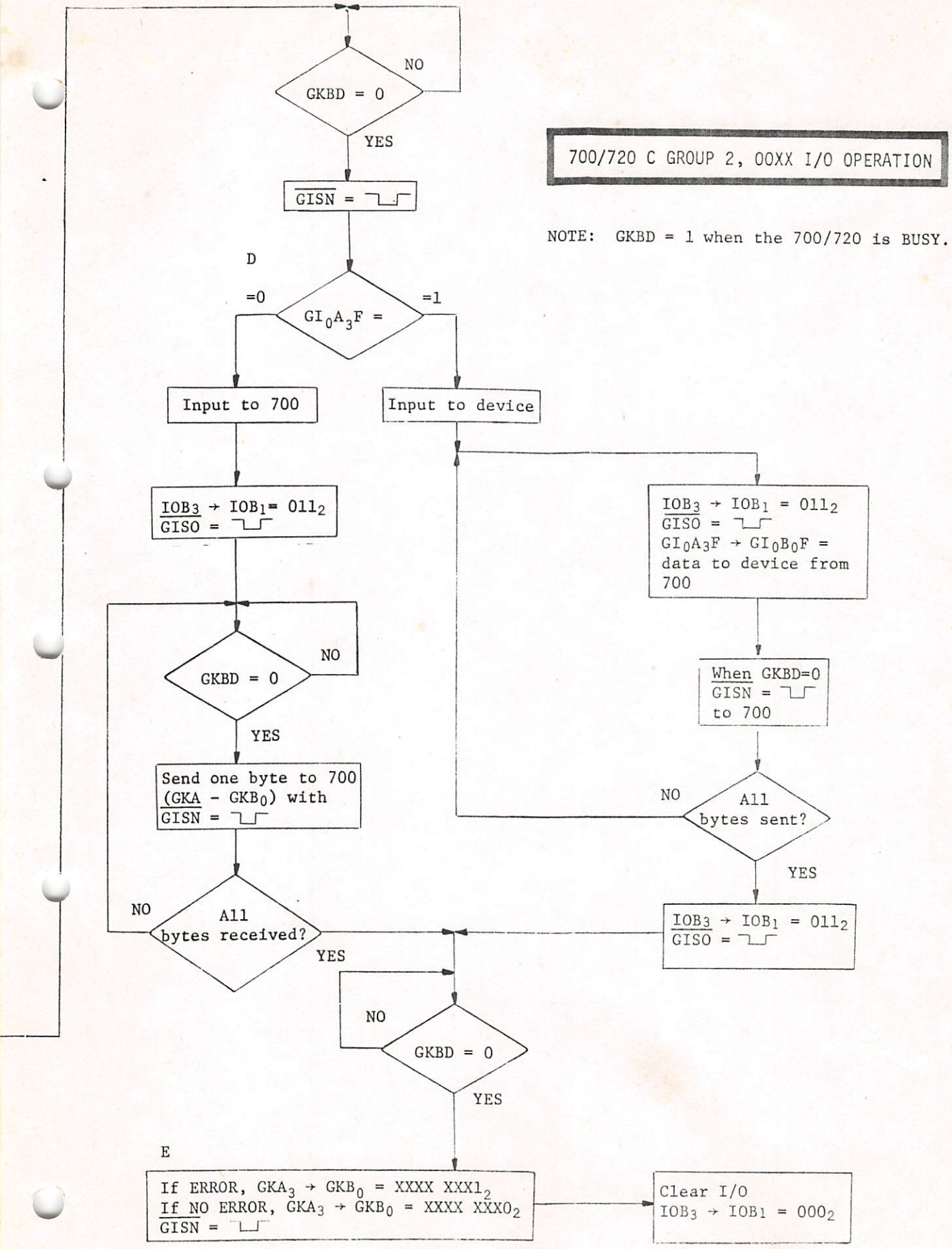
GI_0A_3F through $GI_0A_0F = 0000_2$

GI_0B_3F through $GI_0B_0F = XXXX_2$

3. OUTPUT STROBE

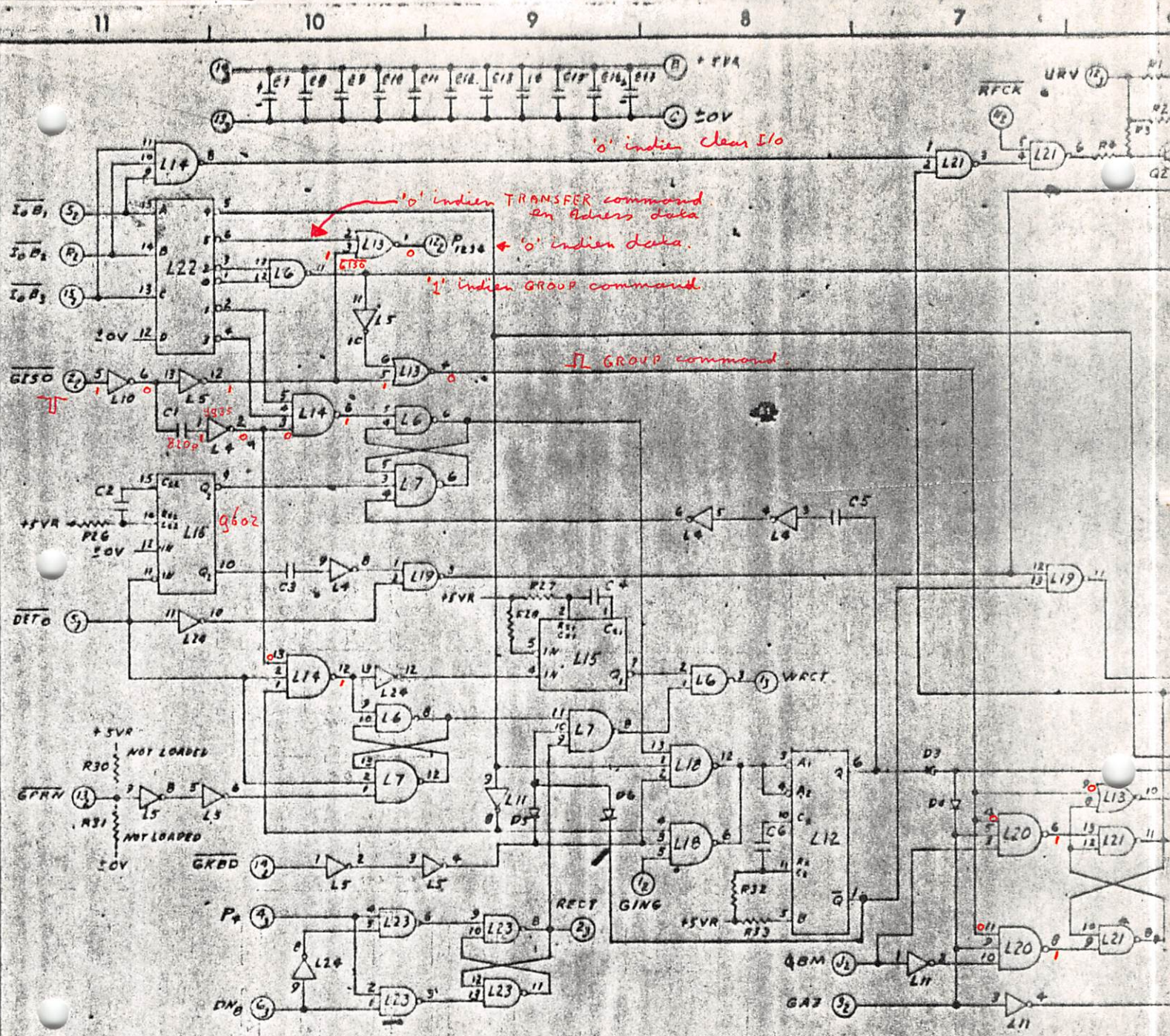
$\overline{GISO}$ = ⌐⌐ (10 µs)





700/720 C GROUP 2, 00XX I/O OPERATION

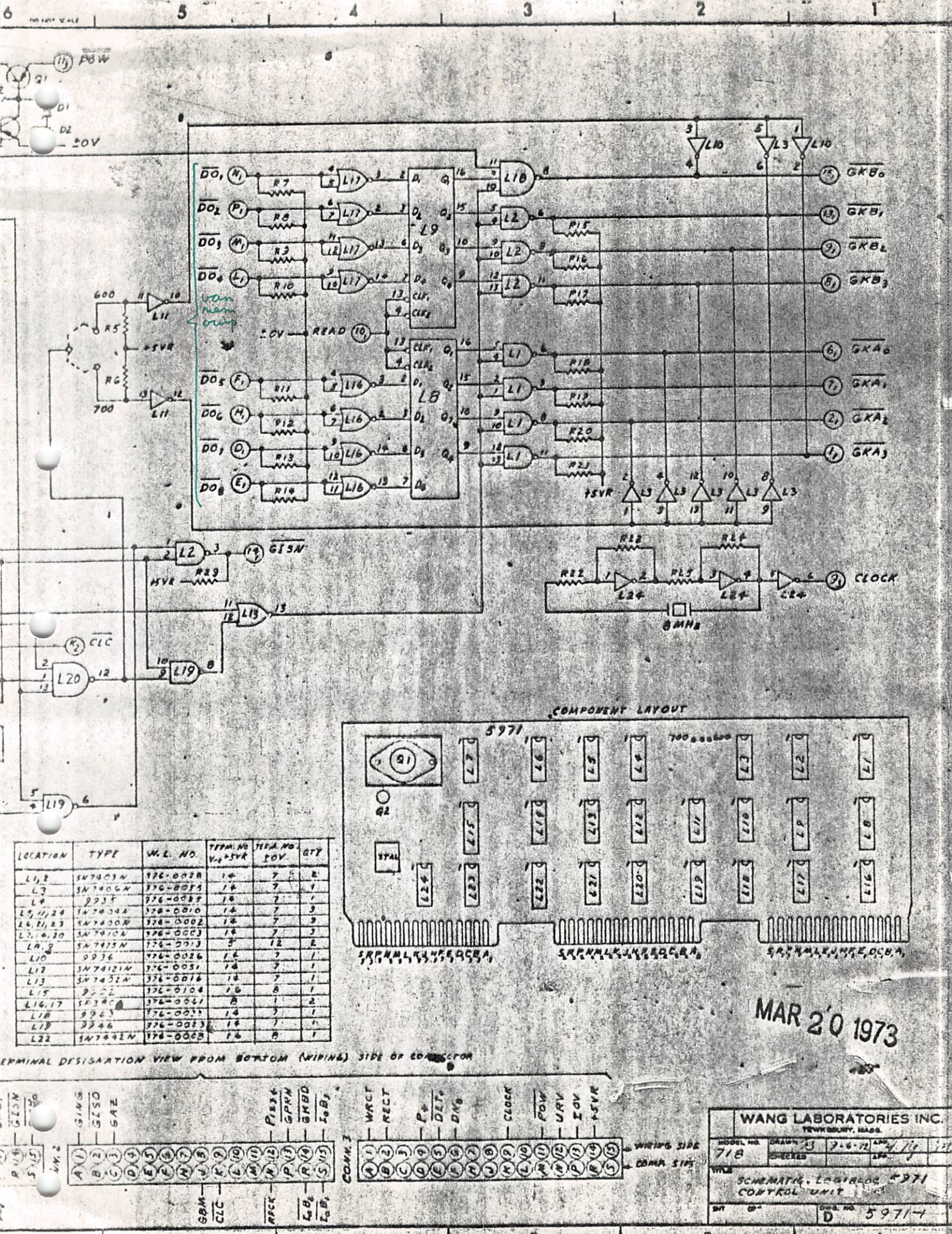
NOTE: GKBD = 1 when the 700/720 is BUSY.



COMPONENT	SIZE/TYPE	W.L. NO.	QTY
R1	3.92Ω 1/4W	333-3039	1
R2	196Ω 1/4W	333-3019	1
R3	3.3K 1/4W	333-3033	1
R4	47KΩ 1/4W	333-3047	1
R5,6,28,33	10K 1/4W	333-4010	4
R7 THRU 17, R29	4.7K 1/4W	333-3047	16
R22, 23	220Ω 1/4W	333-3022	2
R25	180Ω 1/4W	333-3018	1
R24, 31	1.8K 1/4W	333-3018	2
R26	39K 1/4W	333-4039	1
R27	22K 1/4W	333-4022	1
R30	1K 1/4W	333-3010	1
R32	15K 1/4W	333-4015	1

COMPONENT	SIZE/TYPE	W.L. NO.	QTY
C1, 8	820 pF	332-1820	2
C3	.002 μF	300-1202	1
C2	3-30 pF MT	340-0003	1
C5	330 pF	300-1331	1
C7, 17	10 μF 16VDC	340-3006	2
C9 THRU 16	.05 μF	300-1205	8
D2	1N753A 6.2V	380-2062	1
D1	1N758A 10V	380-2068	1
D3, 4, 5, 6	SILICON DIODE	380-0000	4
Q1	2N4124	315-1091	1
Q2	GTS44	315-1811	1
XTAL	8 MHz	321-0007	1
C6	.0015 μF	300-1901	1

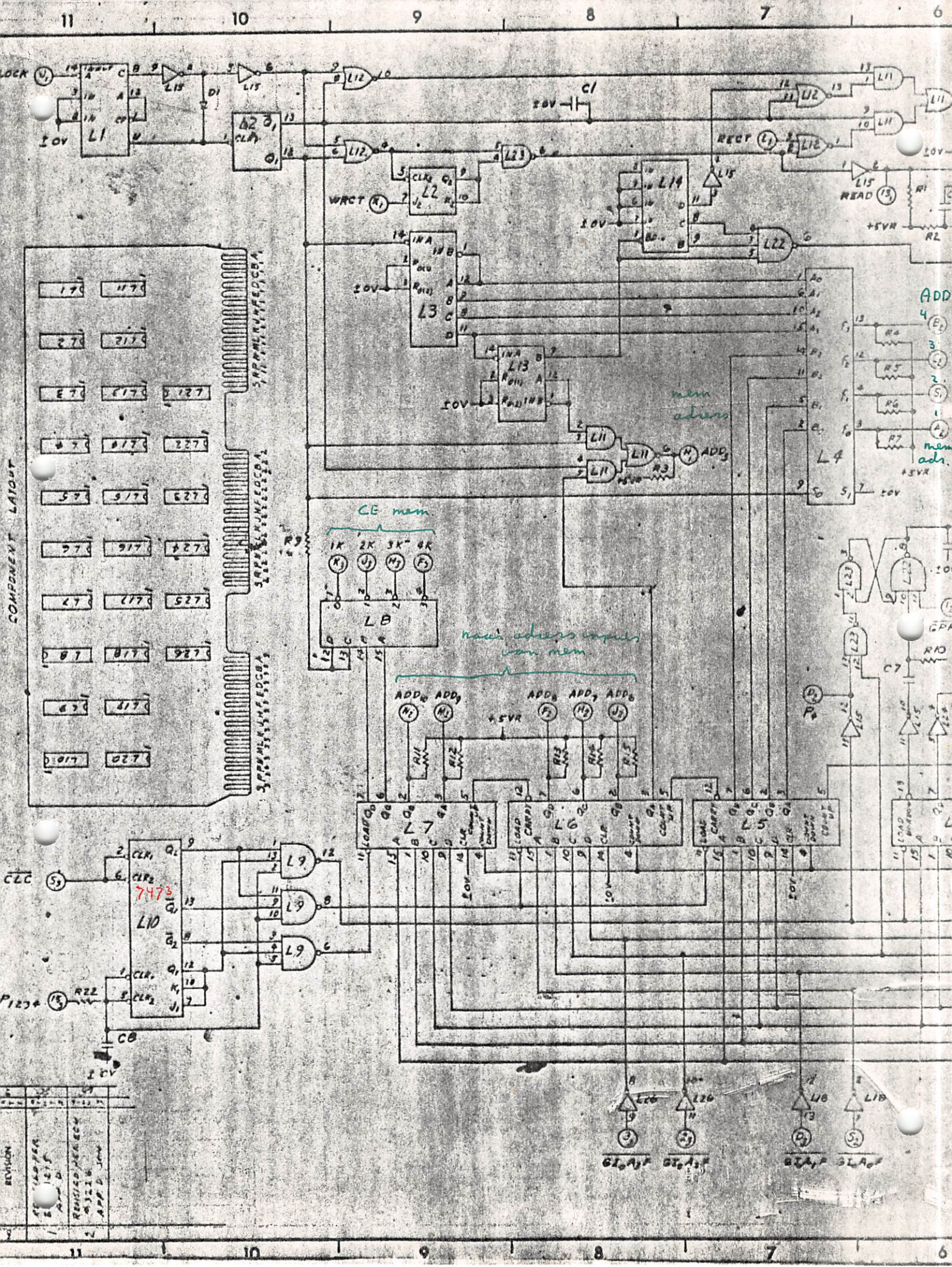
REVISION	BY	CHK	DATE
1	PC 3218 REVISED PER ECN 3526 AND D SCHG	C.H.A.	3-18-75
2	REVISED PER ECN 3526 APPROVED		

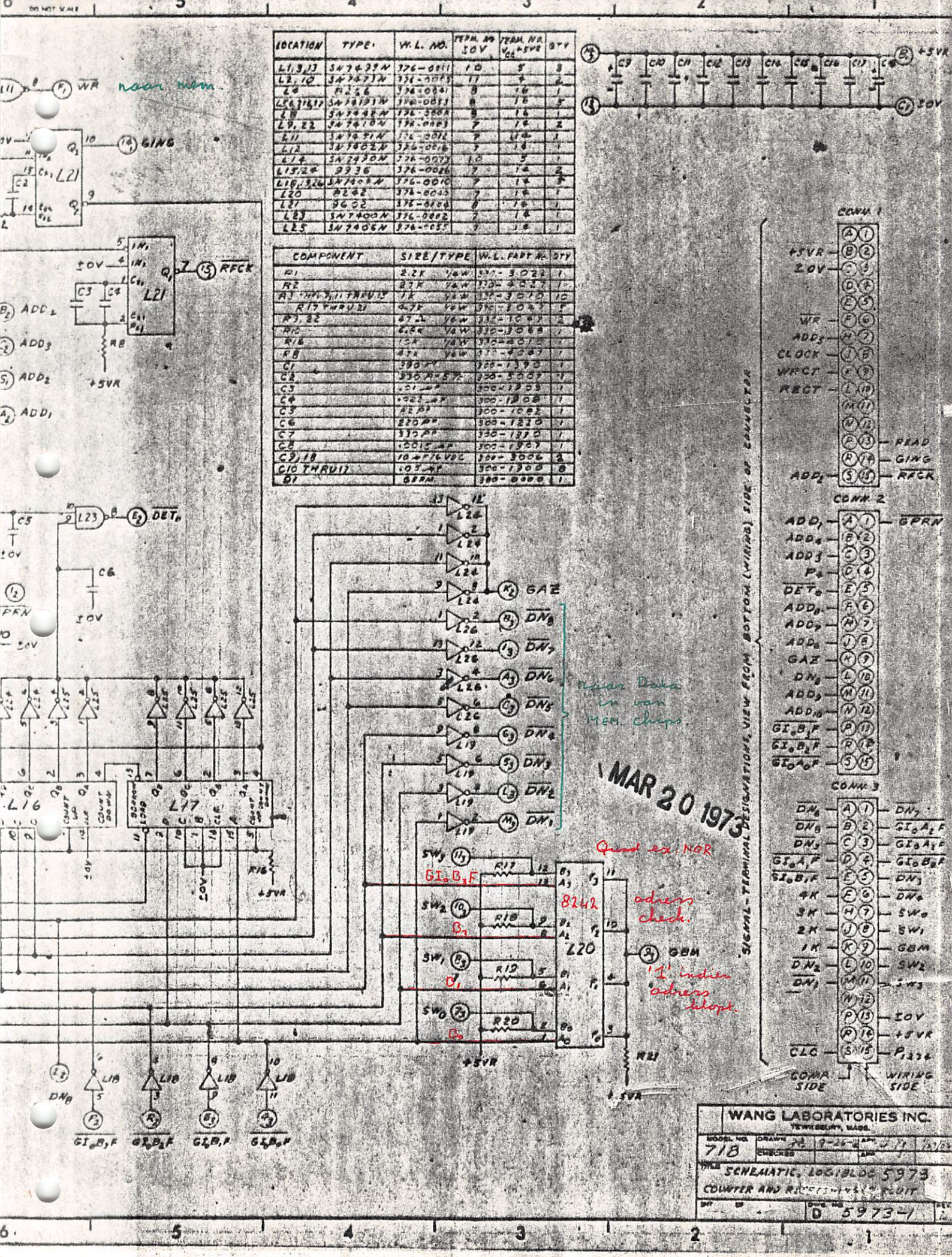


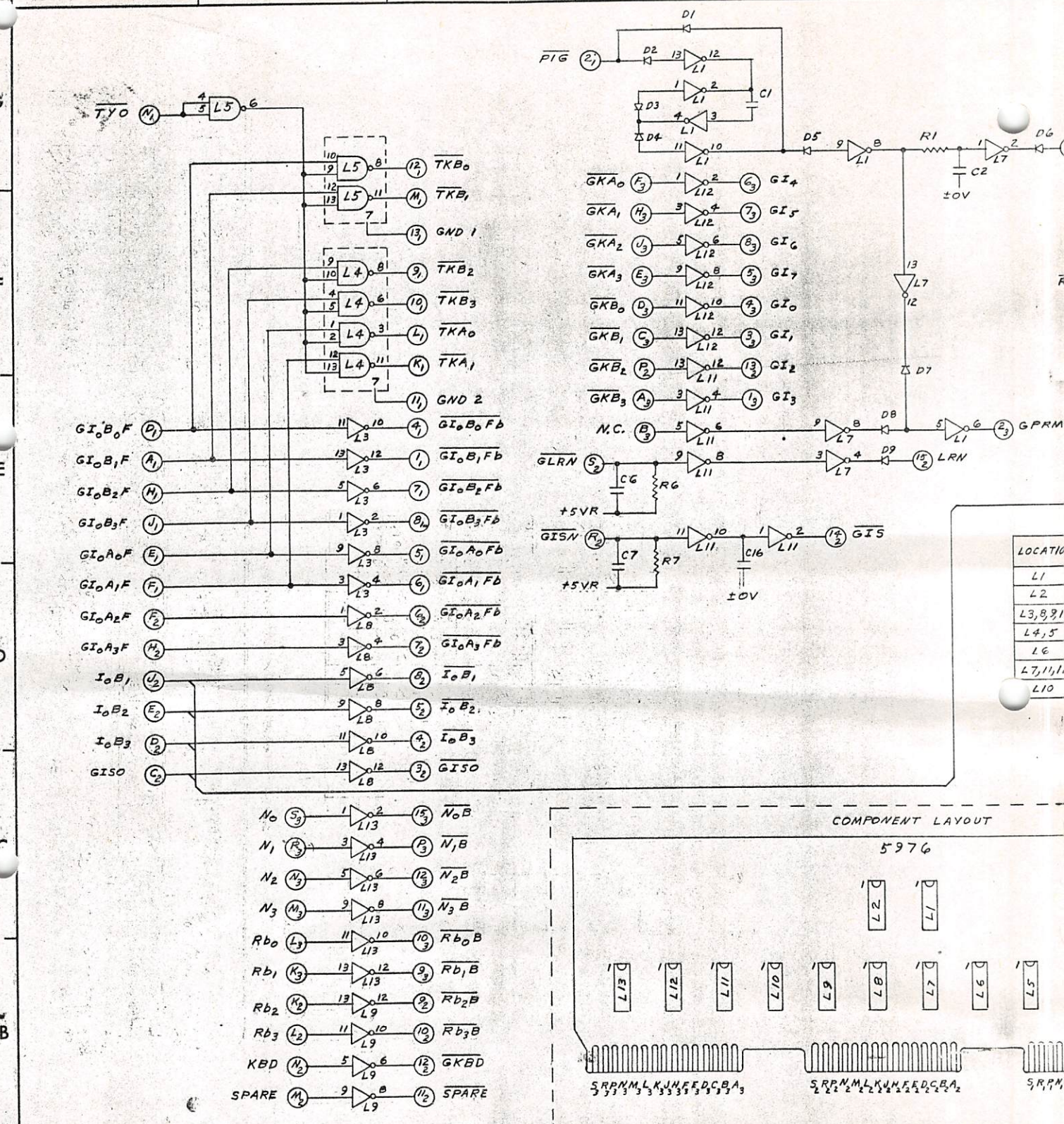
LOCATION	TYPE	W. L. NO.	TERM. NO. V+ +5VR	TERM. NO. +5V	QTY
L1, 2	SN7403N	376-0028	14	7	2
L3	SN7406N	376-0075	14	7	1
L4	9935	376-0085	16	7	1
L5, 11, 24	SN7404N	376-0010	14	7	3
L6, 21, 23	SN7430N	376-0002	14	7	3
L7, 16, 20	SN7410N	376-0003	14	7	3
L8, 9	SN7475N	376-0073	5	12	2
L10	9936	376-0026	16	7	1
L12	SN74121N	376-0031	14	7	1
L13	SN7452N	376-0016	14	7	1
L15	9602	376-0104	16	8	1
L16, 17	SN380	376-0061	8	1	2
L18	9963	376-0033	14	7	1
L19	9946	376-0023	14	7	1
L22	SN7442N	376-0008	16	8	1

TERMINAL DESIGNATION VIEW FROM BOTTOM (WIRING) SIDE OF CONNECTOR

MAR 20 1973

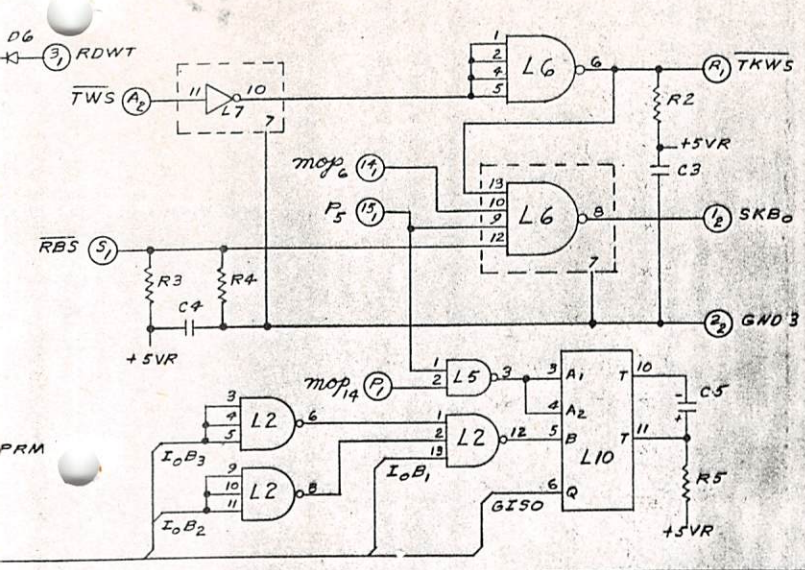

11
10
9
8
7
6
LOCK
+5V
C1
RECT
WRCT
READ
+5VR
ADD
4
3
2
1
mem adrs
mem adrs.
CE mem
1K
2K
3K
4K
mem adrs inputs from mem
ADD10
ADD9
ADD8
ADD7
ADD6
L1
L2
L3
L4
L5
L6
L7
L8
L9
L10
L11
L12
L13
L14
L15
L22
L23
L26
R1
R2
R3
R4
R5
R6
R7
R9
R10
R11
R12
R13
R14
R15
R22
C7
C8
7473
CLC
P1234
COMPONENT LAYOUT
REVISION


LOCATION | TYPE | W. L. NO. | TERM. NO. +10V | TERM. NO. +5VR | QTY
L1, 3, 13 | SN7439N | 376-0011 | 10 | 5 | 3
L2, 12 | SN7473N | 376-0013 | 11 | 4 | 2
L4 | R2L6 | 376-0041 | 8 | 16 | 1
L5, 6, 16, 17 | SN74191N | 376-0053 | 8 | 16 | 4
L8 | SN7442N | 376-3004 | 8 | 16 | 1
L9, 22 | SN7410N | 376-0003 | 7 | 14 | 2
L11 | SN7451N | 376-0012 | 7 | 14 | 1
L12 | SN7402N | 376-0016 | 7 | 14 | 1
L14 | SN7430N | 376-0007 | 10 | 5 | 1
L15, 24 | 9936 | 376-0026 | 7 | 14 | 2
L18, 19, 26 | SN7408N | 376-0010 | 7 | 14 | 3
L20 | 8242 | 376-0043 | 7 | 14 | 1
L21 | 9602 | 376-8168 | 8 | 16 | 1
L23 | SN7400N | 376-0002 | 7 | 14 | 1
L25 | SN7406N | 376-0035 | 7 | 14 | 1
COMPONENT | SIZE/TYPE | W.L. PART NO. | QTY
R1 | 2.2K 1/4W | 330-3022 | 1
R2 | 27K 1/4W | 330-3027 | 1
R3 THRU 11 | 1K 1/4W | 330-3010 | 10
R17 THRU 21 | 4.7K 1/4W | 330-3047 | 5
R3, 22 | 67.1 1/4W | 330-1069 | 2
R15 | 6.8K 1/4W | 330-3068 | 1
R16 | 1K 1/4W | 330-3010 | 1
R18 | 47K 1/4W | 330-3047 | 1
C1 | 330 PF | 300-1330 | 1
C2 | 330 PF 5% | 300-5330 | 1
C3 | .01 MF | 300-1103 | 1
C4 | .022 MF | 300-1223 | 1
C5 | 82 PF | 300-1082 | 1
C6 | 220 PF | 300-1221 | 1
C7 | 330 PF | 300-1331 | 1
C8 | .0015 MF | 300-1152 | 1
C9, 18 | 10 MFD 10VDC | 300-3006 | 2
C10 THRU 17 | .03 MF | 300-1303 | 8
D1 | GERM. | 380-0000 | 1
WR
READ
GING
RFCK
ADD2
ADD3
ADD1
DET0
GAZ
DN8
DN7
DN6
DN5
DN4
DN3
DN2
DN1
SW3
SW2
SW1
SW0
GBM
GI0B3F
GI0B2F
GI0A2F
GI0B1F
GI0A1F
R17
R18
R19
R20
R21
L20
8242
L16
L17
L21
L23
L24
L25
L26
L18
L19
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
C12
C13
C14
C15
C16
C17
+5VR
+10V
CONN 1
CONN 2
CONN 3
CLOCK
WFCT
RECT
ADD2
ADD3
ADD1
GAZ
DN8
DN7
DN6
DN5
DN4
DN3
DN2
DN1
SW3
SW2
SW1
SW0
GBM
GI0B3F
GI0A2F
GI0B2F
GI0A1F
GI0B1F
CLC
P1234
IOV
+5VR
READ
GING
RFCK
GPRN
SIGNAL-TERMINAL DESIGNATIONS, VIEW FROM BOTTOM (WIRING) SIDE OF CONNECTOR
COMP SIDE
WIRING SIDE
near Data in com MEM. chips
Quad ex. NOR
address check.
'1' indicates address select.
MAR 20 1973
WANG LABORATORIES INC.
TEWKSBURY, MASS.
MODEL NO. 718
DRAWN
CHECKED
APP'D
TITLE SCHEMATIC, LOGIBLOC 5973
COUNTER AND REGISTER CIRCUIT
SHT
DWG NO. D 5973-1
REV
DO NOT SCALE



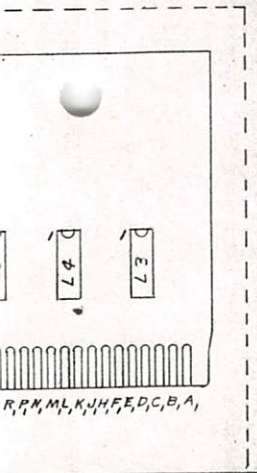
NO.	REVISION	BY	DATE
1	PER ECN #2621 ADDED C16, .022μF MYLAR TO L11-PIN 10 AND 1 TO GND. APPD - JK HO	JB	12/10/71

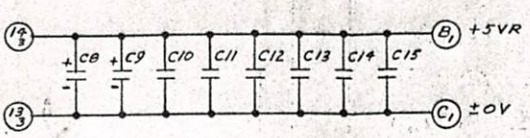
5976

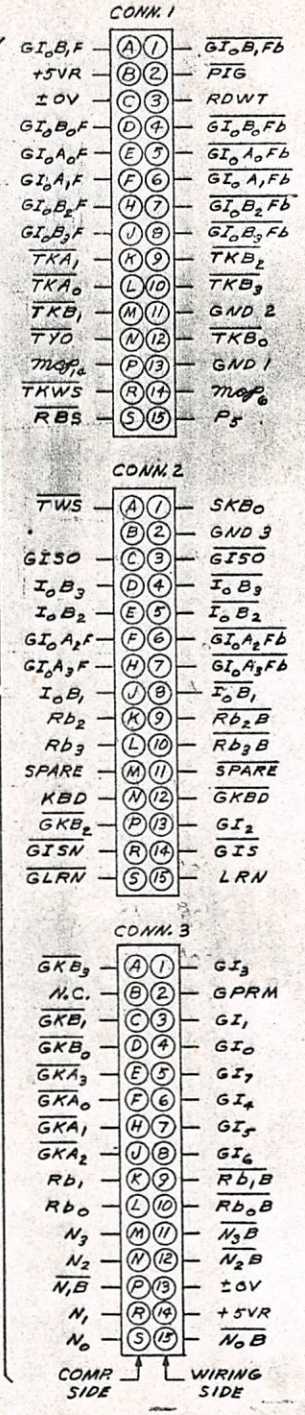


CATION	TYPE	WANG LAB. No.	TERM. No. V_{CC} +5VR	TERM. No. ±0V	QTY
L1	9935	376-0025	14	7	1
L2	SN7410N	376-0003	14	7	1
1,8,9,13	SN74H04N	376-0045	14	7	4
4,5	SN74H00N	376-0037	14	7	2
L6	MC884	376-0024	14	7	1
7,11,12	SN7404N	376-0010	14	7	3
L10	74121N	376-0051	14	7	1

COMPONENT	SIZE/TYPE	WANG LAB. No.	QTY
R1	33Ω 1/4W	330-1033	1
R2	270Ω 1/4W	330-2027	1
R3	180Ω 1/4W	330-2018	1
R4	220Ω 1/4W	330-2022	1
R5	39K 1/4W	330-4039	1
R6,7	1K 1/4W	330-3010	2
C1	.22 MF	300-1902	1
C2,3, C10 THRU C15	.05 MF	300-1900	8
C4	.1 MF	300-1901	1
C5	330PF 5%	300-1331	1
C6,7	680PF	300-1680	1
C8,9	10 MF 15VDC	300-3006	2
D1,6,9	GERM. DIODE	380-0000	3
D2,3,4,5,7,8	SIL DIODE	380-1001	6
C16	.022 MF	300-2122	1







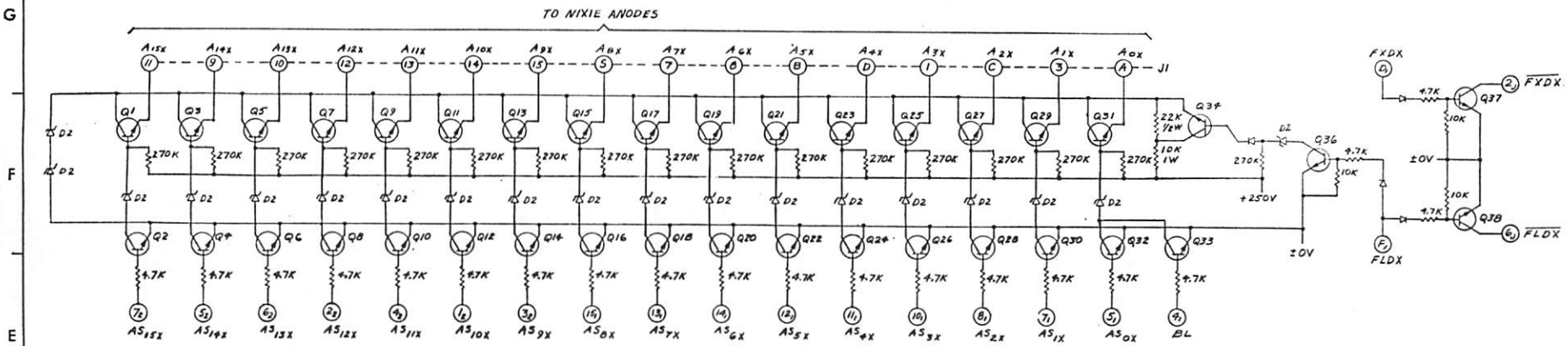
WANG LABORATORIES INC. TEWKSBURY, MASS.				
MODEL NO. 700C/720C	DRAWN	9/15/71	APP.	9/20/71
	CHECKED	9/20/71	APP.	
TITLE SCHEMATIC LOGIBLOC #5976 BUFFER BOARD				
SHT OF	DWG. NO. D 5976-1		REV.	

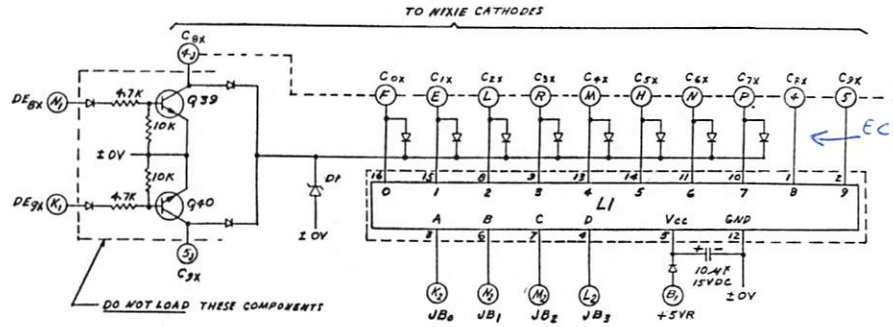
5976

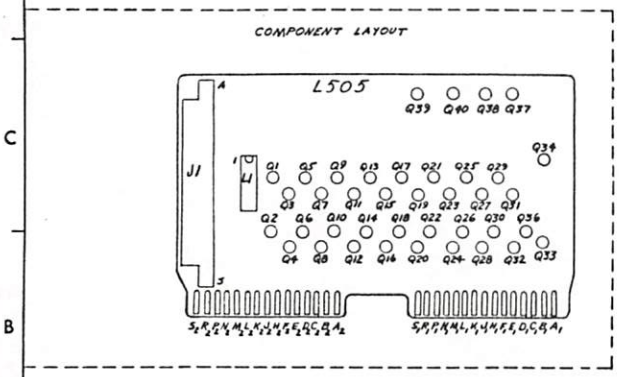
MODEL 700/720

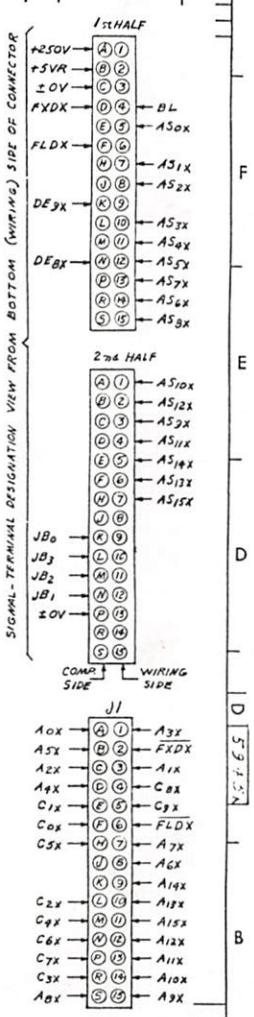
The following schematics are contained in this section in the following order:

TITLE	DRAWING #	NO. OF SHEETS	TITLE	DRAWING #	NO. OF SHEETS
335	C6046	1	5928	D5928	1
336	C6051	1	5929	D5929	1
L505X	D5945-X	1	5930	D5930	1
L505Y	D5945-Y	1	5936	D5936	1
5887	D5887	1	5937	D5937	1
5917	D5917	1	5938	D5938	1
5918	D5918	1	5975	D5975	1
5919	D5919	1	5976	D5976	1
5920	D5920	1	6038	E6038	1
5921	D5921	1	6052	D6052-1	1
5922	D5922	1	I/O CONN	B6088	1
5923	D5923	1	TY CONN	B6089	1
5924	D5924	1	P1 CONN	B6090	1
5926	D5926	1	P2 CONN	C6091	1
5927	D5927	1	P3 CONN	C6092	1









LOCATION	TYPE & MFR'R	TERM No. VCC +5VR	TERM No. ±0V	QTY
L1	SN7941N SIGNETICS CORP.	5	12	1

NOTE:
1. ALL DIODES EM403 UNLESS SPECIFIED OTHERWISE (QTY 15) D1 WL380-2127 (1) (60V) D2 WL 380-2125 (19) (110V)
2. ALL TRANSISTORS ARE 2N3490 (QTY 39)

REVISION	BY	DATE	APPROVED BY	DATE
1 PER ECN 1510 REMOVED 16-10K RESISTORS, ADDED 16-390PF CAP. APP. RJT		6/30/70		
2 PER ECN 1512 ADDED .0047 uF 50V CAP. CER. APP. RJT		8/6/70		
3 PER ECN "1756 ADDED 390 PF TO Q36 AND CHANGED 10K TO 180K APP. RJT		9/10/70		
4 PER ECN "1637 ADDED TRANSFER JUMPER WIRE APP. RJT		9/17/70		
5 PER ECN "1835 REVISED WITH EXTENSIVE CHANGE APP. RJT		11-20-70		
6 PER ECN 2480 REMOVED D1 210V ZENER ADDED 2-D2 110V ZENERS APP.- SKH		8/31/71		
7 PER ECN "2595 D1 WAS 380-2125 ADDED 10 DIODES EM403 APP.- SKH		11/29/71		

WANG PART NO	ITEM	QTY	NAME
			WANG LABORATORIES INC. TEWKSBURY, MASS. U.S.A.
MATERIAL			MODEL NO 700
FINISH			SCALE SHT OF

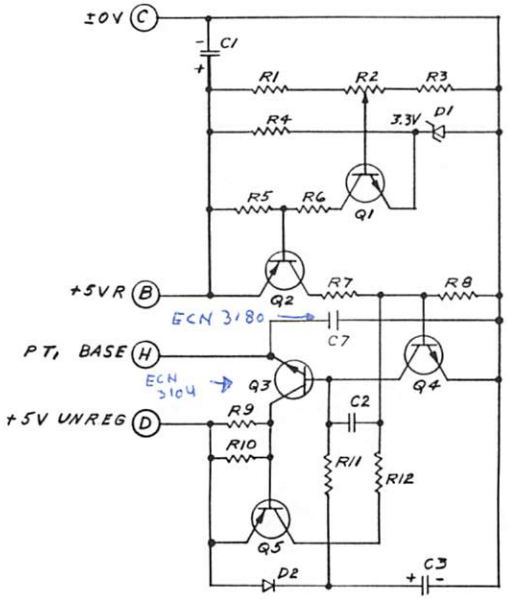
MATERIAL	DESCRIPTION
DWN JRB	12/4/69 E ENGR SKH 5/17/71
CHK RJT	9-13-70 M ENGR
E C CONTROL	MFG ENGR
TITLE SCHEMATIC LOGIBLOC "X" NIXIE CONTROL ANODE SWITCH	
WANG PART NUMBER	D 5945-X 7 SIZE DRAWING NUMBER REV

DO NOT SCALE

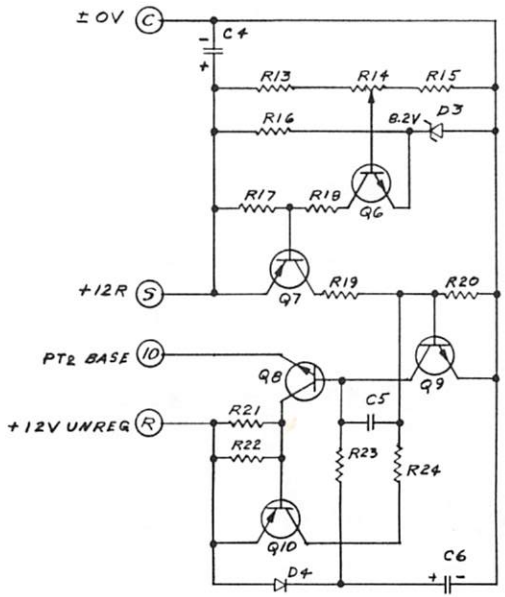
HOLE LEGEND		
DRILLED OR PUNCHED HOLE TOLERANCES:	HOLE DIA. .0135 to .125	TOL. +.003 −.001
	.126 to .250	+.004 −.001
	.251 to .500	+.005 −.001

IDENT.	DESCRIPTION	QTY.
A		

+5 V REGULATOR

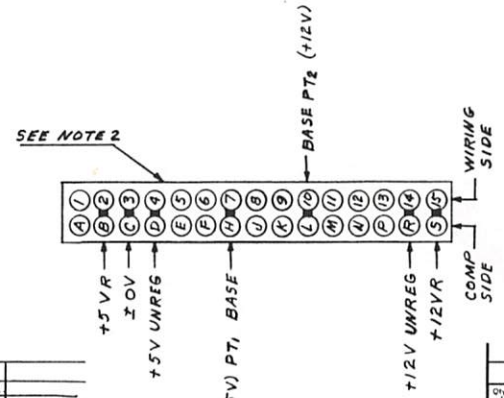


+12 V REGULATOR



COMP.	SIZE / TYPE	QTY
R1, 13	220 Ω	2
R2, 14	1K POT.	2
R3, 15	1-5 K	2
R4	100 Ω ½W	1
R5, 8, 17, 20	4.7K	4
R6, 18	1K	2
R7, 19	2.2K	2
R21, 22	10 Ω	2
R23	330 Ω ½ W	1
R12, 24	470 Ω ½W	2
C1, 3	100 MF/15V	2
C2	0.1 MF 10V CER.	1
C4	50 MF/50 V	1
C6	50 MF/50 V	1
D1	IN746A	1
D2, 4	IN3253	2
Q1, 4, 6, 9	RCA35224	4
Q2, 5, 7, 10	GT544	4
Q8	2N5189	1
D3	IN756A	1
R9	2.7 Ω	1
R16	330 Ω 10%	1
Q3	2N3725	1
C7	.01 MF 25V	1
R11	100 Ω ½W	1
C5	.1 MF 20V CER.	1

ECN 2010



NOTE:
1. ALL RESISTORS ¼ W UNLESS SPECIFIED OTHERWISE
2. PC SOCKET IS SOLDERED ON TO PC6052

NO	REVISION	DATE	BY
1	PER ECN 1197 R9 WAS 10Ω DELETED R10 Ω	4-29-70	LEG
2	PER ECN 2010 R16 WAS 330 Ω ½W NOW 330 Ω ¼W 10% APP. RJT	3/15/71	PD
3	PER ECN #3104 Q3 WAS 2N5189 APP'D SKH	8/2/72	PD
4	PER ECN #3180 ADDED C7 CAP. APP'D. SKH	9-25-72	PD
5	PER ECN #3359 R11 WAS 330 Ω ¼W APP'D SKH	11-3-72	PD

WANG PART NO.	ITEM	QTY.	NAME
Qty. Per Unit	FIRST USED ON	ASSY USED ON	

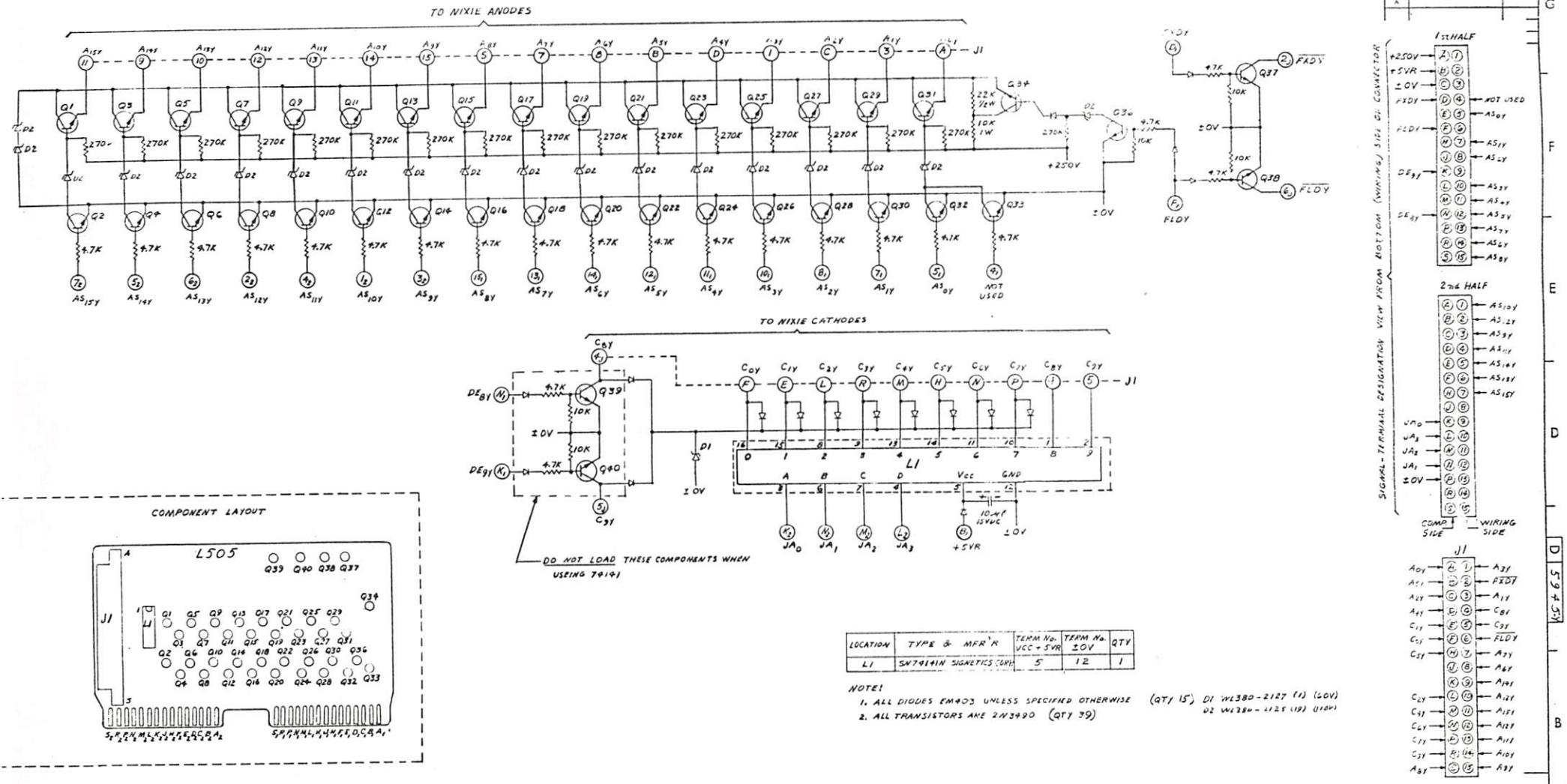
WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.

MATERIAL	MODEL NO. 700 SYSTEM
	SEE ENGRG SPECIFICATIONS No.
FINISH	TOL. EX. AS NOTED XX ± .010 FRAC ± 1/64 XXX ± .005 ANG ± 1°30' FINISH √
	SCALE SHT. OF

MATERIAL	DESCRIPTION		
BY	DATE	APPROVED BY	DATE
DWN PD	9/29/69	E ENGR SKH	2/18/70
CHK RJT	3/13/70	M ENGR	
E. C. CONTROL		MFG ENGR	

TITLE +5V REGULATOR P.C. #335
+12V REGULATOR

WANG PART NUMBER	SIZE	DRAWING NUMBER	REV
	C	6046	5

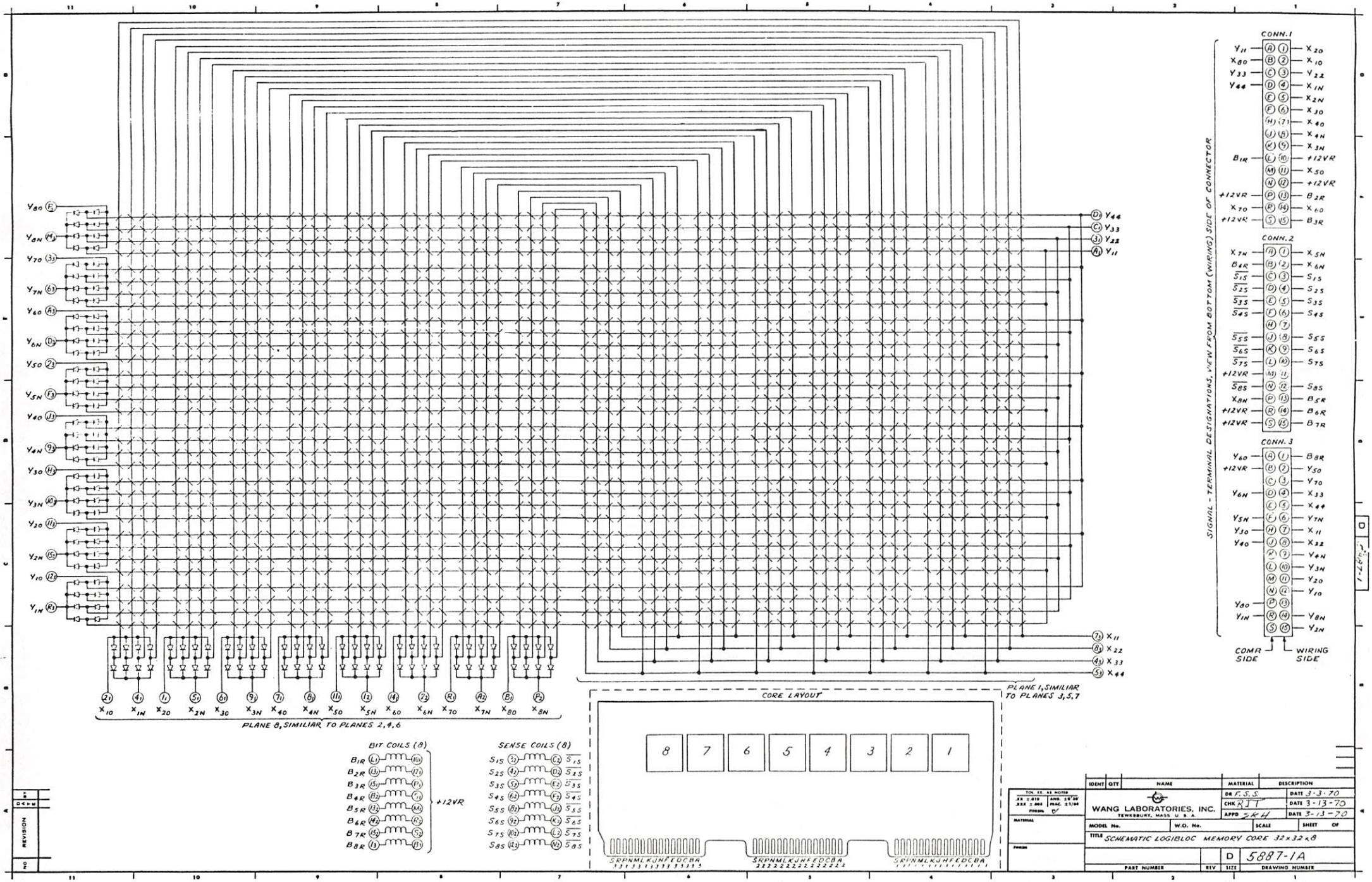


LOCATION	TYPE & MFR'R	TERM No. VCC + 5VR	TERM No. ±0V	QTY
L1	SN7941N SIGNETICS CORP	5	12	1

NOTE!
1. ALL DIODES EM403 UNLESS SPECIFIED OTHERWISE (QTY 15) D1 WE380-2127 (1) (60V) D2 WE380-2125 (19) (10V)
2. ALL TRANSISTORS ARE 2N3490 (QTY 39)

ZONE	LTR	DESCRIPTION	DATE	APPROVED
	1	PER ECN 1570 REMOVED 16-10K RESISTORS, ADDED 16-390Ω RES. APP. RJT	5/1/70	
	2	PER ECN 1573 ADDED .0047 MF 500V CER. ANA. RJT	5/13/70	
	3	PER ECN 1634 ALTERED SHAPE TO 5" AND CHANGED INK TO 180K ANA RJT	7/17/70	
	4	PER ECN 1635 ALTERED HOLE SIZE AT PIN 1 AND 2 AND ADDED NOTE TO L1 APP. RJT	7/17/70	
	5	PER ECN 1835 REVISED TITLE & CHANGED ANA RJT	10/14/70	
	6	PER ECN 2440 REMOVED D3 220V DIODE ADDED D2 10V ZENER) APP. RJT	9/21/71	
	7	PER ECN 2528 ADDED 16 DIODES ANA - J.K.H	11/29/71	

WANG PART NO	ITEM	QTY	NAME	MATERIAL	DESCRIPTION
			WANG LABORATORIES, INC.	DWN	DATE / APPROVED BY / DATE
				CHK	
				E.C. CONTROL	MFG ENGR
			MODEL NO 700	TITLE SCHEMATIC LOGIBLOC L505 Y NIXIE CONTROL ANODE SWITCH	
			FINISH	SIZE D	DRAWING NUMBER 5945-Y	REV 6



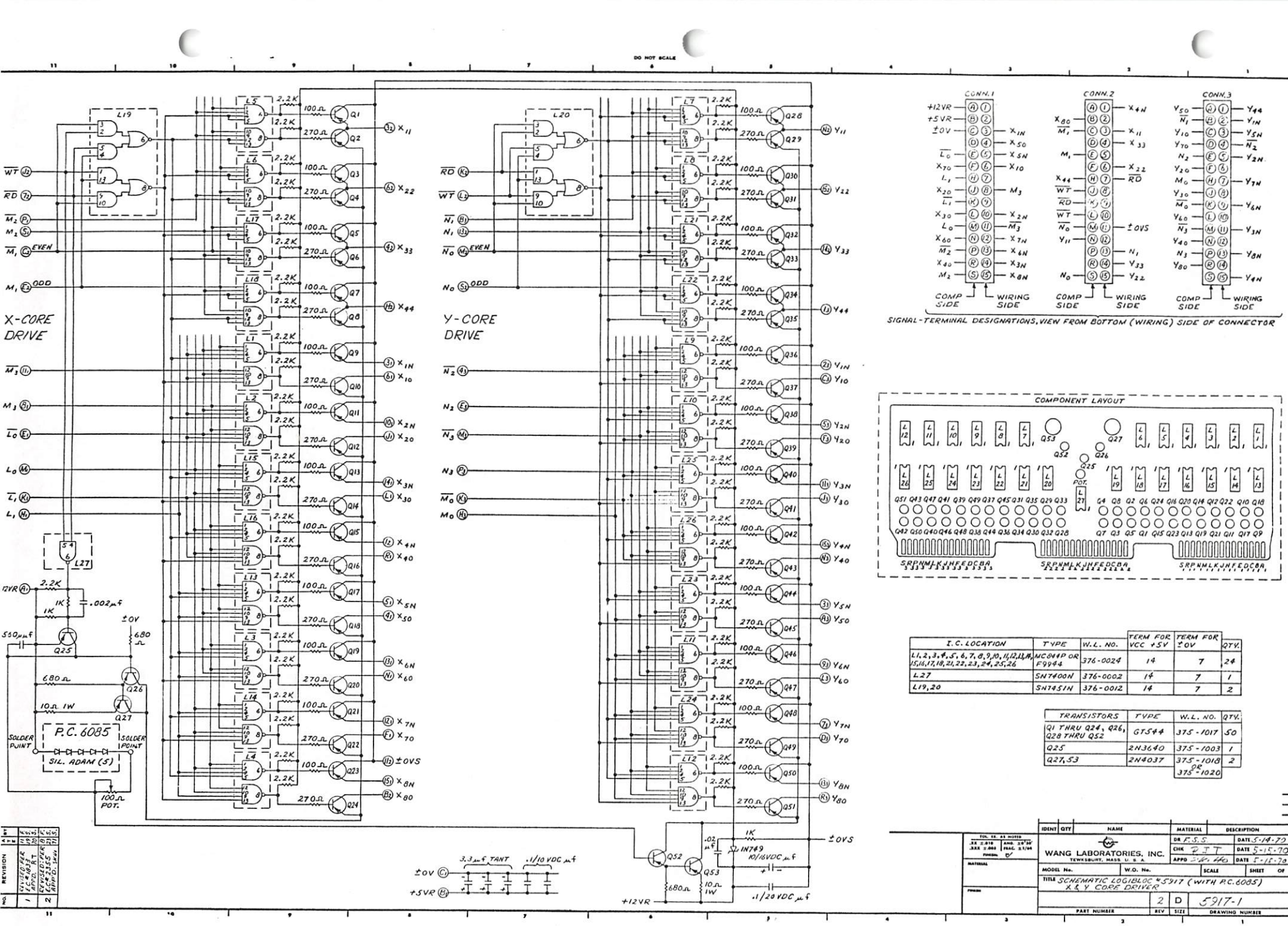
IDENT	QTY	NAME	MATERIAL	DESCRIPTION
			DR F.S.S.	DATE 3-3-70
		WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.	CHK R.J.T.	DATE 3-13-70
			APPD S.K.H.	DATE 3-13-70

MODEL No. | W.O. No. | SCALE | SHEET OF

TITLE SCHEMATIC LOGIBLOC MEMORY CORE 32×32×8

PART NUMBER | REV | SIZE D | DRAWING NUMBER 5887-1A

REVISION | BY | NO



SIGNAL-TERMINAL DESIGNATIONS, VIEW FROM BOTTOM (WIRING) SIDE OF CONNECTOR

I.C. LOCATION	TYPE	W.L. NO.	TERM FOR VCC +5V	TERM FOR ±0V	QTY.
L1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,21,22,23,24,25,26	MC844P OR F9944	376-0024	14	7	24
L27	SN7400N	376-0002	14	7	1
L19,20	SN7451N	376-0012	14	7	2

TRANSISTORS	TYPE	W.L. NO.	QTY.
Q1 THRU Q24, Q26, Q28 THRU Q52	GT544	375-1017	50
Q25	2N3640	375-1003	1
Q27,53	2N4037	375-1018 OR 375-1020	2

NO.	REVISION	BY	DATE
1	REVISED PER ECR 1085	W.S.	11-5-69
2	REVISED PER ECR 2355 APPD. 3-4-70	P.C.	3-5-70

IDENT	QTY	NAME	MATERIAL	DESCRIPTION

WANG LABORATORIES, INC.
TEWKSBURY, MASS. U.S.A.

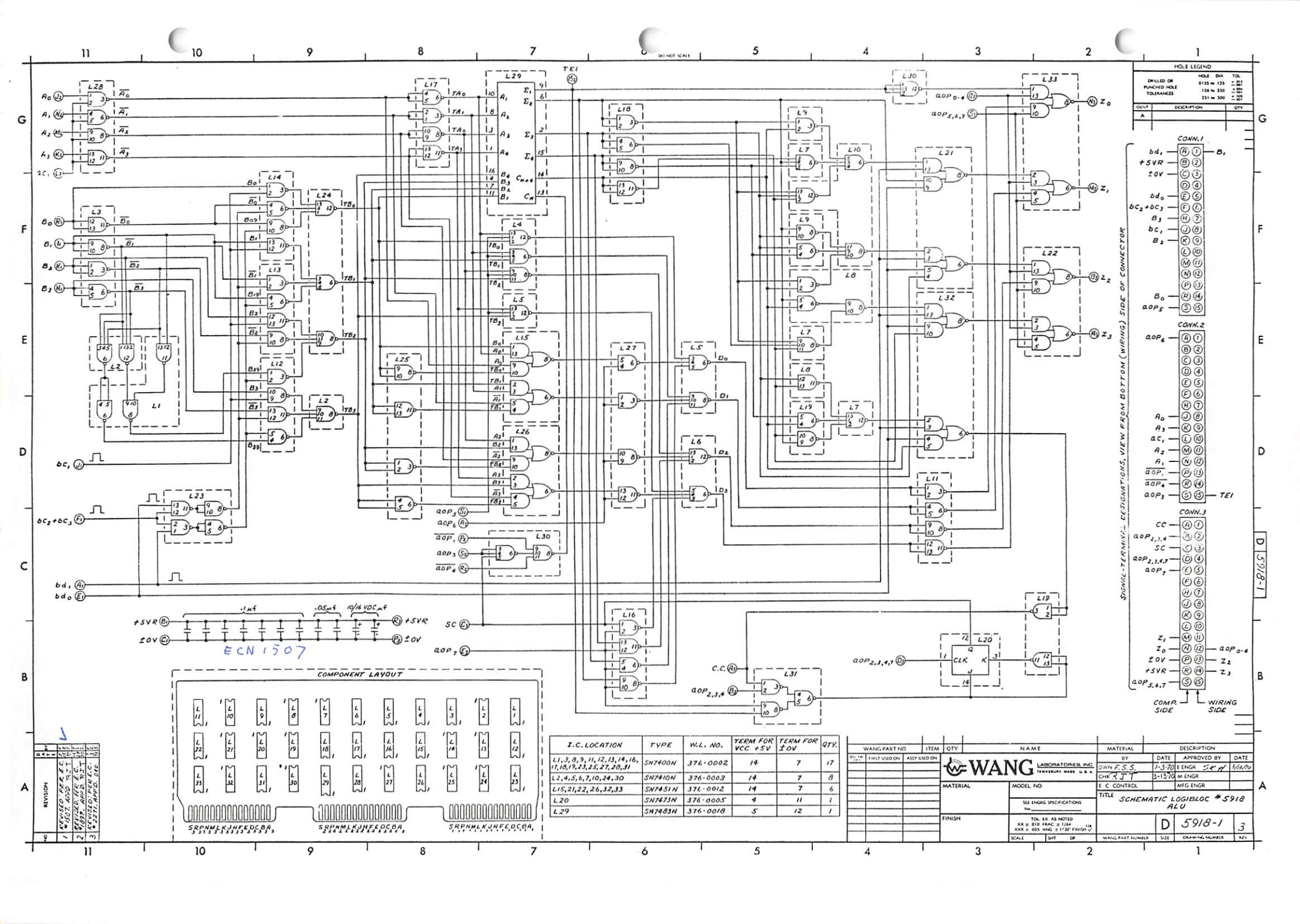
DR F.S.S. DATE 5-14-70
CHK D.J.T DATE 5-15-70
APPD D.J.T. DATE 5-15-70

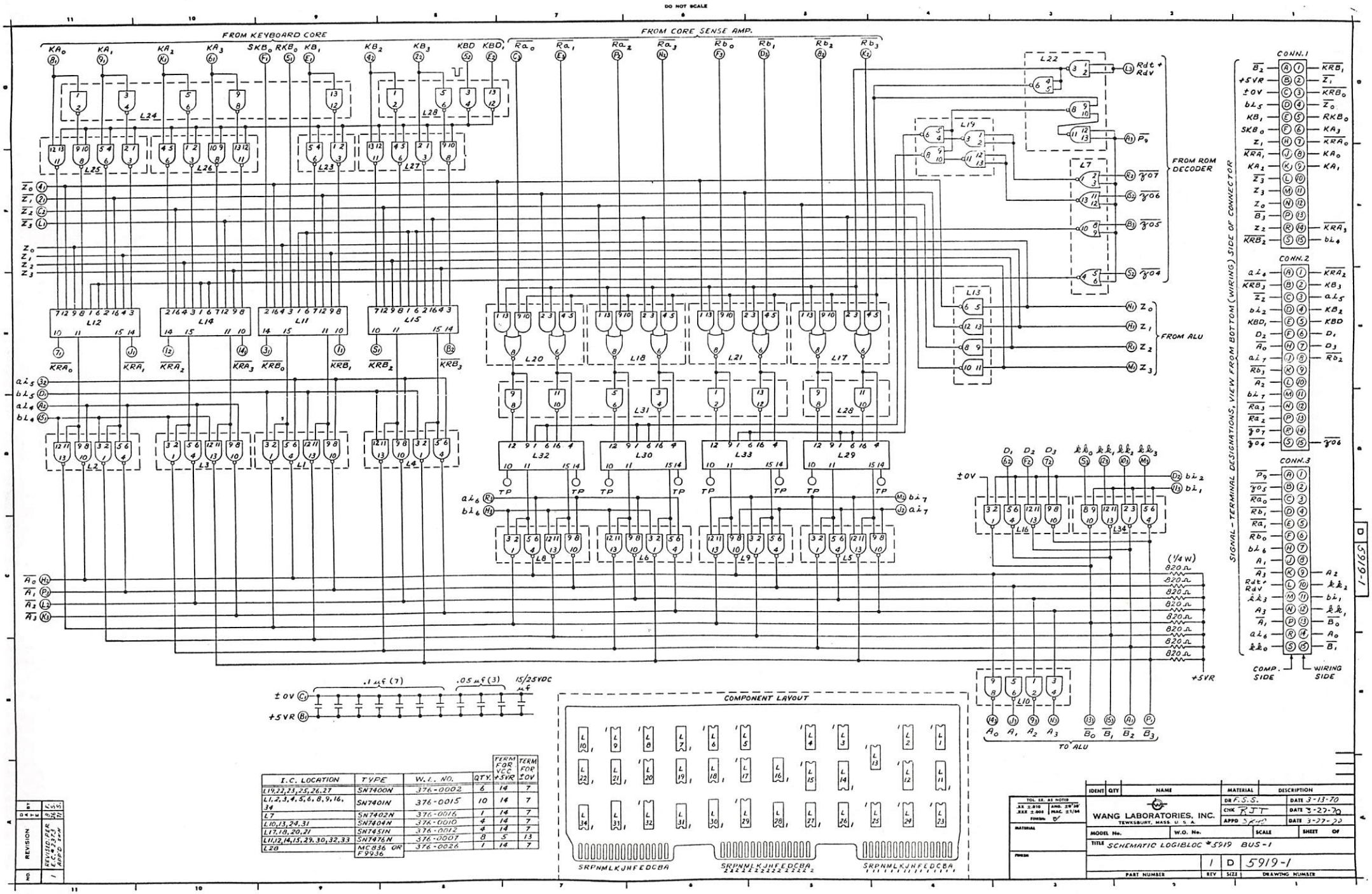
MODEL No. | W.O. No. | SCALE | SHEET OF

TITLE SCHEMATIC LOGIBLOC 5917 (WITH P.C.6085)
X & Y CORE DRIVER

PART NUMBER | REV 2 | SIZE D | DRAWING NUMBER 5917-1

D 5917-1


DO NOT SCALE
HOLE LEGEND
DRILLED OR PUNCHED HOLE TOLERANCES
HOLE DIA TOL
0135 to 125 ± .003
128 to 250 ± .004
251 to 500 ± .005
IDENT DESCRIPTION QTY
A
CONN.1
CONN.2
CONN.3
SIGNAL-TERMINAL DESIGNATIONS, VIEW FROM BOTTOM (WIRING) SIDE OF CONNECTOR
COMP. SIDE
WIRING SIDE
.1μf
.05μf
10/16 VDC μf
+5VR
±0V
SC
aop7
ECN 1507
COMPONENT LAYOUT
SRPNMLKJHFEDCBA
I.C. LOCATION | TYPE | W.L. NO. | TERM FOR VCC +5V | TERM FOR ±0V | QTY.
L1,3,8,9,11,12,13,14,16,17,18,19,23,25,27,28,31 | SN7400N | 376-0002 | 14 | 7 | 17
L2,4,5,6,7,10,24,30 | SN7410N | 376-0003 | 14 | 7 | 8
L15,21,22,26,32,33 | SN7451N | 376-0012 | 14 | 7 | 6
L20 | SN7473N | 376-0005 | 4 | 11 | 1
L29 | SN7483N | 376-0018 | 5 | 12 | 1
REVISION
1 REVISED PER E.C. #1201, ADD TEST
2 REVISED PER E.C. #1261, ADD TEST
3 REVISED PER E.C. #2271, APPD. 5-13-70
WANG PART NO | ITEM | QTY | NAME
FIRST USED ON | ASSY USED ON
WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.
MATERIAL | DESCRIPTION
MODEL NO
SEE ENGRG SPECIFICATIONS No.
FINISH
TOL. EX. AS NOTED XX ± .010 FRAC. ± 1/64 XXX ± .005 ANG. ± 1°30' FINISH √
SCALE SHT OF
BY | DATE | APPROVED BY | DATE
DWN F.S.S. 1-3-70 E ENGR. 5/13/70
CHK R.J.T. 3-13-70 M ENGR
E C CONTROL | MFG ENGR
TITLE SCHEMATIC LOGIBLOC #5918 ALU
WANG PART NUMBER D 5918-1 SIZE DRAWING NUMBER 3 REV



I.C. LOCATION	TYPE	W.I. NO.	QTY.	TERM FOR VCC +5VR	TERM FOR ±0V
L19,22,23,25,26,27	SN7400N	376-0002	6	14	7
L1,2,3,4,5,6,8,9,16,34	SN7401N	376-0015	10	14	7
L7	SN7402N	376-0016	1	14	7
L10,13,24,31	SN7404N	376-0010	4	14	7
L17,18,20,21	SN7451N	376-0012	4	14	7
L11,12,14,15,29,30,32,33	SN7476N	376-0007	8	5	13
L28	MC836 OR F9936	376-0026	1	14	7

NO	REVISION	DATE	BY
1	REVISED PER E.C.#3343 APPD 3/27	3/27	R.S.

IDENT	QTY	NAME	MATERIAL	DESCRIPTION
			DR F.S.S.	DATE 3-13-70
			CHK RJT	DATE 3-22-70
			APPD JKH	DATE 3-27-70

TOL. EX. AS NOTED .XX ±.010 .XXX ±.005 ANG. 20' PRAC. ±1/64 FINISH: 8/

MATERIAL

FINISH

WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.

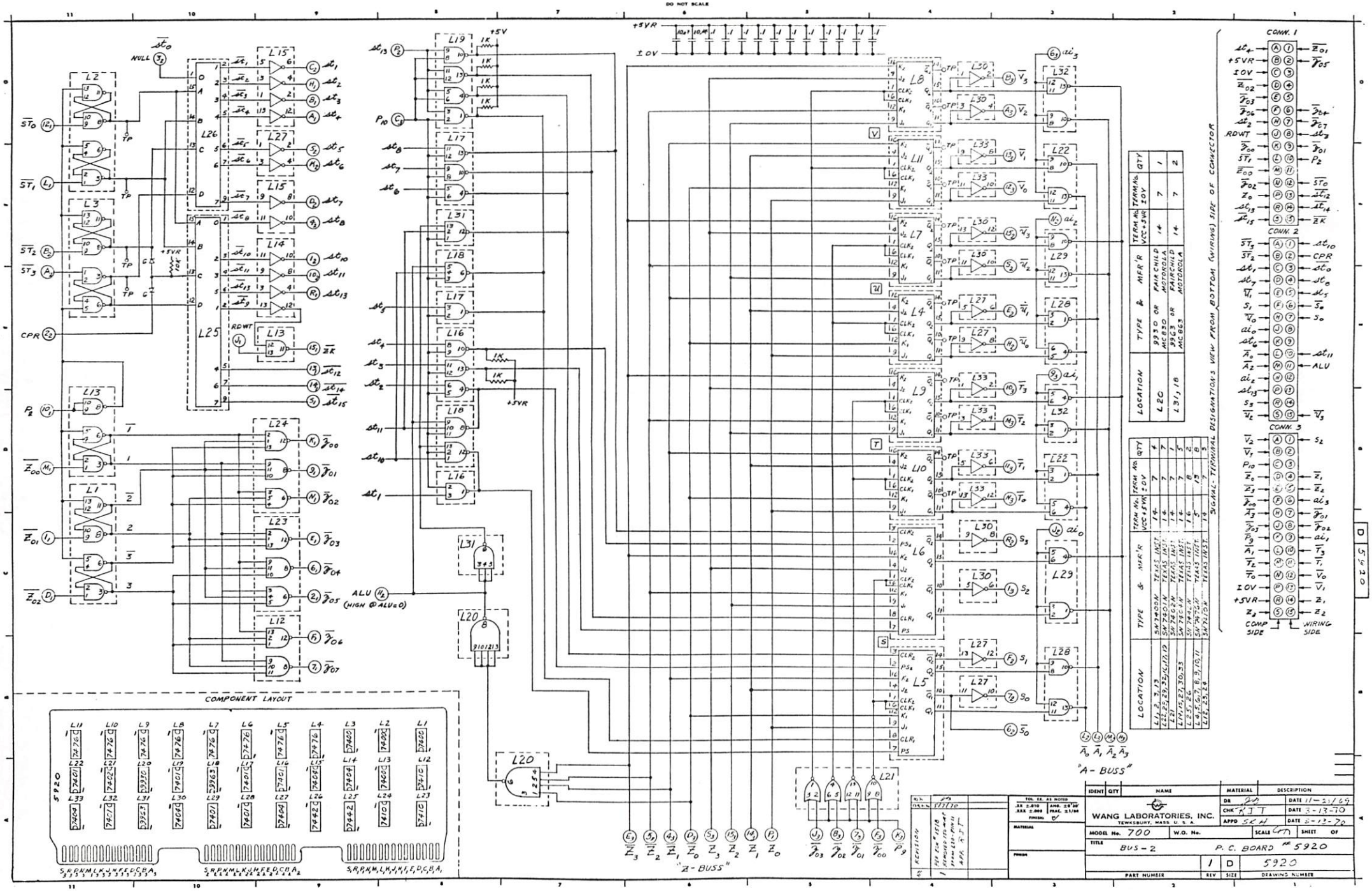
MODEL No. | W.O. No. | SCALE | SHEET OF

TITLE SCHEMATIC LOGIBLOC #5919 BUS-1

PART NUMBER | REV | SIZE | DRAWING NUMBER

1 | D | 5919-1

D 5919-1



LOCATION	TYPE & MFR'R	TERM NO. VCC=+5VR	TERM NO. -10V	QTY
L2C	9930 OR FAIRCHILD MC830	14	7	1
L3I, 1B	9963 OR FAIRCHILD MC863	14	7	2

LOCATION	TYPE & MFR'R	TERM NO. VCC=+5VR	TERM NO. -10V	QTY
L1, 2, 3, 13	SN7400N TEXAS INST.	14	7	4
L22, 25, 29, 32, L7, L12, L20	SN7401N TEXAS INST.	14	7	7
L21	SN7402N TEXAS INST.	14	7	1
L14, 15, L24, 30, 33	SN7404N TEXAS INST.	14	7	5
L23, 26	SN7410N TEXAS INST.	14	7	2
L4, 5, L17, 18, 19, 10, 11	SN7420N TEXAS INST.	14	7	6
L16, 27, 28	SN7438N TEXAS INST.	14	7	3

IDENT	QTY	NAME	MATERIAL	DESCRIPTION
		WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.	DR	DATE 11-21-69
			CHK KJT	DATE 3-13-70
			APPD SRH	DATE 3-13-70
MODEL No. 700		W.O. No.	SCALE 4/1	SHEET OF
TITLE BUS-2			P.C. BOARD № 5920	
PART NUMBER			REV 1	SIZE D
			DRAWING NUMBER 5920	

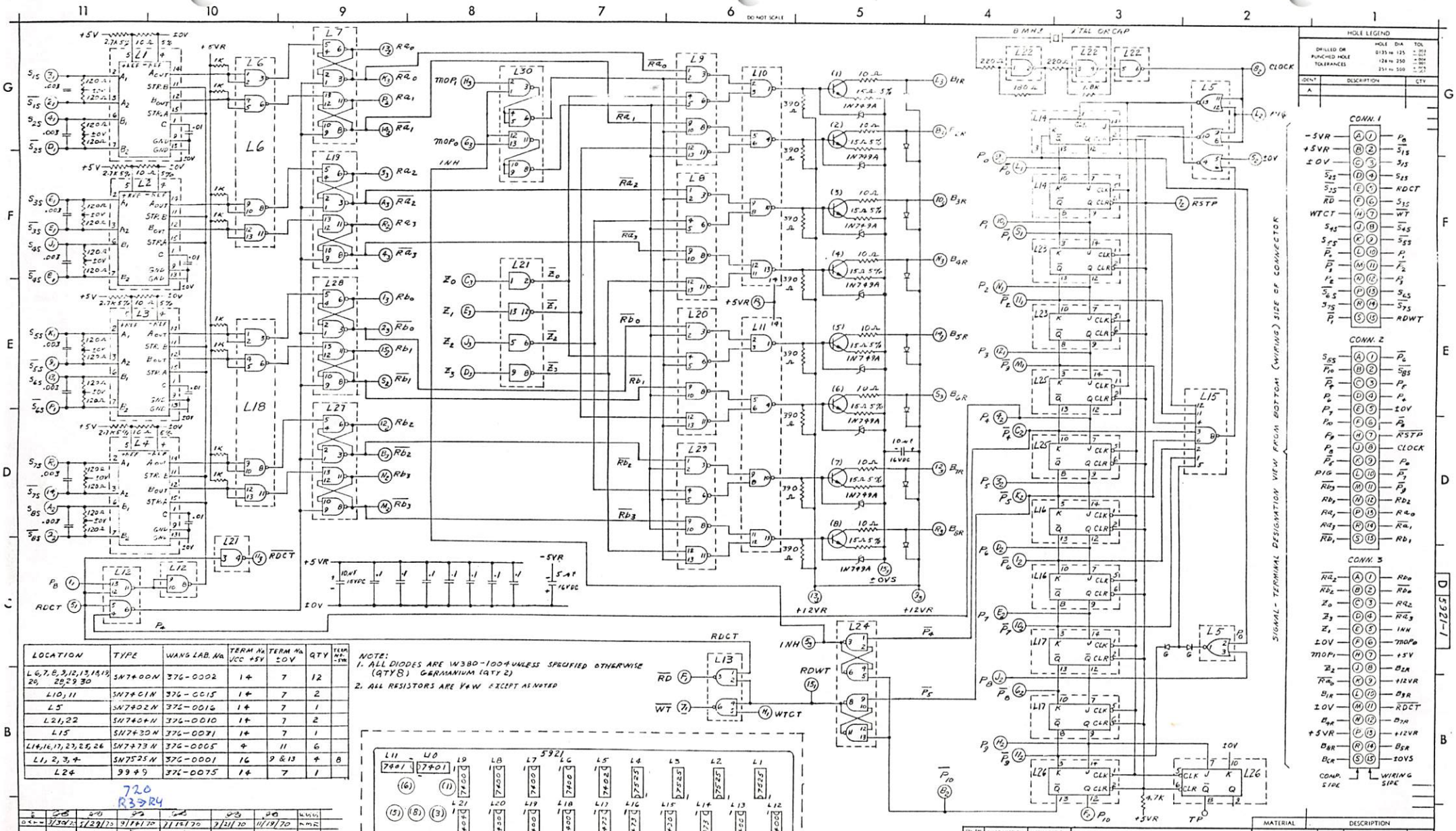
TOL. EX. AS NOTED .XX ± .010 .XXX ± .005 FINISH

MATERIAL

FINISH

REVISIONS

NO.	DATE	REVISION
1	3/13/70	PER ECO #1518 ADD L20 & L21 PER ECO #1519 ADD L22 PER ECO #1520 ADD L23 PER ECO #1521 ADD L24 PER ECO #1522 ADD L25 PER ECO #1523 ADD L26 PER ECO #1524 ADD L27 PER ECO #1525 ADD L28 PER ECO #1526 ADD



LOCATION	TYPE	WANG LAB. No	TERM No +5V	TERM No ±0V	QTY	TERM No. -5V
L6,7,8,3,12,13,18,19, 20, 28,29 30	SN7400N	376-0002	14	7	12	
L10, 11	SN7401N	376-0015	14	7	2	
L5	SN7402N	376-0016	14	7	1	
L21,22	SN7404N	376-0010	14	7	2	
L15	SN7430N	376-0031	14	7	1	
L14,16,17,23,25,26	SN7473N	376-0005	4	11	6	
L1, 2, 3, 4	SN75251N	376-0001	16	9 & 13	4	8
L24	9949	376-0075	14	7	1	

NOTE:
1. ALL DIODES ARE W380-1004 UNLESS SPECIFIED OTHERWISE (QTY 8) GERMANIUM (QTY 2)
2. ALL RESISTORS ARE ¼W EXCEPT AS NOTED

720
R3R4

REVISION	ECO	DWN	CHK	ENGR	MFG ENGR	QA
PER ECN #1757 DIODES WERE W380-1001 AMP. RJT	3/30/70	5/29/70	9/16/70	11/25/70	2/23/70	11/19/70
PER ECN #1710 CIRCUITS CONSTANT CHANGES. APP. RJT						
PER ECO #1737 CHANGE PRINTED WIRING TO ETCH AND SILK ASSIGNATE APP. RJT						
PER ECO #1736 WAS .003 NOW .003 TO. APP. RJT						
PER ECO #1837 DATE REVISED FROM 6-8-70 TO 8-20-70 APP. RJT						
REVISED PER ECO #2276 APP. [illegible]						

HOLE LEGEND

DRILLED OR PUNCHED HOLE TOLERANCES	HOLE DIA	TOL
	0135 to 125	±.003
	126 to 250	±.004
	251 to 500	±.005

IDENT	DESCRIPTION	QTY
A		

MATERIAL	DESCRIPTION
BY	DATE / APPROVED BY / DATE
DWN [illegible]	1/5/70 E ENGR SKH 3/11/70
CHK RJT	3-13-70 M ENGR
E C CONTROL	MFG ENGR

WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.

MATERIAL

MODEL NO 700/720

SEE ENGRG SPECIFICATIONS No. ____

TOL EX AS NOTED
XX ± .010 FRAC ± 1/64
XXX ± .005 ANG ± 1°30' FINISH √

SCALE (1/1) SHT OF

TITLE SCHEMATIC LOGIBLOC "5921
SENCE AMP OSC COREBIT DRIVE

D 5921-1 7

WANG PART NUMBER 5921 DRAWING NUMBER REV

D 5921-1

DO NOT SCALE

I.C. LOCATION	TYPE	W.L. NO.	QTY.	TERM FOR VCC +5V	TERM FOR ±0V
L1,2,3,4,5,6,7,8,9,10, 11,22	SN7400N	37-0002	12	14	7
L21	SN7402N	37-0011	1	14	7
L17,23,24,25,26,29,30	SN7404N	37-0012	7	14	7
L18,19,31	SN7410N	37-0013	3	14	7
L14	SN7420N	37-0004	1	14	7
L12,15,16,20	SN7442N	37-0008	4	16	8

COMPONENT LAYOUT

SRPNMLKJHFEDCBA

SIGNAL-TERMINAL DESIGNATIONS VIEW, FROM BOTTOM (WIRING) SIDE OF CONNECTOR

CONN.1

$\overline{MOP_0}$	A	1	$\overline{AL_1}$
+5VR	B	2	$\overline{MOP_3}$
±0V	C	3	$\overline{mop_7}$
$\overline{MOP_1}$	D	4	mop_{10}
mop_9	E	5	mop_{11}
Rdt + Rdv	F	6	mop_{12}
$\overline{MOP_2}$	H	7	mop_8
bd_0	J	8	ac_1
$\overline{BD}$	K	9	$a\dot{k}_4$
bd_1	L	10	$a\dot{k}_7$
mop_{13}	M	11	mop_1
	N	12	mop_2
P_2	P	13	mop_3
RDCT	R	14	mop_4
Wt CT	S	15	mop_5

CONN.2

$\overline{AL_0}$	A	1	mop_0
$\overline{kk_3}$	H	2	$a\dot{k}_5$
$\overline{KK_3}$	C	3	$a\dot{k}_4$
$k\dot{k}_3$	D	4	$a\dot{k}_3$
$\overline{KK_2}$	E	5	$a\dot{k}_0$
$\overline{KK_1}$	F	6	$a\dot{k}_1$
$k\dot{k}_2$	H	7	$a\dot{k}_2$
$\overline{k\dot{k}_2}$	J	8	$\overline{mop_{15}}$
$k\dot{k}_1$	K	9	
$\overline{k\dot{k}_1}$	L	10	
$k\dot{k}_3$	M	11	
$\overline{k\dot{k}_0}$	N	12	
$\overline{KK_0}$	P	13	mop_{14}
$\overline{BL_0}$	R	14	
BIT	S	15	bi_7

CONN.3

$\overline{AL_2}$	H	1	bi_5
$\overline{BL_2}$	B	2	bi_6
bi_4	C	3	bi_1
bi_2	D	4	$\overline{AOP_1}$
$\overline{BL_1}$	E	5	$\overline{aop_4}$
aop_7	F	6	aop_5
aop_6	H	7	aop_3
$\overline{aop_1}$	J	11	aop_2
bc_1	K	9	$\overline{AOP_0}$
$\overline{BC_0}$	L	10	$aop_{5,6,7}$
aop_{0-4}	M	11	$\overline{AOP_2}$
$bc_2 + bc_3$	N	12	$\overline{AC}$
±0V	P	13	$aop_{2,3,4}$
$bc_0 + bc_2$	R	14	$\overline{Pq}$
$\overline{BC_1}$	S	15	$aop_{2,3,4,7}$

COMP. SIDE — WIRING SIDE

±0V — .01 µf — .01 µf — .01 µf — .01 µf — +5VR

10/16 VDC µf

ECN 1552

ECN 1707

ECN 1507

NO	REVISION	BY	DATE	APPD
1	REVISED PER ECN-LOGIBLOC APPR. KJT	K.S.S.	9/25/69	KJT
2	REVISED PER ECN-1707 APPD. KJT	K.S.S.	10/28/69	KJT
3	REVISED PER ECN-LOGIBLOC APPD. KJT	K.S.S.	3/13/70	KJT

IDENT	QTY	NAME	MATERIAL	DESCRIPTION
		WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.		DR K.S.S. DATE 9/25/69
				CHK K.J.T. DATE 3/13/70
				APPD S.K.H. DATE 3/13/70

MODEL No. | W.O. No. | SCALE | SHEET OF

TITLE SCHEMATIC LOGIBLOC #5922 ROM DECODER

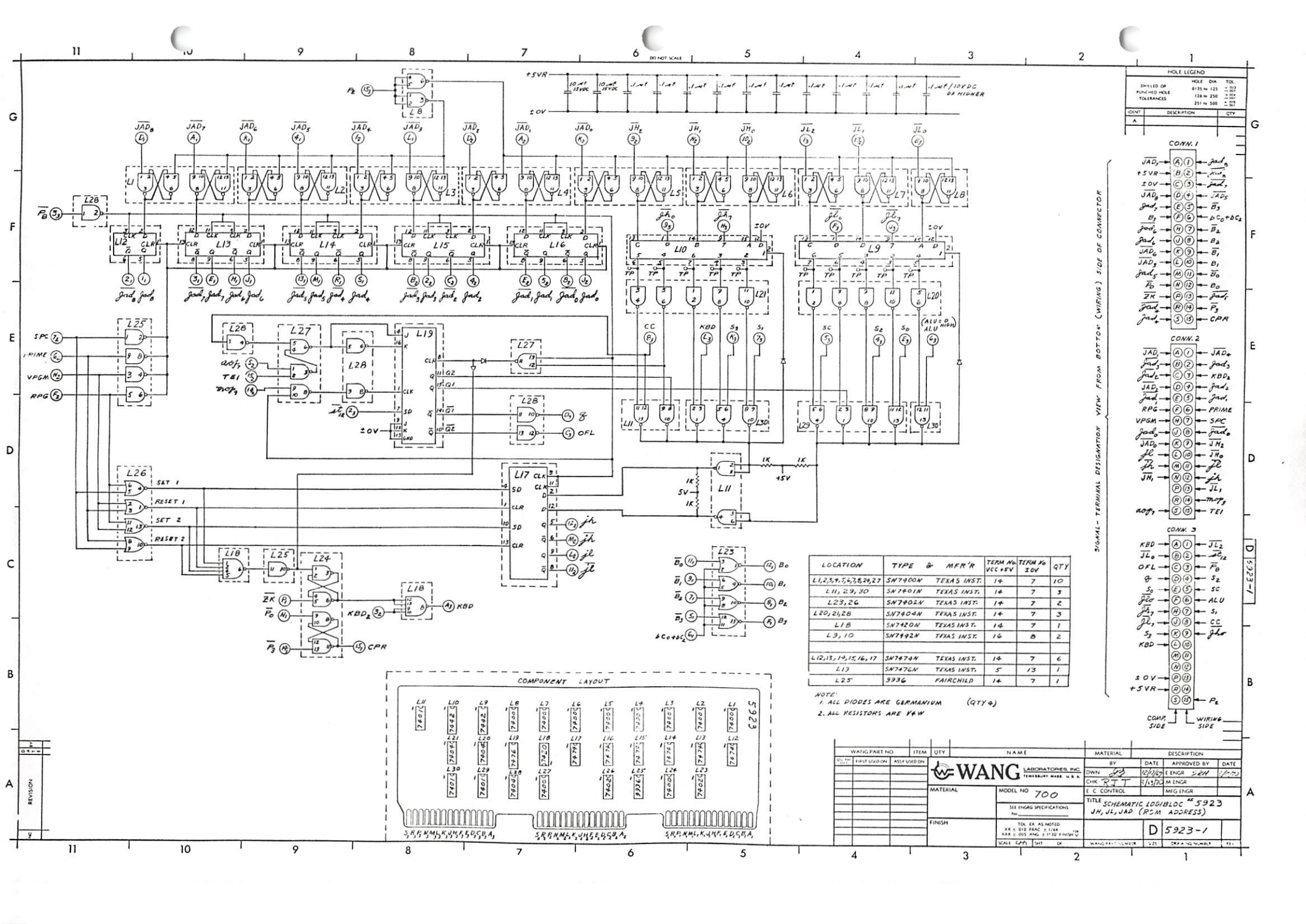
PART NUMBER 3 | REV D | SIZE 5922-1 | DRAWING NUMBER

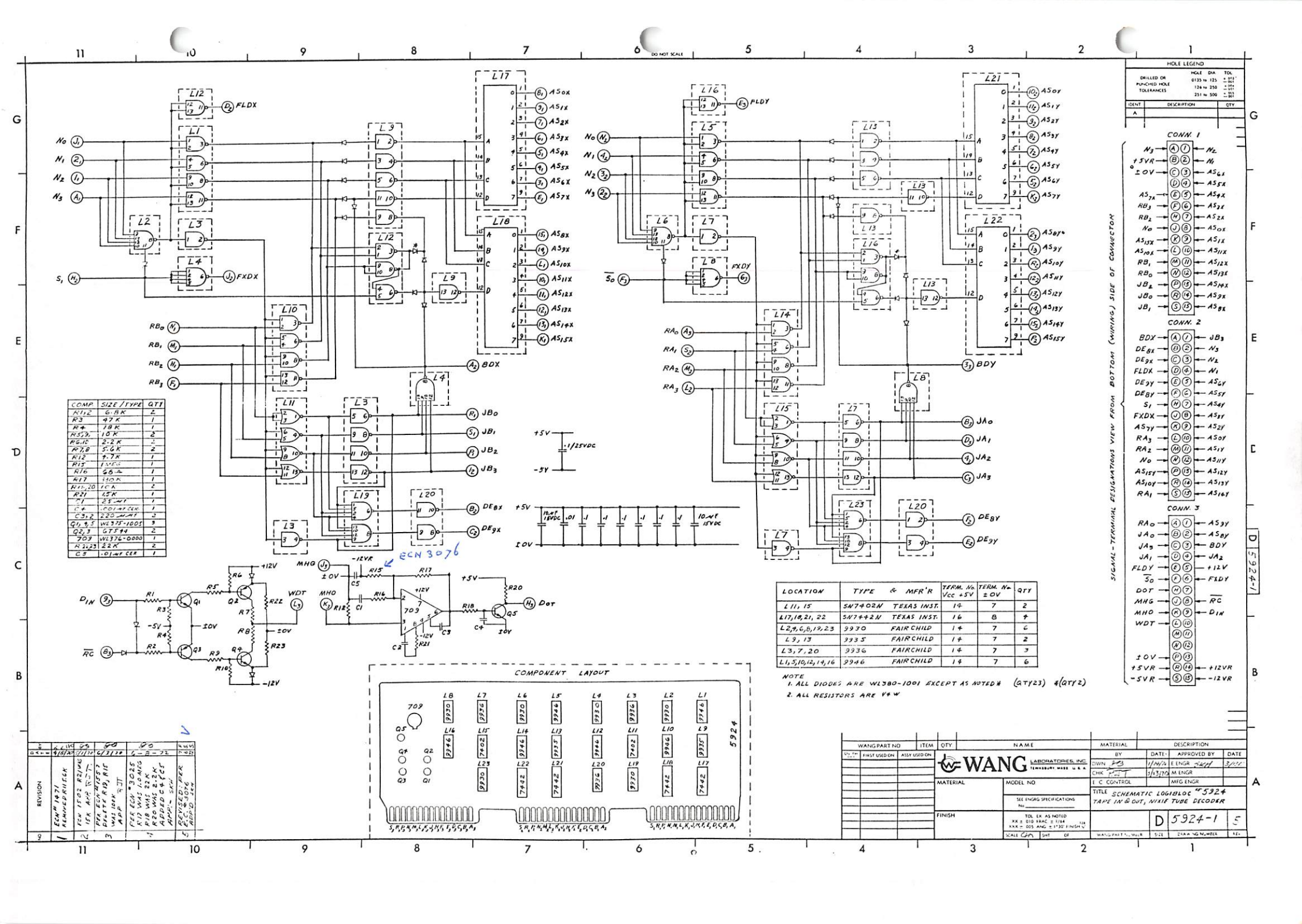
TOL. EE AS NOTED .XX ± .010 ANG. ± 0° 30' .XXX ± .005 FRAC. ± 1/64 FINISH 0

MATERIAL

FINISH

D 5922-1


DO NOT SCALE
+5VR
±0V
L1 L2 L3 L4 L5 L6 L7 L8
L10 L9
L11
L12 L13 L14 L15 L16
L17
L18
L19
L20 L21
L23
L24
L25
L26
L27
L28
L29 L30
SET 1
RESET 1
SET 2
RESET 2
KBD
CPR
OFL
PRIME
SPC
VPGM
RPG
TEI
CC
KBD
SC
ALU
(ALU=0 ALU HIGH)
HOLE LEGEND
DRILLED OR PUNCHED HOLE TOLERANCES
HOLE DIA TOL
.0135 to .125 ±.002
.126 to .250 ±.004
.251 to .500 ±.005
IDENT DESCRIPTION QTY
CONN. 1
CONN. 2
CONN. 3
SIGNAL-TERMINAL DESIGNATION VIEW FROM BOTTOM (WIRING) SIDE OF CONNECTOR
COMP. SIDE
WIRING SIDE
COMPONENT LAYOUT
5923
LOCATION TYPE & MFR'R TERM No VCC +5V TERM No IOV QTY
L1,2,3,4,5,6,7,8,26,27 SN7400N TEXAS INST. 14 7 10
L11, 29, 30 SN7401N TEXAS INST. 14 7 3
L23, 26 SN7402N TEXAS INST. 14 7 2
L20, 21, 28 SN7404N TEXAS INST. 14 7 3
L18 SN7420N TEXAS INST. 14 7 1
L9, 10 SN7442N TEXAS INST. 16 8 2
L12,13,14,15,16,17 SN7474N TEXAS INST. 14 7 6
L19 SN7476N TEXAS INST. 5 13 1
L25 9336 FAIRCHILD 14 7 1
NOTE:
1. ALL DIODES ARE GERMANIUM (QTY 4)
2. ALL RESISTORS ARE ¼W
WANG PART NO ITEM QTY NAME MATERIAL DESCRIPTION
WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.
MATERIAL MODEL NO 700
FINISH
BY DATE APPROVED BY DATE
DWN 12/12/69 E ENGR 12/12/69
CHK RJT 12/12/69 M ENGR
E C CONTROL MFG ENGR
TITLE SCHEMATIC LOGIBLOC # 5923 JH, JL, JAD (RSM ADDRESS)
D 5923-1
REVISION



COMP	SIZE / TYPE	QTY
R11,12	6.8K	2
R3	47K	1
R4	18K	1
R5,9	10K	2
R6,10	2.2K	2
R7,8	5.6K	2
R12	4.7K	1
R13	1 MEG	1
R14	68 Ω	1
R17	150K	1
R15,20	10K	2
R21	15K	1
C1	25 MF	1
C4	.001 MF CER	1
C3,2	220 MMF	2
Q1,4,5	WL375-1005	3
Q2,3	GT594	2
709	WL376-0000	1
R22,23	22K	2
C5	.01 MF CER	1

LOCATION	TYPE & MFR'R	TERM. No Vcc +5V	TERM. No ±0V	QTY
L11, 15	SN7402N TEXAS INST.	14	7	2
L17,18,21, 22	SN7442N TEXAS INST.	16	8	4
L2,4,6,8,19,23	9930 FAIRCHILD	14	7	6
L9, 13	9935 FAIRCHILD	14	7	2
L3, 7, 20	9936 FAIRCHILD	14	7	3
L1, 5, 10, 12, 14, 16	9946 FAIRCHILD	14	7	6

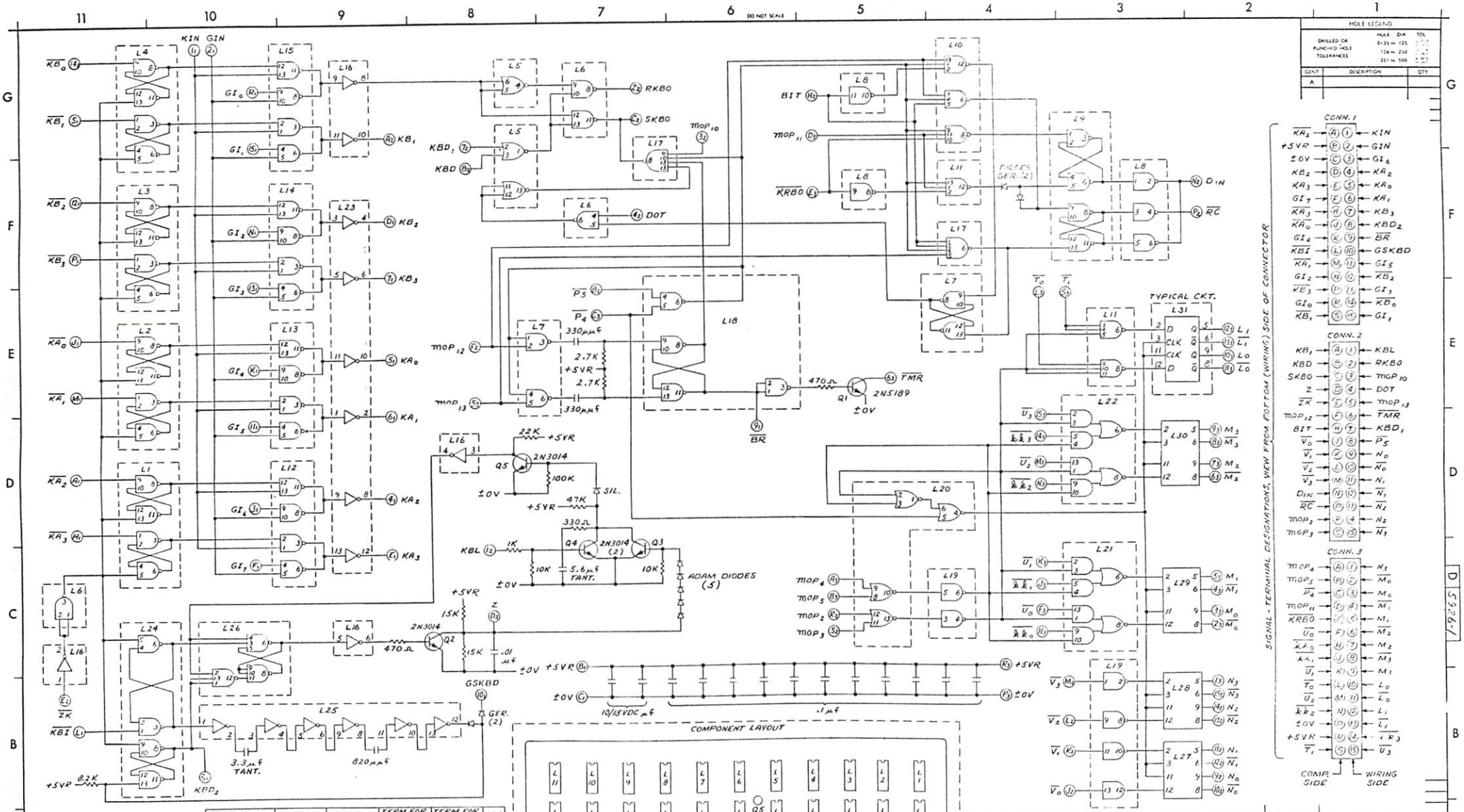
NOTE
1. ALL DIODES ARE WL380-1001 EXCEPT AS NOTED # (QTY 23) # (QTY 2)
2. ALL RESISTORS ARE ¼ W

HOLE LEGEND	HOLE DIA	TOL
DRILLED OR PUNCHED HOLE TOLERANCES	0135 to 125	± .003
	126 to 250	± .004
	251 to 500	± .005

IDENT	DESCRIPTION	QTY
A		

REVISION	BY	DATE	APPROVED BY	DATE
A ECN# 1871 REMOVE R11 & R6		9/6/72		9/11/72
B ECN# 1502 REMOVE R2 & R3 ECN# 1514 ADD R2 & R3		11/11/72		11/17/72
C PER ECN# 1502 & 1514 R11 & R6 WAS 10K R2 & R3 WAS 22K ADDED C4 & C5 APPR. SKH		6/31/73		6-8-73
D PER ECN# 3025 R10 WAS 10 MEG R18 WAS 22K R20 WAS 5.6K ADDED C4 & C5 APPR. SKH		6/31/73		6-8-73
E REVISED PER ECN# 3076 ADD 709		7/25/73		7-25-73

WANG PART NO	ITEM	QTY	NAME	MATERIAL	DESCRIPTION
WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.					
MATERIAL	MODEL NO		DWN [illegible] 11/14/72 E ENGR [illegible] 3/1/73		
	SEE ENGR'G SPECIFICATIONS No.		CHK [illegible] 3/13/73 M ENGR		
FINISH	TOL ER AS NOTED XX ± .010 XXX ± .005 ANG ± 1°30'		I C CONTROL	MFG ENGR	
			TITLE SCHEMATIC LOGIBLOC #5924 TAPE IN & OUT, NIXIE TUBE DECODER		
	SCALE [illegible]	SHT OF	WANG PART NO. 5924	SIZE D	DRAWING NUMBER 5924-1



HOLE LEGEND

DRILLED OR PUNCHED HOLE TOLERANCES	HOLE DIA	TOL
	0135 in 125	±.003
	126 in 250	±.005
	251 in 500	±.007

AGENT	DESCRIPTION	QTY
A		

T.C. LOCATION	TYPE	W.L. NO.	TERM FOR VCC +5V	TERM FOR ±0V	QTY.
L17	F9310	376-0022	14	7	1
L25	F9935	376-0025	14	7	1
L6,16,23	F9312	376-0026	14	7	3
L1,2,3,4,6,7,9, 10,13,14,15,24	F9944	376-0023	14	7	12
L10	F9368	376-0033	14	7	1
L18	SN7400N	376-0002	14	7	1
L5,20	SN7402N	376-0001A	14	7	2
L19	SN7404N	376-0010	14	7	1
L11,26	SN7410N	376-0003	14	7	2
L21,22	SN7451N	376-0012	14	7	2
L27,28,29,30,31	SN7474N	376-0006	14	7	5

REVISION	BY	DATE	APPROVED BY	DATE
1	ADDED PREP. FOR ADAM			
2	REVISED PER E.C. #1235			
3	REVISED PER E.C. #1235, ADD'L RST			
4	REVISED PER E.C. #1413 MISC. SYS			

WANG PART NO	ITEM	QTY.	NAME

WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.

MATERIAL	DESCRIPTION

	BY	DATE	APPROVED BY	DATE
DWN	F.S.S.	11-[illegible]	ENGR	[illegible]
CHK	M.T.	[illegible]	M ENGR	
E. C. CONTROL			MFG ENGR	

MATERIAL — MODEL NO. — SEE ENGRG SPECIFICATIONS No. —

FINISH — TOL. EX. AS NOTED .XX ± .010 FRAC. ± 1/64 .XXX ± .005 ANG. ± 1°30' FINISH √

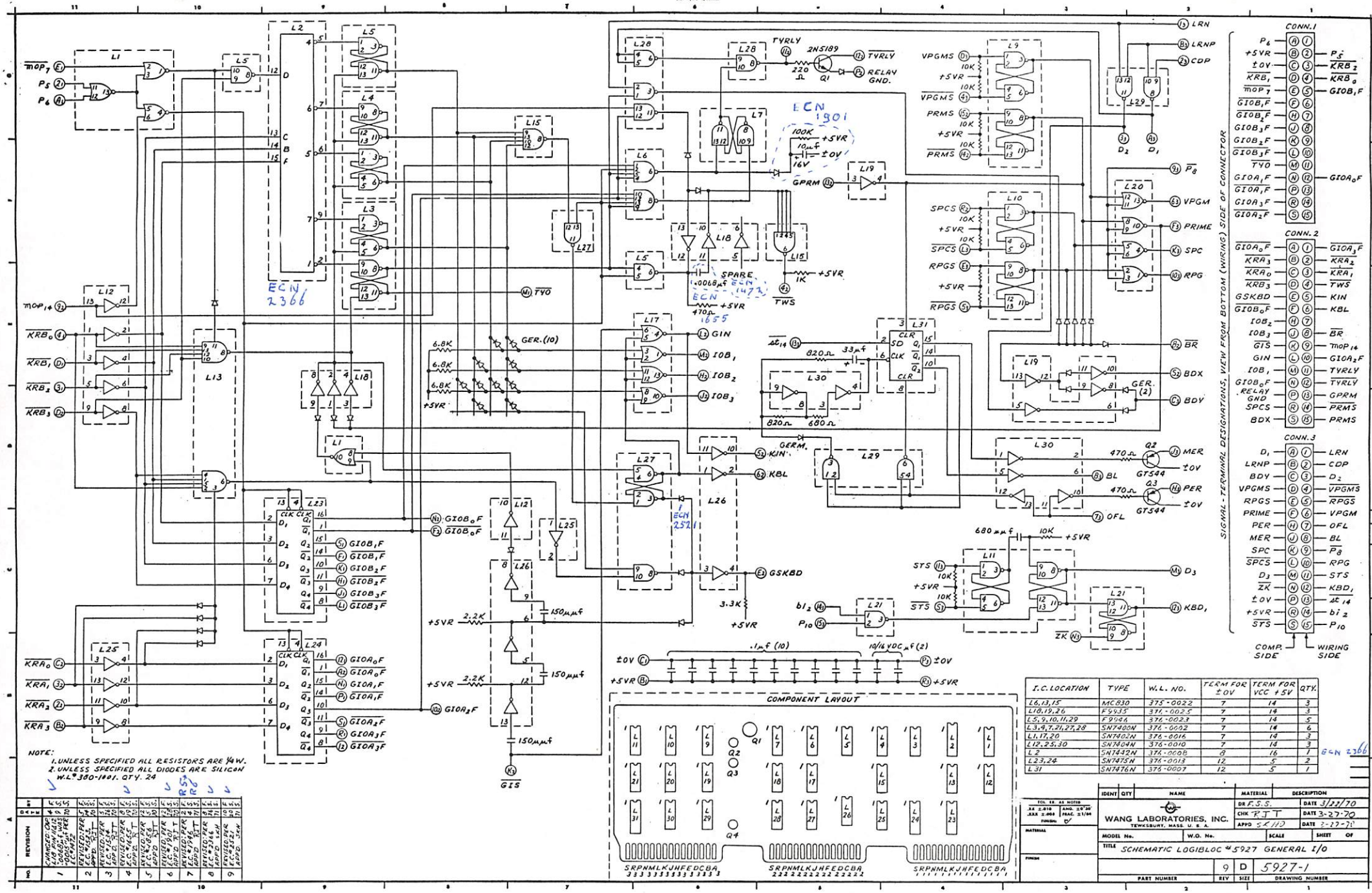
SCALE — SHT — OF

TITLE: SCHEMATIC LOGIBLOC #5926 KBD, CORE ADDRESS, KEYBOARD REG.

D 5926-1 4

WANG PART NO. — DRAWING NUMBER — REV.

D 5926/1



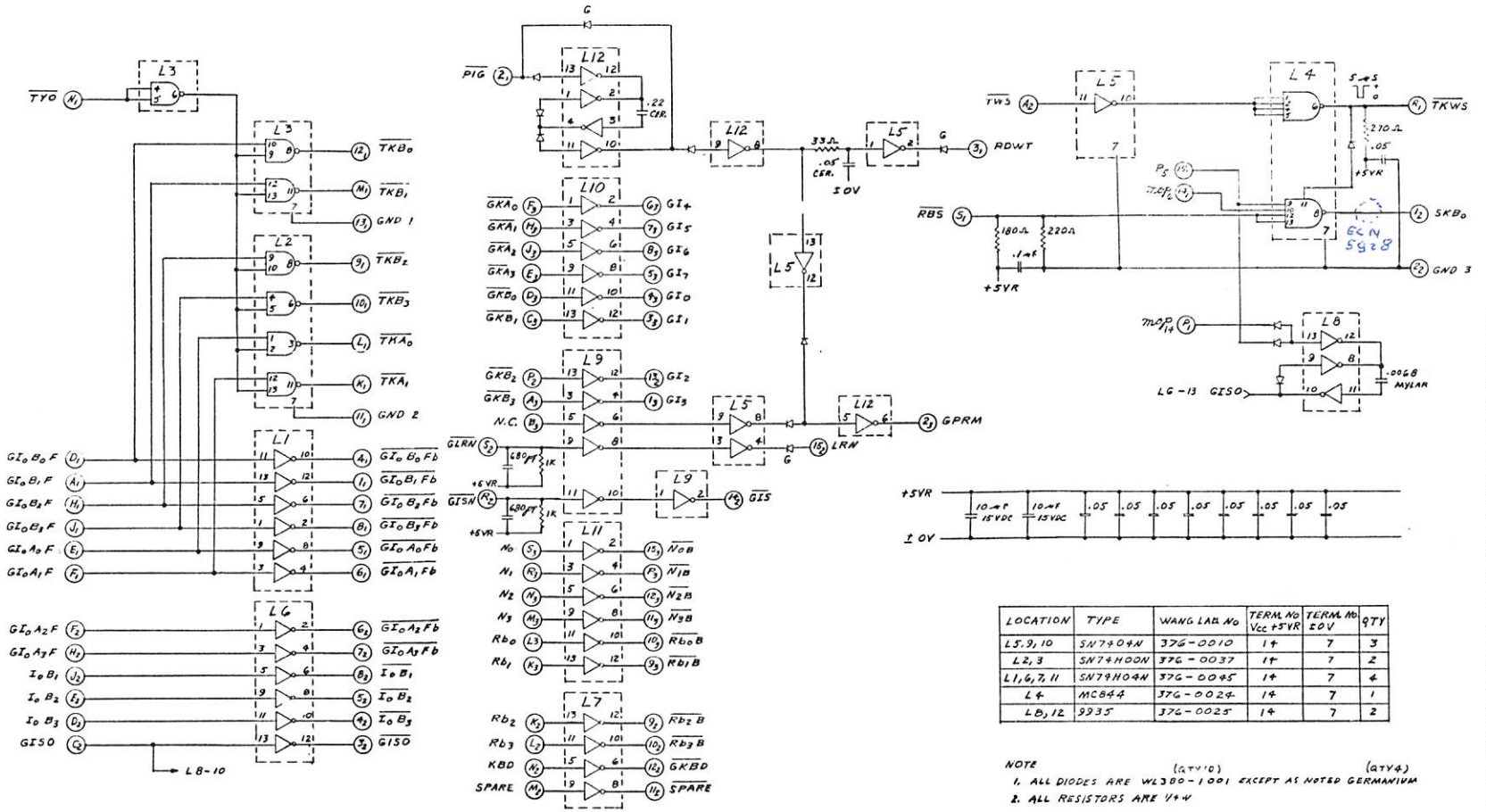
I.C. LOCATION	TYPE	W.L. NO.	TERM FOR ±0V	TERM FOR VCC +5V	QTY.
L6,13,15	MC830	375-0022	7	14	3
L10,19,26	F9935	376-0025	7	14	3
L5,9,10,11,29	F9946	376-0023	7	14	5
L3,4,7,21,27,28	SN7400N	376-0002	7	14	6
L1,17,20	SN7402N	376-0016	7	14	3
L12,25,30	SN7404N	376-0010	7	14	3
L2	SN7420N	376-0008	8	16	1
L23,24	SN7475N	376-0013	12	5	2
L31	SN7476N	376-0007	12	5	1

ECN 2366

NO.	REVISION	BY	DATE
1	CHANGED CAP. L10 FM 1412 .0068 MF 1000 V MIL TO ECN #2521	E.S.S.	4-15-70
2	REVISED PER E.C.# 1477 ADDED R.T.T.	E.S.S.	3-16-70
3	REVISED PER E.C.# 1524 ADDED R.T.T.	E.S.S.	3-16-70
4	REVISED PER E.C.# 1655 ADDED R.T.T.	E.S.S.	3-16-70
5	REVISED PER E.C.# 1901 ADDED R.T.T.	E.S.S.	3-16-70
6	REVISED PER E.C.# 1901 T ADDED R.T.T.	E.S.S.	3-16-70
7	REVISED PER E.C.# 1946 ADDED R.T.T.	E.S.S.	4-15-70
8	REVISED PER E.C.# 1946 ADDED VAM	E.S.S.	10-15-70
9	REVISED PER E.C.# 2521 ADDED SMW	E.S.S.	11-5-70

IDENT	QTY	NAME	MATERIAL	DESCRIPTION
		WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.	DR F.S.S.	DATE 3/22/70
			CHK R.T.T.	DATE 3-27-70
			APPD S.K.W.	DATE 3-27-70
		MODEL No.	W.O. No.	SCALE	SHEET OF
		TITLE SCHEMATIC LOGIBLOC #5927 GENERAL I/O		
		PART NUMBER	REV 9	SIZE D	DRAWING NUMBER 5927-1

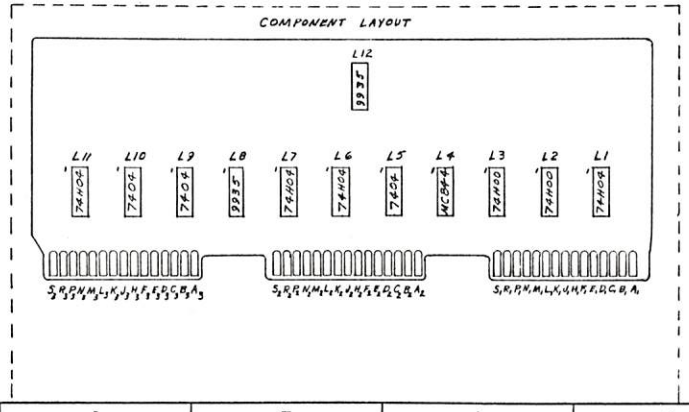
D 5927-1



LOCATION	TYPE	WANG LAB. NO	TERM. NO Vcc +5VR	TERM. NO ±0V	QTY
L5, 9, 10	SN7404N	376-0010	14	7	3
L2, 3	SN7400N	376-0037	14	7	2
L1, 6, 7, 11	SN7404N	376-0045	14	7	4
L4	MC844	376-0024	14	7	1
L8, 12	9935	376-0025	14	7	2

NOTE
1. ALL DIODES ARE WL300-1001 EXCEPT AS NOTED GERMANIUM (QTY 10) (QTY 4)
2. ALL RESISTORS ARE ¼W

COMPONENT LAYOUT



HOLE LEGEND

DRILLED OR PUNCHED HOLE TOLERANCES	HOLE DIA	TOL
	0135 to 125	± .003
	126 to 250	± .004
	251 to 500	± .005

IDENT	DESCRIPTION	QTY
A		

CONN. 1

GI2B1F	A 1	GI0B1Fb
+5VR	B 2	PIG
±0V	C 3	RDWT
GI0B0F	D 4	GI0B0Fb
GI0A0F	E 5	GI0A0Fb
GI0A1F	F 6	GI0A1Fb
GI0B2F	H 7	GI0B2Fb
GI0B1F	J 8	GI0B3Fb
TKA1	K 9	TKB2
TKA0	L 10	TKB3
TKB1	M 11	GND 2
TY0	N 12	TKB0
mop14	P 13	GND 1
TKWS	R 14	mop16
RBS	S 15	P5

CONN. 2

TWS	A 1	SKB0
	B 2	GND 3
GISO	C 3	GISO
I0B3	D 4	I0B3
I0B2	E 5	I0B2
GI0A2F	F 6	GI0A2Fb
GI0A3F	H 7	GI0A3Fb
I0B1	J 8	I0B1
Rb2	K 9	Rb2B
Rb3	L 10	Rb3B
SPARE	M 11	SPARE
KBD	N 12	GKBD
GKB2	P 13	GI2
GISN	R 14	GIS
GLRN	S 15	LRN

CONN. 3

GKB3	A 1	GI3
NC	B 2	GPRM
GKB1	C 3	GI1
GKB0	D 4	GI0
GKA3	E 5	GI7
GKA0	F 6	GI4
GKA1	H 7	GI5
GKA2	J 8	GI6
Rb1	K 9	Rb1B
Rb0	L 10	Rb0B
N3	M 11	N3B
N2	N 12	N2B
N1B	P 13	±0V
N1	R 14	+5VR
N0	S 15	N0B

COMP. SIDE — WIRING SIDE

SIGNAL-TERMINAL DESIGNATION VIEW FROM (WIRING) SIDE OF CONNECTOR

D 5928-1

REVISION	BY	DATE	APPROVED
1 PER ECN #1683 ADDED IN RESISTOR AND CAP FROM PIN 9 TO PIN 8 OF L9, PIN 8 OF L8 ADDED IN RESISTOR AND CAP FROM GROUND TO PIN 11 OF L8 PIN 11 OF L8 TO +5VR	RJT	4/11/70	
2 PC BOARD IS 1935 REVISE DIGIT FROM 4 TO 5	RJT	4-23-70	
3 PER ECN #1695 ADDED 2.2K Ω RESIS TO PIN 6 (GND) AND CHANGE 270Ω TO 180Ω APR - RJT	RJT	5-19-70	
4 PER ECN #1732 DELETED IN RANGE ADDED 270Ω RESISTOR APR	RJT	7-2-70	

WANG PART NO	ITEM	QTY	NAME
QTY PER UNIT	FIRST USED ON	ASSY USED ON	

WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.

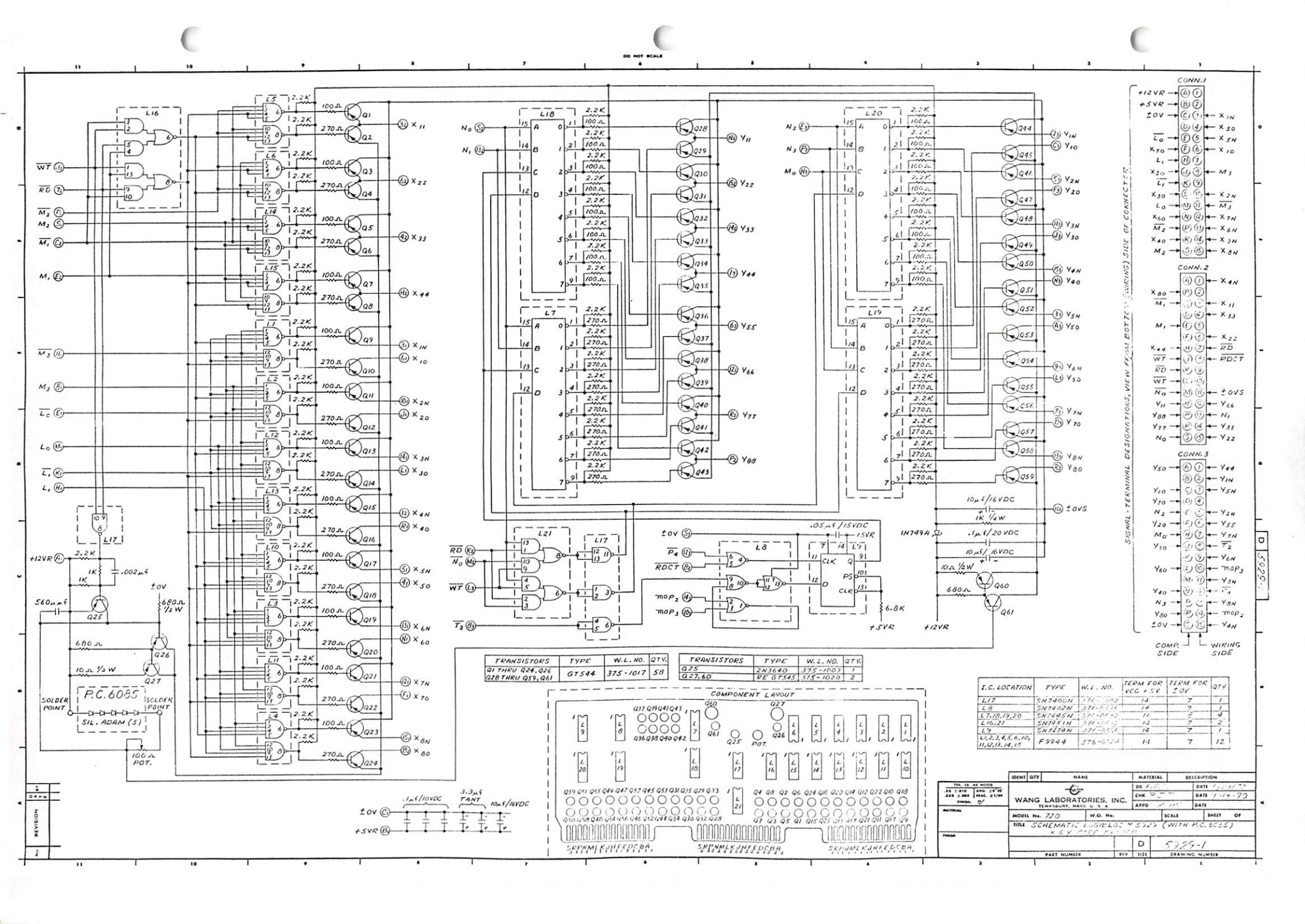
MATERIAL	DESCRIPTION
DWN SB 2/23/70	APPROVED BY / DATE: M ENGR
CHK RJT 3-27-70	M ENGR
E.C. CONTROL	MFG ENGR

MATERIAL — MODEL NO. — SEE ENGRG SPECIFICATIONS No. —
FINISH — TOL EX AS NOTED XX ± 010 FRAC ± 1/64 XXX ± 005 ANG ± 1°30' FINISH
SCALE — SHT OF

TITLE: SCHEMATIC LOGIBLOC # 5928 BUFFER BOARD

D 5928-1 4

WANG PART NUMBER — SIZE — DRAWING NUMBER — REV.



TRANSISTORS	TYPE	W.L. NO.	QTY.
Q1 THRU Q24, Q26 Q28 THRU Q59, Q61	GT544	375-1017	58

TRANSISTORS	TYPE	W.L. NO.	QTY.
Q25	2N3640	375-1003	1
Q27, 60	RE GT545	375-1020	2

I.C. LOCATION	TYPE	W.L. NO.	TERM FOR VCC + 5V	TERM FOR ±0V	QTY
L17	SN7400N	376-0002	14	7	1
L8	SN7402N	376-0016	14	7	1
L7,18,19,20	SN7495N	376-0042	14	5	4
L16,21	SN7451N	376-0012	14	7	2
L9	SN7474N	376-0054	14	7	1
L1,2,3,4,5,6,10, 11,12,13,14,15	F9944	376-0034	14	7	12

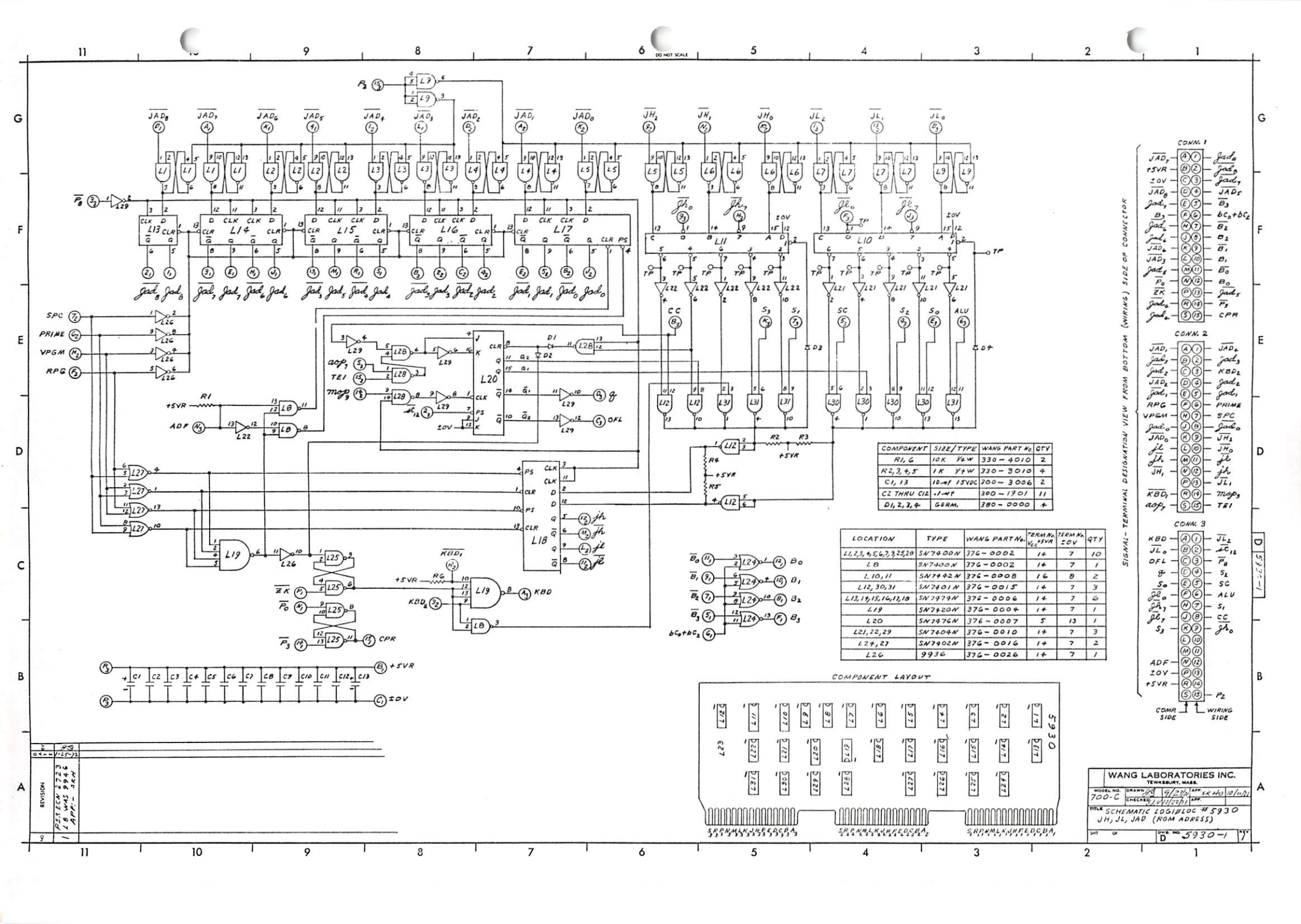
IDENT	QTY	NAME	MATERIAL	DESCRIPTION
		WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.	DR	DATE 1-14-70
			CHK	DATE 1-14-70
			APPD	DATE
		MODEL No. 720	W.O. No.	SCALE SHEET OF
		TITLE SCHEMATIC LOGIC DIAGRAM 5929 (WITH P.C. 6085) X-Y CORE PLANE		
			D	5929-1
		PART NUMBER	REV	SIZE DRAWING NUMBER

TOL. EX AS NOTED .XX ± .010 .XXX ± .005 FINISH

MATERIAL

FINISH

D 5929-1



COMPONENT	SIZE/TYPE	WANG PART No.	QTY
R1, 6	10K ¼W	330-4010	2
R2, 3, 4, 5	1K ¼W	330-3010	4
C1, 13	10 MF 15VDC	200-3006	2
C2 THRU C12	.1 MF	300-1901	11
D1, 2, 3, 4	GERM.	380-0000	4

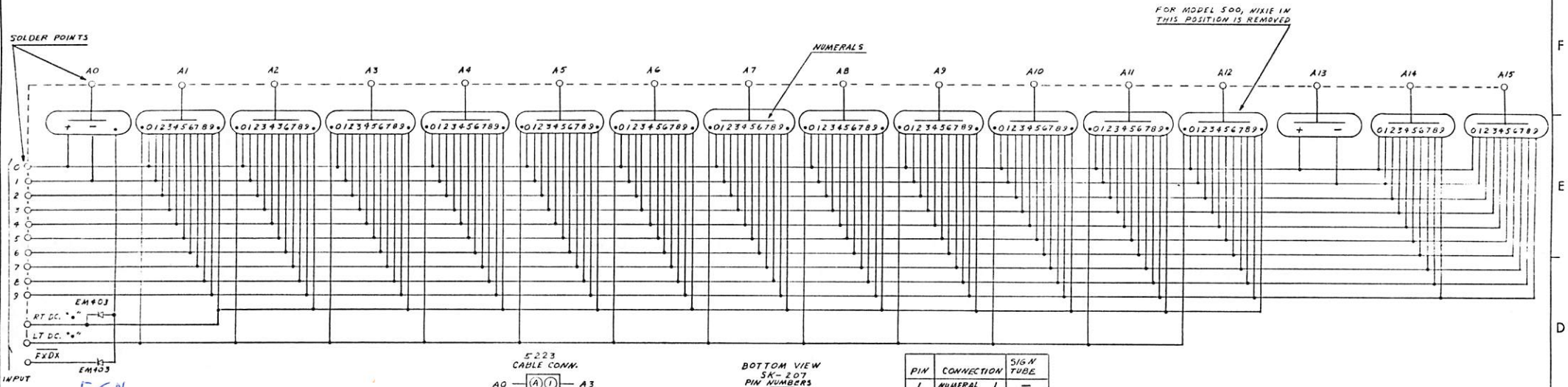
LOCATION	TYPE	WANG PART No.	TERM No. V_{CC} +5VR	TERM No. ±0V	QTY
L1, 2, 3, 4, 5, 6, 7, 8, 25, 28	SN7400N	376-0002	14	7	10
L8	SN7400N	376-0002	14	7	1
L10, 11	SN7442N	376-0008	16	8	2
L12, 30, 31	SN7401N	376-0015	14	7	3
L13, 14, 15, 16, 17, 18	SN7474N	376-0006	14	7	6
L19	SN7420N	376-0004	14	7	1
L20	SN7476N	376-0007	5	13	1
L21, 22, 29	SN7404N	376-0010	14	7	3
L24, 27	SN7402N	376-0016	14	7	2
L26	9936	376-0026	14	7	1

REVISION	BY	DATE	APP'D
1	PST	ECN 2723	
	LB	WAS 9946	
	APP'L	WAS 9944	

DO NOT SCALE

HOLE LEGEND		
DRILLED OR PUNCHED HOLE TOLERANCES	HOLE DIA	TOL
	.0135 to .125	± .003
	.126 to .250	± .004
	.251 to .500	± .005

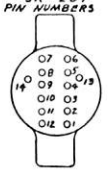
IDENT	DESCRIPTION	QTY
A		



ECN 2149

5223 CABLE CONN.

A0	A	A3
A5	B	FXDX
A2	C	A1
A4	D	B
1	E	9
0	F	RT DC
5	H	A7
	J	A6
	K	A14
2	L	A13
4	M	A15
6	N	A12
7	P	A11
3	R	A10
A8	S	A9



PIN	CONNECTION	SIGN TUBE
1	NUMERAL 1	—
2	" 2	
3	" 3	
4	" 4	
5	" 5	
6	" 6	+
7	ANODE	ANODE
8	NUMERAL 7	
9	" 8	
10	ANODE	
11	NUMERAL 9	
12	" 0	
13	RT. DC. PT.	RT. DC. PT.
14	LT. DC. PT	

TUBE TYPE	WANG. NO.	QTY
SIGN	340-0010	2
NUMERAL	340-0011	14

D5936-1

REVISION	BY	DATE
FOR ECN # 2149 ADDED FXDX CKT, ADDED 5223 CABLE CONN. ADDED NOTE FOR MODEL 500 APPS.	KS	5/14/71

WANG PART NO	ITEM	QTY	NAME	MATERIAL	DESCRIPTION
QTY PER UNIT	FIRST USED ON	ASSY USED ON			

WANG LABORATORIES INC. TEWKSBURY, MASS. U.S.A.

	BY	DATE	APPROVED BY	DATE
DWN	KS	5/21/71	E ENGR	5/21/71
CHK	RTT	5-13-71	M ENGR	
E C CONTROL			MFG ENGR	

MATERIAL — MODEL NO 700/500 — SEE ENGAG SPECIFICATIONS No.

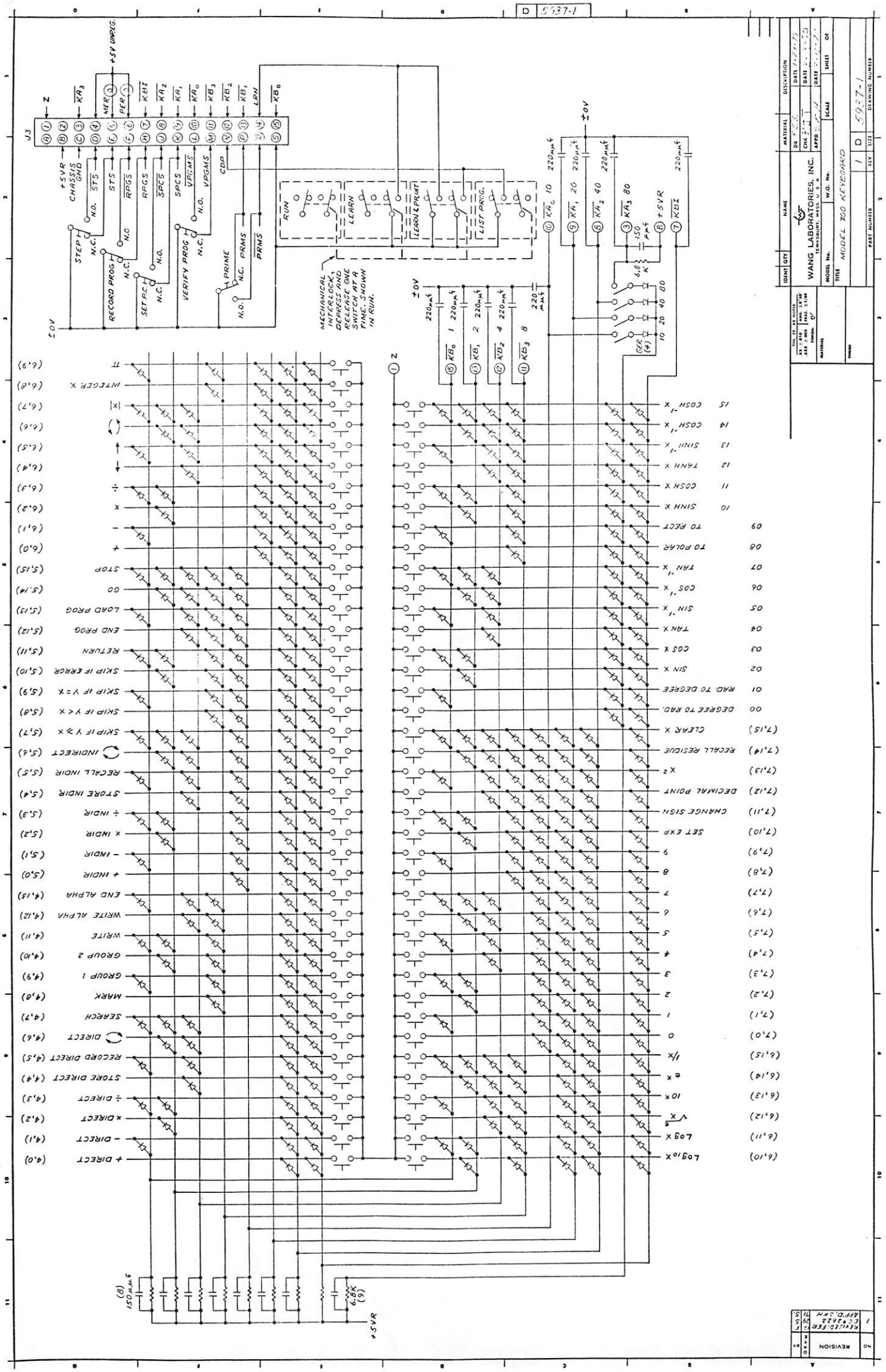
FINISH — TOL EX AS NOTED XX ± .010 FRAC ± 1/64 XXX ± .005 ANG ± 1°30' FINISH V

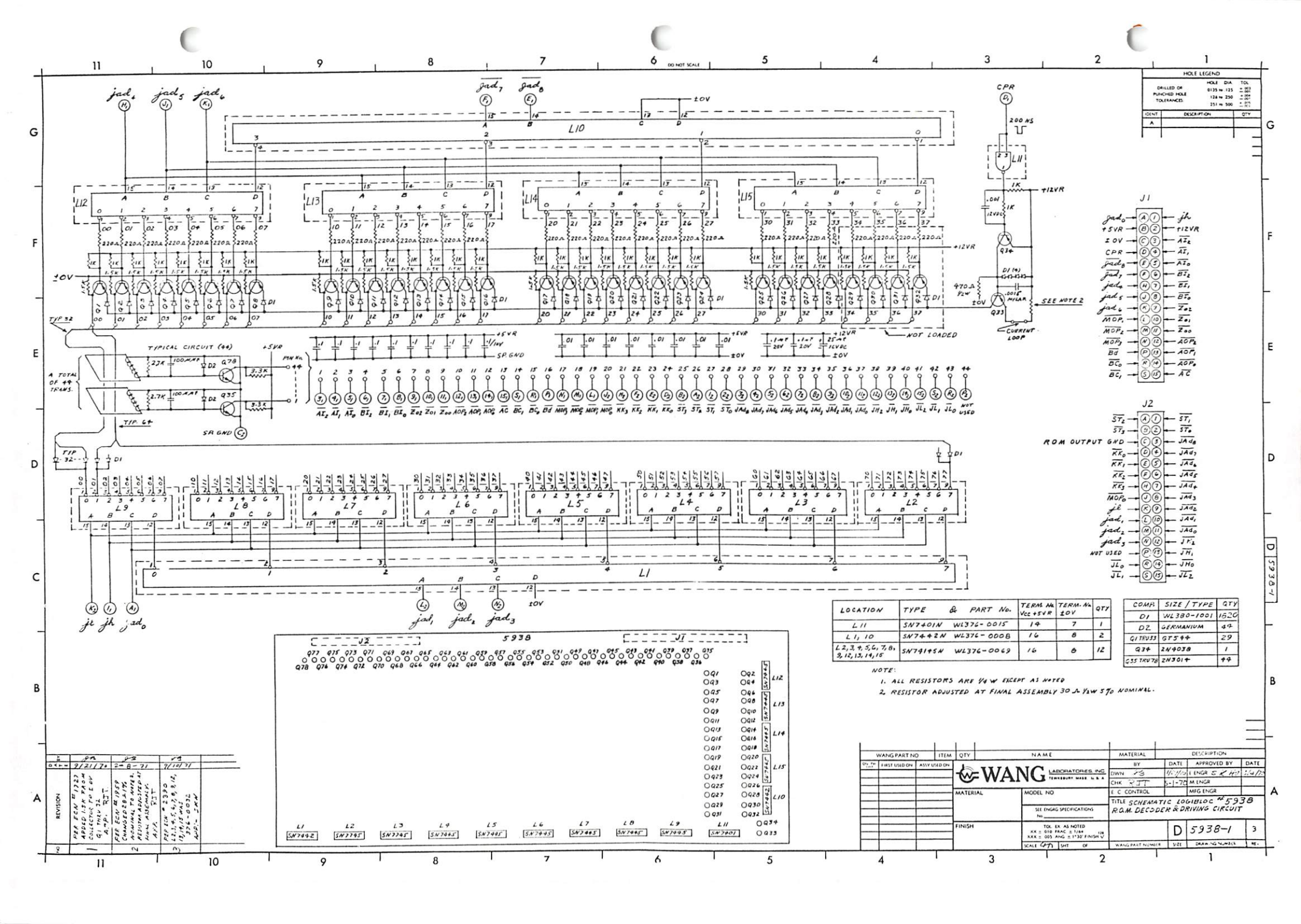
TITLE SCHEMATIC LOGIBLOC No 5936 NIXIE DISPLAY

SCALE FULL SHT OF

WANG PART NUMBER D 5936-1 SIZE 1

DRAW NO. 1 WEB


D 5937-1
WANG LABORATORIES, INC.
TEWKSBURY, MASS. U.S.A.
MODEL 300 KEYBOARD
5937-1
REVISION
RECORD PROG
SET PC
VERIFY PROG
PRIME
RUN
LEARN
LEARN & PRINT
LIST PROG.
MECHANICAL INTERLOCK, DEPRESS AND RELEASE ONE SWITCH AT A TIME. SHOWN IN RUN.
+5V UNREG.
+5V
CHASSIS GND
STEP
CDP
PRMS
LPRN
+ DIRECT
- DIRECT
× DIRECT
÷ DIRECT
STORE DIRECT
RECORD DIRECT
DIRECT
SEARCH
MARK
GROUP 1
GROUP 2
WRITE
WRITE ALPHA
END ALPHA
+ INDIR
- INDIR
× INDIR
÷ INDIR
STORE INDIR
RECALL INDIR
INDIRECT
SKIP IF Y ≥ X
SKIP IF Y < X
SKIP IF Y = X
SKIP IF ERROR
RETURN
END PROG
LOAD PROG
GO
STOP
+
-
×
÷
=
INTEGER X
π
SIN X
COS X
TAN X
SIN⁻¹ X
COS⁻¹ X
TAN⁻¹ X
TO POLAR
TO RECT
SINH X
COSH X
TANH X
SINH⁻¹ X
COSH⁻¹ X
DEGREE TO RAD.
RAD TO DEGREE
CLEAR X
RECALL RESIDUE
X²
DECIMAL POINT
CHANGE SIGN
SET EXP
0
1
2
3
4
5
6
7
8
9
1/X
eˣ
10ˣ
Log X
Log₁₀X



LOCATION	TYPE & PART No.	TERM. No. Vcc +5V	TERM. No. ±0V	QTY
L11	SN7401N WL376-0015	14	7	1
L1, 10	SN7442N WL376-0008	16	8	2
L2,3,4,5,6,7,8,9,12,13,14,15	SN7445N WL376-0069	16	8	12

COMP.	SIZE / TYPE	QTY
D1	WL380-1001	1620
D2	GERMANIUM	44
GI TRU33	GT544	29
Q34	2N4038	1
Q35 TRU78	2N3014	44

NOTE:
1. ALL RESISTORS ARE 1/4 W EXCEPT AS NOTED
2. RESISTOR ADJUSTED AT FINAL ASSEMBLY 30 Ω 1/2W 5% NOMINAL.

REVISION	BY	DATE	APPROVED BY	DATE
A	PER ECN #1323 ADDED 1.5K FROM COLLECTOR TO E ON Q1 THRU 72. A. JR. RJT.	9/21/70		
B	PER ECN #1959 CHANGED BASING RESISTOR TO MATCH NOMINAL TO MATCH RESISTOR ADJUSTED AT FINAL ASSEMBLY. A.J.R. RJT	2-8-71		
C	PER ECN #2330 L2,3,4,5,6,7,8,9,12, 13,14,15 WAS 376-0032 NOW 376-0069 A.J.R. RJT	9/10/71		

WANG PART NO	ITEM	QTY	NAME
			WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.
MATERIAL			MODEL NO
FINISH			SEE ENGRG SPECIFICATIONS No.
			TOL EX. AS NOTED XX = .010 FRAC ± 1/64 XXX = .005 ANG ± 1°30' FINISH √
			SCALE 4/1 SHT OF

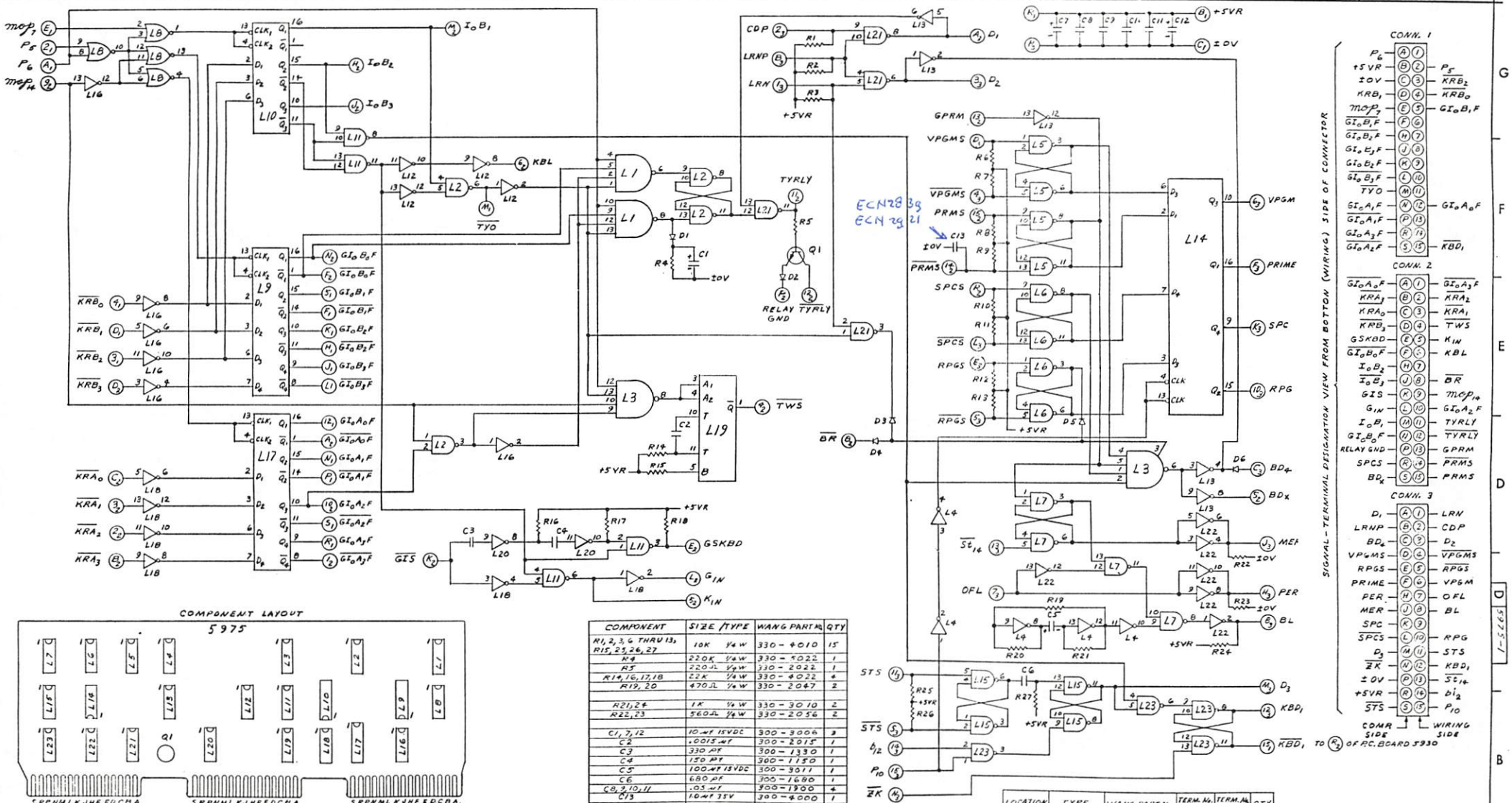
MATERIAL	DESCRIPTION			
	BY	DATE	APPROVED BY	DATE
	DWN RS	11/10/70	L ENGR S. K. HO	11/12/70
	CHK RJT	5-1-70	M ENGR	
	E C CONTROL		MFG ENGR	

TITLE SCHEMATIC LOGIBLOC # 5938 R.O.M. DECODER & DRIVING CIRCUIT

D 5938-1 3

WANG PART NUMBER SIZE DRAWING NUMBER REV.

D 5938-1



COMPONENT LAYOUT

5975

COMPONENT	SIZE / TYPE	WANG PART NO.	QTY
R1, 2, 3, 6 THRU 13, R15, 23, 26, 27	10K ¼ W	330 - 4010	15
R4	220K ¼ W	330 - 5022	1
R5	220Ω ¼ W	330 - 2022	1
R14, 16, 17, 18	22K ¼ W	330 - 4022	4
R19, 20	470Ω ¼ W	330 - 2047	2
R21, 24	1K ¼ W	330 - 3010	2
R22, 23	560Ω ¼ W	330 - 2056	2
C1, 7, 12	10 MF 15VDC	300 - 3006	3
C2	.0015 MF	300 - 2015	1
C3	330 PF	300 - 1330	1
C4	150 PF	300 - 1150	1
C5	100 MF 15VDC	300 - 3011	1
C6	680 PF	300 - 1680	1
C8, 9, 10, 11	.05 MF	300 - 1500	4
C13	10 MF 35V	300 - 4000	1
D1, 6	GERM. DIODE	380 - 0000	2
D2, 3, 4, 5	SIL. DIODE	380 - 1001	4
Q1	2N5189	375 - 1021	1

LOCATION	TYPE	WANG PART NO.	TERM. NO. V_{CC} +5VR	TERM. NO. ±0V	QTY
L1, 3	MC830P	376 - 0022	14	7	2
L2, 5, 6, 7, 11, 15, 23	9946	376 - 0023	14	7	7
L4, 16, 18	SN7404N	376 - 0010	14	7	3
L8	SN7402N	376 - 0016	14	7	1
L14, 9, 10, 17	SN74175N	376 - 0013	5	12	4
L12, 13	9936	376 - 0026	14	7	2
L19	SN74121N	376 - 0051	14	7	1
L20	9935	376 - 0025	14	7	1
L22	SN7400N	376 - 0055	14	7	1
L21	SN7410N	376 - 0002	14	7	1

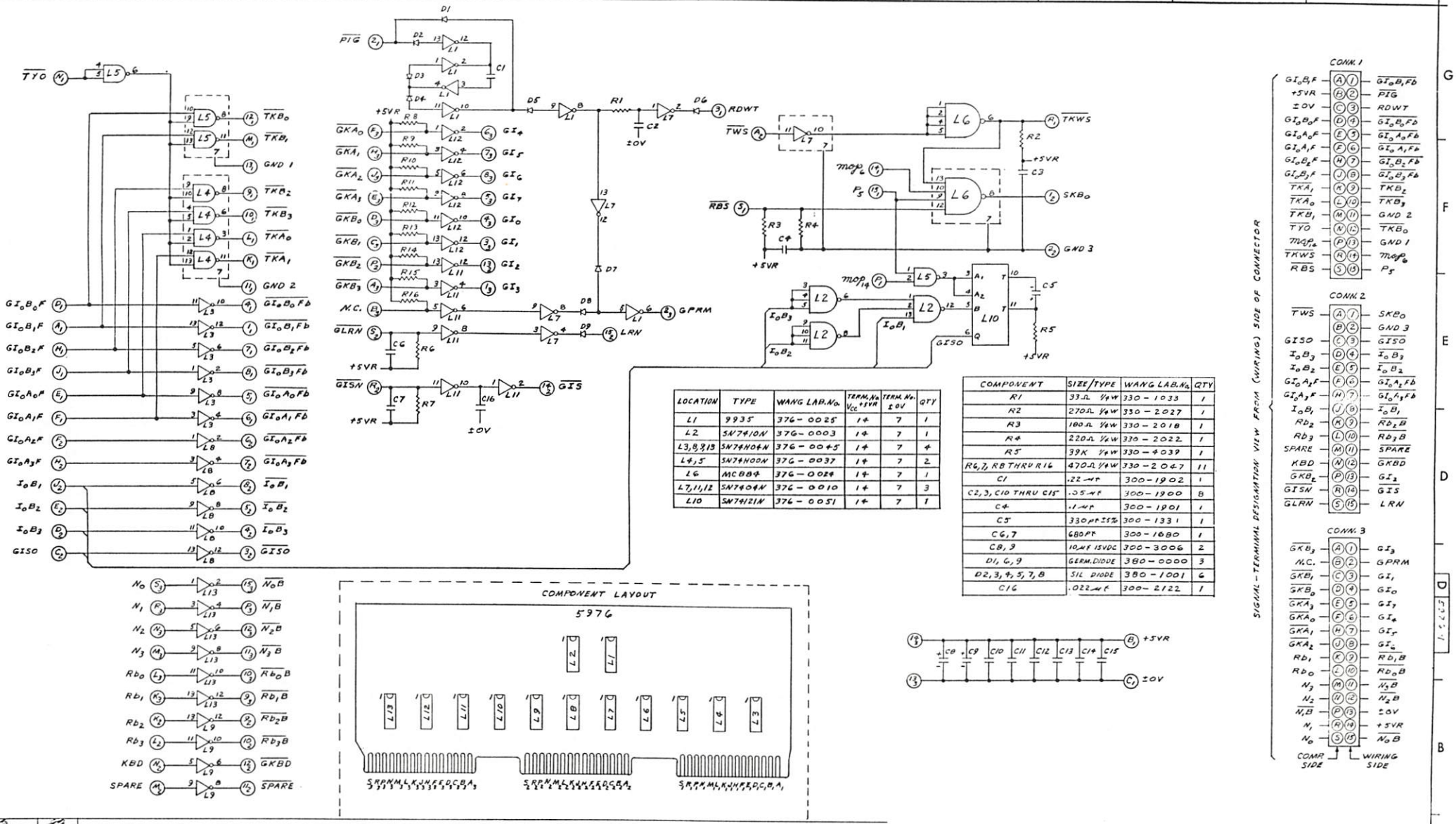
REV	2	3	4
DATE	2-3-72	3-23-72	9-26-72
REVISION	FIX ECN # 2751 L14 WAS 7402, P.C. CIRCUIT CHANGE A.P.D. SKH	PER ECN # 2751 ADDED C13 (.1 MF) SKH APD	PER ECN 2921 C13 WAS .22 MF APD

WANG LABORATORIES INC.
TEWKSBURY, MASS.

MODEL NO.	DRAWN	DATE	APP.	DATE
2200 700 C	CHECKED		APP.	

TITLE: SCHEMATIC LOGIBLOC # 5975
GENERAL I/O

SHT. 1 OF 1 — 5975 — DWG NO. D 5975-1 — REV. 3



LOCATION	TYPE	WANG LAB. No.	TERM. No V_{CC} +5VR	TERM. No. ±0V	QTY
L1	9935	376-0025	14	7	1
L2	SN7410N	376-0003	14	7	1
L3,8,9,13	SN7404N	376-0045	14	7	4
L4,5	SN7400N	376-0037	14	7	2
L6	MC884	376-0024	14	7	1
L7,11,12	SN7404N	376-0010	14	7	3
L10	SN74121N	376-0051	14	7	1

COMPONENT	SIZE/TYPE	WANG LAB. No.	QTY
R1	33Ω ¼W	330-1033	1
R2	270Ω ¼W	330-2027	1
R3	180Ω ¼W	330-2018	1
R4	220Ω ¼W	330-2022	1
R5	39K ¼W	330-4039	1
R6,7,8 THRU R16	470Ω ¼W	330-2047	11
C1	.22 MF	300-1902	1
C2,3,C10 THRU C15	.05 MF	300-1900	8
C4	.1 MF	300-1901	1
C5	330 MF 15%	300-1331	1
C6,7	680 PF	300-1680	1
C8,9	10 MF 15VDC	300-3006	2
D1,6,9	GERM. DIODE	380-0000	3
D2,3,4,5,7,8	SIL DIODE	380-1001	6
C16	.022 MF	300-2122	1

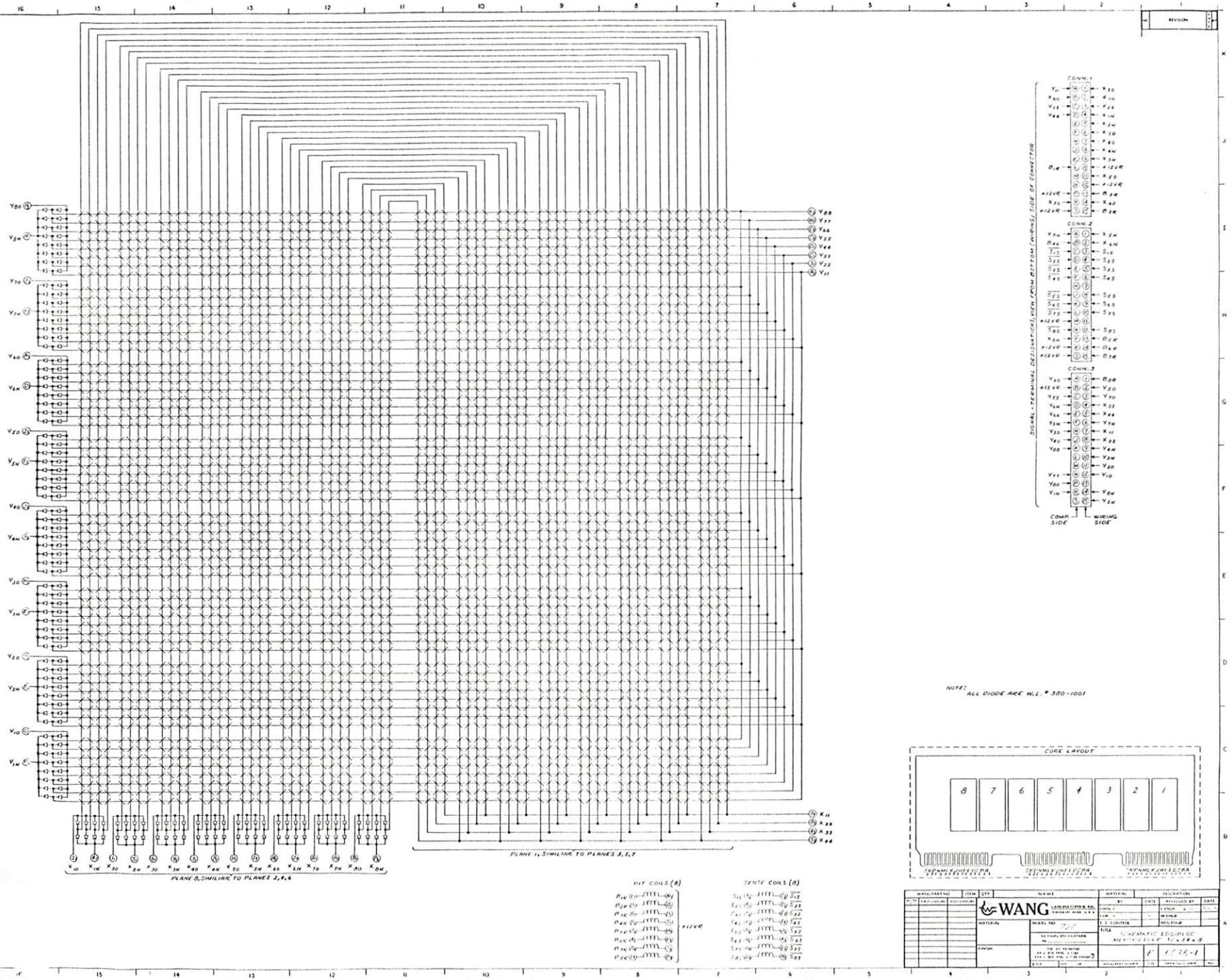
REVISION	BY	DATE	APP'D
PER ECN #2621 ADDED C16, CHANGED WIRING TO GND. APP'D - SK 9/10		12/10/71	
PER ECN #3128 ADDED R16 THRU R16 APP'D SHK		8-10-72	

WANG LABORATORIES INC.
TEWKSBURY, MASS.

MODEL NO.	DRAWN	DATE	APP.
700C/720C		9/15/71	
	CHECKED		APP.

TITLE: SCHEMATIC LOGIBLOC #5976
BUFFER BOARD

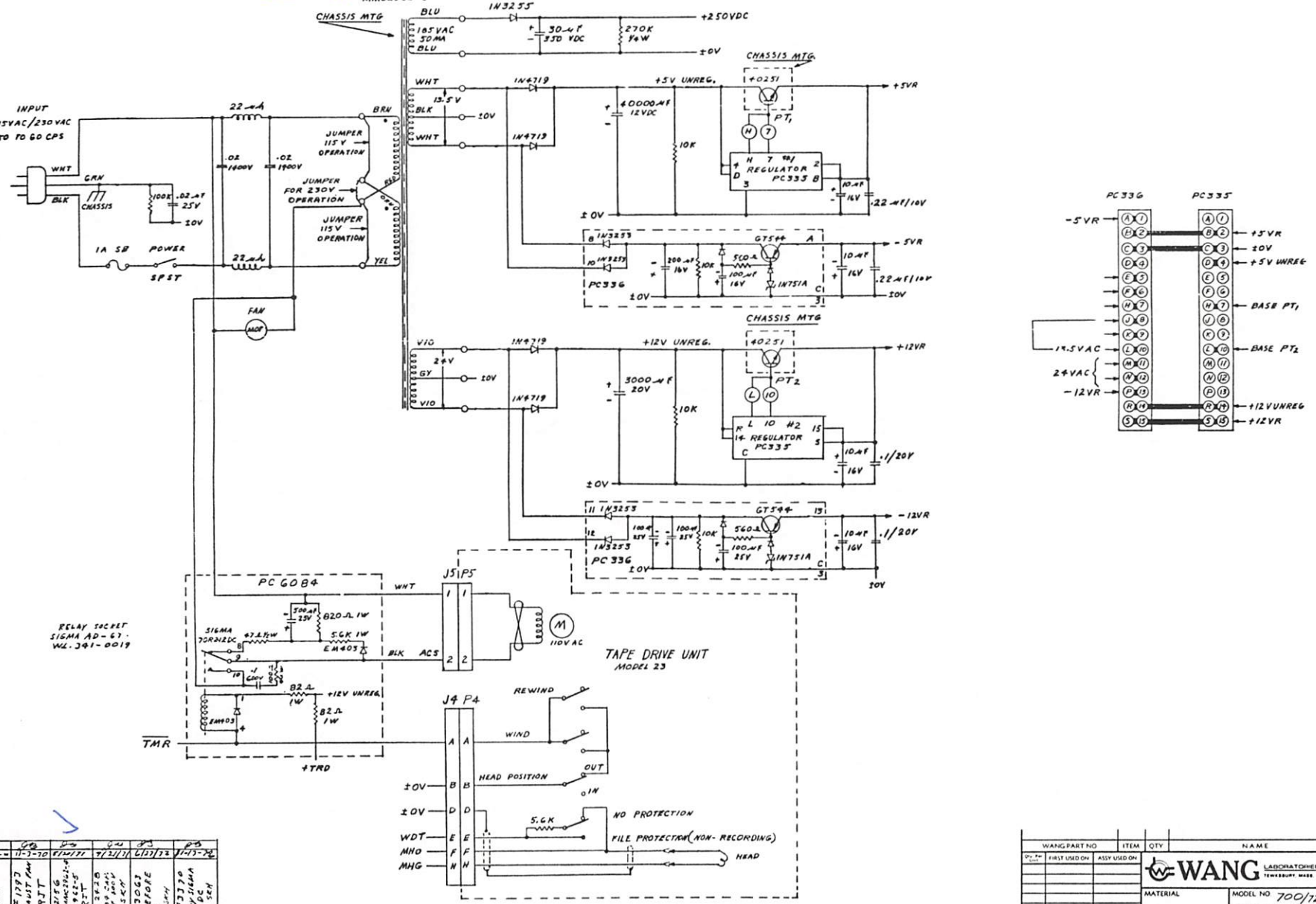
SHT OF | DWG. NO. D 5976-1 | REV



NOTE:
ALL DIODE ARE W.L.* 380-1001

| | | |

ECN 2156



HOLE LEGEND

DRILLED OR PUNCHED HOLE TOLERANCES	HOLE DIA	TOL
	.0135 to .125	± .002
	.126 to .250	± .003
	.251 to .500	± .004
		± .005

IDENT	DESCRIPTION	QTY
A		

REVISION					
DWN	GB	GB	GB	GB	GB
DATE	11-3-70	1/14/71	9/21/71	6/23/72	11-15-72
9	1 PER ECN # 1787 ADDED EXHAUST FAN APP.- RJT	PER ECN #2156 MISCELLANEOUS CHANGES APP.- RJT	2 PER ECN #2430 CHANGED CABLE W45 TO W47 APP.- SKH	3 PER ECN 3063 FUSE BEFORE SWITCH APP.- SKH	4 PER ECN #3370 CHANGED DIAGRAM APP.- SKH

WANG PART NO	ITEM	QTY	NAME

WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.

MATERIAL	DESCRIPTION
MATERIAL	MODEL NO 700/720
	SEE ENGRG SPECIFICATIONS No.
FINISH	TOL EX. AS NOTED XX ± .010 FRAC ± 1/64 XXX ± .005 ANG. ± 1°30' FINISH √
SCALE (1/2)	SHT OF

BY	DATE	APPROVED BY	DATE
DWN FB	1/21/70	E ENGR S. K. HA	2/15/72
CHK RJT	5-15-70	M ENGR	
E C. CONTROL		MFG ENGR	

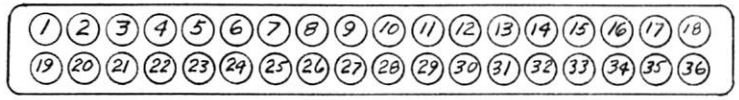
TITLE SCHEMATIC PC6052 & PC6084 POWER DISTRIBUTION & TAPE CONTROL CIRCUIT

WANG PART NUMBER	SIZE	DRAWING NUMBER	REV
	D	6052-1	5

D 6052-1

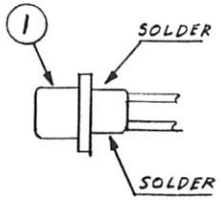
DO NOT SCALE

AMPHENOL 57-40360



PIN No.	WIRE COLOR	LENGTH
1	8	9 1/2"
2	9	9 1/2"
3	71	9 1/2"
4	7	9 1/2"
5	6	9"
6	5	9"
7	97	6"
8	3	8"
9	98	7"
10	4	8 1/2"
11	901	7"
12	81	10 1/2"
13	72	10"
14	76	10"
15	75	10"
16	74	10"
17	73	10"
18	908	6 1/2"

PIN No.	WIRE COLOR	LENGTH
19	1	7"
20	92	8"
21	91	8 1/2"
22	95	8"
23	96	7"
24	93	8"
25	94	8 1/2"
26	905	5"
27	906	5"
28	907	5 1/2"
29	904	6
30	903	5"
31	902	4 1/2"
32	2	6 1/2"
33	80	10"
34	83	11 1/2"
35	82	12"
36	54	15"



IDENT	QTY	NAME	WANG LAB. No.	DESCRIPTION
	1	WIRE (908)	600-3908	#26 WH/BLK/GRY
	1	WIRE (907)	600-3907	#26 WH/BLK/VIO
	1	WIRE (906)	600-3906	#26 WH/BLK/BLU
	1	WIRE (905)	600-3905	#26 WH/BLK/GRN
	1	WIRE (904)	600-3904	#26 WH/BLK/YEL
	1	WIRE (903)	600-3903	#26 WH/BLK/ORN
	1	WIRE (902)	600-3902	#26 WH/BLK/RED
	1	WIRE (901)	600-3901	#26 WH/BLK/BRN
	1	WIRE (98)	600-3098	#26 WH/GRY
	1	WIRE (97)	600-3097	#26 WH/VIO
	1	WIRE (96)	600-3096	#26 WH/BLU
	1	WIRE (95)	600-3095	#26 WH/GRN
	1	WIRE (94)	600-3094	#26 WH/YEL
	1	WIRE (93)	600-3093	#26 WH/ORN
	1	WIRE (92)	600-3092	#26 WH/RED
	1	WIRE (91)	600-3091	#26 WH/BRN
	1	WIRE (54)	600-3054	#26 GRN/YEL
	1	WIRE (83)	600-3083	#26 GRY/ORN
	1	WIRE (82)	600-3082	#26 GRY/RED
	1	WIRE (81)	600-3081	#26 GRY/BRN
	1	WIRE (80)	600-3080	#26 GRY/BLK
	1	WIRE (76)	600-3076	#26 VIO/BLU
	1	WIRE (75)	600-3075	#26 VIO/GRN
	1	WIRE (74)	600-3074	#26 VIO/YEL
	1	WIRE (73)	600-3073	#26 VIO/ORN
	1	WIRE (72)	600-3072	#26 VIO/RED
	1	WIRE (71)	600-3071	#26 VIO/BRN
	1	WIRE (9)	600-3009	#26 WHITE
	1	WIRE (8)	600-3008	#26 GRAY
	1	WIRE (7)	600-3007	#26 VIOLET
	1	WIRE (6)	600-3006	#26 BLUE
	1	WIRE (5)	600-3005	#26 GREEN
	1	WIRE (4)	600-3004	#26 YELLOW
	1	WIRE (3)	600-3003	#26 ORANGE
	1	WIRE (2)	600-3002	#26 RED
	1	WIRE (1)	600-3001	#26 BROWN
	1	WIRE (10)	600-3000	#26 BLACK
1	1	CONNECTOR	350-1026	

NO.	REVISION	DATE	BY
1	PER ECN #3228 PIN 36 WIRE WAS GRY/BLU APP'D SKH	9-16-72	

TOL. EX. AS NOTED
.XX ±.010 FRAC. ±1/64
.XXX ±.005 ANG. ±0°30'
FINISH: ▽

MATERIAL

FINISH

WANG LABORATORIES, INC.
TEWKSBURY, MASS. U. S. A.

DR		DATE 5/4/70
CHK		DATE 6-9-70
APPD	RIT	DATE 6-10-70

MODEL No. 7004/B | W.O. No. | SCALE 4/1 | SHEET OF

TITLE IN / OUT CONNECTOR

PART NUMBER	REV	SIZE	DRAWING NUMBER
220-0022	1	B	6088

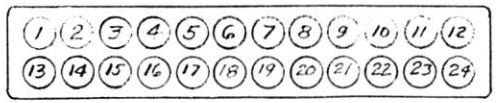
DO NOT SCALE

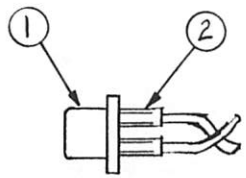
F/W

WIRE COLOR	LENGTH	PIN No.
91	5"	1
10	5"	13
9	6"	2
10	5"	14
93	6"	3
10	5"	15
94	6"	4
10	5"	16
95	6"	5
10	7"	17
96	6"	6
10	8"	18
97	7"	7
10	8"	19
98	7"	8
10	8"	20
10	5"	11
902	14"	12
54	10"	23
1	11"	24

TWISTED PAIR

AMPHENOL 57-90240





5 TWIST PER INCH

IDENT	QTY	NAME	WANG LAB. No.	DESCRIPTION
	1	WIRE (902)	600-3902	#26 WH/BLK/RED
	1	WIRE (98)	600-3098	#26 WH/GRY
	1	WIRE (97)	600-3097	#26 WH/VIO
	1	WIRE (96)	600-3096	#26 WH/BLU
	1	WIRE (95)	600-3095	#26 WH/GRN
	1	WIRE (94)	600-3094	#26 WH/YEL
	1	WIRE (93)	600-3093	#26 WH/ORN
	1	WIRE (91)	600-3091	#26 WH/BRN
	1	WIRE (9)	600-3009	#26 WHITE
	1	WIRE (54)	600-3054	#26 GREEN/YELLOW
	ASSD	WIRE (10)	600-3000	#26 BLACK
2	20	SLEEVE	605-0002	#15 3/8 LONG
1	1	CONNECTOR	350-1025	

NO.	REVISION	DATE	BY
1	PER ECN # 3227 PIN23 WIRE WAS GREEN APP'D SKH	9-16-72	SKH

TOL. EX. AS NOTED
.XX ±.010 FRAC. ±1/64
.XXX ±.005 ANG. ±0°30'
FINISH: 63

MATERIAL

FINISH

WANG LABORATORIES, INC.
TEWKSBURY, MASS. U. S. A.

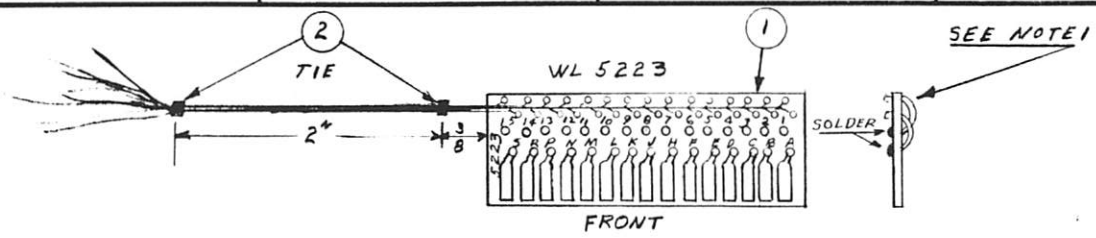
DR		DATE 5/4/70
CHK		DATE 6/9/70
APPD	RJT	DATE 6-10-70

MODEL No. 700 | W.O. No. | SCALE 4/1 | SHEET OF

TITLE TYPEWRITER CONNECTOR ASSEMBLY

PART NUMBER	REV	SIZE	DRAWING NUMBER
220-0021	1	B	6089

DO NOT SCALE



PIN NO.	WIRE COLOR	LENGTH
A	901	26½"
B	2	29½"
C	0	29½"
D	902	26"
E	903	28"
F	904	28"
H	905	26"
J	906	26"
K	907	26½"
L	908	29½"
M	71	29"
N	72	29½"
P	73	27½"
R	74	31"
S	75	33"

PIN No.	WIRE COLOR	LENGTH
1	1	29½"
2	92	32"
3	3	31"
4	4	29½"
5	5	28"
6	6	28"
7	7	32"
8	8	30"
9	91	27½"
10	93	27½"
11	94	26"
12	95	33½"
13	96	33"
14	97	33"
15	98	33½"

NOTE
1. ALL WIRES ARE FED THRU CLEARENCE HOLES ON FRONT OF CONNECTOR AND SOLDERED AT FRONT AS SHOWN.

IDENT	QTY	NAME	WANG LAB. No.	DESCRIPTION
	1	WIRE (908)	600-3908	#26 WH/BLK/GRY
	1	WIRE (907)	600-3907	#26 WH/BLK/VIO
	1	WIRE (906)	600-3906	#26 WH/BLK/BLU
	1	WIRE (905)	600-3905	#26 WH/BLK/GRN
	1	WIRE (904)	600-3904	#26 WH/BLK/YEL
	1	WIRE (903)	600-3903	#26 WH/BLK/ORN
	1	WIRE (902)	600-3902	#26 WH/BLK/RED
	1	WIRE (901)	600-3901	#26 WH/BLK/BRN
	1	WIRE (98)	600-3098	#26 WH/GRY
	1	WIRE (97)	600-3097	#26 WH/VIO
	1	WIRE (96)	600-3096	#26 WH/BLU
	1	WIRE (95)	600-3095	#26 WH/GRN
	1	WIRE (94)	600-3094	#26 WH/YEL
	1	WIRE (93)	600-3093	#26 WH/ORN
	1	WIRE (92)	600-3092	#26 WH/RED
	1	WIRE (91)	600-3091	#26 WH/BRN
	1	WIRE (75)	600-3075	#26 VIO/GRN
	1	WIRE (74)	600-3074	#26 VIO/YEL
	1	WIRE (73)	600-3073	#26 VIO/ORN
	1	WIRE (72)	600-3072	#26 VIO/RED
	1	WIRE (71)	600-3071	#26 VIO/BRN
	1	WIRE (8)	600-3008	#26 GRAY
	1	WIRE (7)	600-3007	#26 VIOLET
	1	WIRE (6)	600-3006	#26 BLUE
	1	WIRE (5)	600-3005	#26 GREEN
	1	WIRE (4)	600-3004	#26 YELLOW
	1	WIRE (3)	600-3003	#26 ORANGE
	1	WIRE (2)	600-3002	#26 RED
	1	WIRE (1)	600-3001	#26 BROWN
	1	WIRE (0)	600-3000	#26 BLACK
2	AS REQ'D	LACING CORD	605-1000	
1	1	CONNECTOR	WL 5223	

NO.	REVISION	BY	DATE

TOL. EX. AS NOTED
.XX ±.010 FRAC. ±1/64
.XXX ±.005 ANG. ±0°30'
FINISH: ▽

MATERIAL

FINISH

WANG LABORATORIES, INC.
TEWKSBURY, MASS. U. S. A.

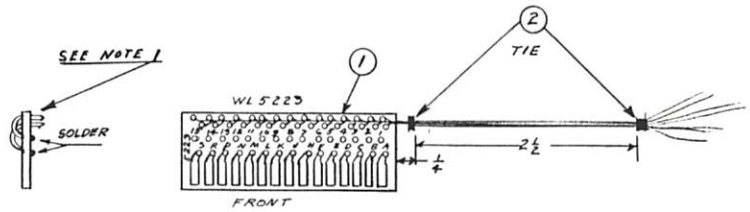
DR	[illegible]	DATE 5/5/70
CHK		DATE
APPD	RJT	DATE 8-7-70

MODEL No. 700 | W.O. No. | SCALE 4/1 | SHEET OF

TITLE CONNECTOR ASSEMBLY P1

PART NUMBER	REV	SIZE	DRAWING NUMBER
		B	6090

DO NOT SCALE



HOLE LEGEND		
DRILLED OR PUNCHED HOLE TOLERANCES:	HOLE DIA. .0135 to .125	TOL. +.003 −.001
	.126 to .250	+.004 −.001
	.251 to .500	+.005 −.001

IDENT.	DESCRIPTION	QTY.
A		

PIN NO.	WIRE COLOR	LENGTH
A	901	26"
B	81	25½"
C	0	30"
D	902	28½"
E	903	27½"
F	904	27½"
H	905	28"
J	906	29"
K	907	27½"
L	908	27"
M	71	27"
N	72	27"
P	—	—
R	74	29½"
S	75	28"

PIN NO.	WIRE COLOR	LENGTH
1	1	28"
2	76	27½"
3	3	28"
4	4	27"
5	5	27"
6	6	28"
7	7	27"
8	8	28"
9	91	27"
10	93	26½"
11	94	27½"
12	95	28"
13	96	27"
14	97	27½"
15	98	30"

NOTE:
1. ALL WIRES ARE FED THRU CLEARENCE HOLES ON FRONT OF CONNECTOR AND SOLDERED AT FRONT AS SHOWN.

WANG PART NO.	ITEM	QTY	NAME	MATERIAL	DESCRIPTION
600 − 3908		1	WIRE (908)		#26 WH/BLK/GRY
600 − 3907		1	WIRE (907)		#26 WH/BLK/VIO
600 − 3906		1	WIRE (906)		#26 WH/BLK/BLU
600 − 3905		1	WIRE (905)		#26 WH/BLK/GRN
600 − 3904		1	WIRE (904)		#26 WH/BLK/YEL
600 − 3903		1	WIRE (903)		#26 WH/BLK/ORN
600 − 3902		1	WIRE (902)		#26 WH/BLK/RED
600 − 3901		1	WIRE (901)		#26 WH/BLK/BRN
600 − 3098		1	WIRE (98)		#26 WH/GRY
600 − 3097		1	WIRE (97)		#26 WH/VIO
600 − 3096		1	WIRE (96)		#26 WH/BLU
600 − 3095		1	WIRE (95)		#26 WH/GRN
600 − 3094		1	WIRE (94)		#26 WH/YEL
600 − 3093		1	WIRE (93)		#26 WH/ORN
600 − 3091		1	WIRE (91)		#26 WH/BRN
600 − 3081		1	WIRE (81)		#26 GRY/BRN
600 − 3076		1	WIRE (76)		#26 VIO/BLU
600 − 3075		1	WIRE (75)		#26 VIO/GRN
600 − 3074		1	WIRE (74)		#26 VIO/YEL
600 − 3072		1	WIRE (72)		#26 VIO/RED
600 − 3071		1	WIRE (71)		#26 VIO/BRN
600 − 3008		1	WIRE (8)		#26 GRAY
600 − 3007		1	WIRE (7)		#26 VIOLET
600 − 3006		1	WIRE (6)		#26 BLUE
600 − 3005		1	WIRE (5)		#26 GREEN
600 − 3004		1	WIRE (4)		#26 YELLOW
600 − 3003		1	WIRE (3)		#26 ORANGE
600 − 3001		1	WIRE (1)		#26 BROWN
600 − 3000		1	WIRE (10)		#26 BLACK
605 − 1000	2	AS'D	LACING		
WL 5223	1	1	CONNECTOR		

Qty. Per Unit	FIRST USED ON	ASSY USED ON

DATE	BY	REVISION	NO

WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.

	BY	DATE	APPROVED BY	DATE
DWN	[illegible]	5/6/70	E ENGR	RJT 8-7-70
CHK			M ENGR	
E. C. CONTROL			MFG ENGR	

MATERIAL

MODEL NO 700

SEE ENGRG SPECIFICATIONS No.

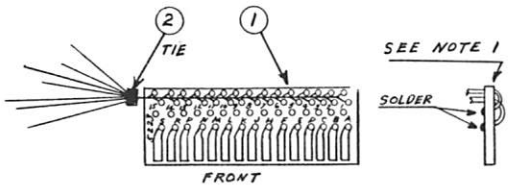
FINISH

TOL. EX. AS NOTED
XX ± .010 FRAC ± 1/64
XXX ± .005 ANG ± 1°30' FINISH √ 128

TITLE CONNECTOR ASSEMBLY P2

SCALE	SHT	OF	WANG PART NUMBER	SIZE	DRAWING NUMBER	REV
			1609	C	6091	D

DO NOT SCALE



PIN No.	WIRE COLOR	LENGTH
A	—	
B	2	22½"
C	54	36"
D	97	31½"
E	98	31"
F	901	30"
H	902	30"
J	903	31"
K	904	27½"
L	905	30½"
M	906	30"
N	72	29"
P	907	28½"
R	908	28½"
S	0	27"

PIN No.	WIRE COLOR	LENGTH
1	1	21½"
2	11	33"
3	3	22"
4	22	36"
5	4	31"
6	5	30"
7	6	21½"
8	7	23"
9	8	21"
10	91	21½"
11	93	21"
12	94	21"
13	95	20"
14	71	29"
15	96	21"

NOTE:
1. ALL WIRES ARE FED THRU CLEARENCE HOLES ON FRONT OF CONNECTOR AND SOLDERED AT FRONT AS SHOWN.

HOLE LEGEND

DRILLED OR PUNCHED HOLE TOLERANCES:	HOLE DIA.	TOL.
	.0135 to .125	+.003 −.001
	.126 to .250	+.004 −.001
	.251 to .500	+.005 −.001

IDENT.	DESCRIPTION	QTY.
A		

NAME	ITEM	QTY.		MATERIAL	DESCRIPTION
600-3011		1	WIRE (11)		#26 TAN
600-3908		1	WIRE (908)		#26 WH/BLK/GRY
600-3907		1	WIRE (907)		#26 WH/BLK/VIO
600-3906		1	WIRE (906)		#26 WH/BLK/BLU
600-3905		1	WIRE (905)		#26 WH/BLK/GRN
600-3904		1	WIRE (904)		#26 WH/BLK/YEL
600-3903		1	WIRE (903)		#26 WH/BLK/ORN
600-3902		1	WIRE (902)		#26 WH/BLK/RED
600-3901		1	WIRE (901)		#26 WH/BLK/BRN
600-3098		1	WIRE (98)		#26 WH/GRY
600-3097		1	WIRE (97)		#26 WH/VIO
600-3096		1	WIRE (96)		#26 WH/BLU
600-3095		1	WIRE (95)		#26 WH/GRN
600-3094		1	WIRE (94)		#26 WH/YEL
600-3093		1	WIRE (93)		#26 WH/ORN
600-3091		1	WIRE (91)		#26 WH/BRN
600-3054		1	WIRE (54)		#26 GRN/YEL
600-3072		1	WIRE (72)		#26 VIO/RED
600-3071		1	WIRE (71)		#26 VIO/BRN
600-3022		1	WIRE (22)		#26 PINK
600-3008		1	WIRE (8)		#26 GRAY
600-3007		1	WIRE (7)		#26 VIOLET
600-3006		1	WIRE (6)		#26 BLUE
600-3005		1	WIRE (5)		#26 GREEN
600-3004		1	WIRE (4)		#26 YELLOW
600-3003		1	WIRE (3)		#26 ORANGE
600-3002		1	WIRE (2)		#26 RED
600-3001		1	WIRE (1)		#26 BROWN
600-3000		1	WIRE (0)		#26 BLACK
605-1000	2	ASS'D	LACING		
WL5223	1	1	CONNECTOR		

Qty. Per Unit	FIRST USED ON	ASSY USED ON

WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.

MATERIAL

MODEL NO. 700

SEE ENGRG SPECIFICATIONS No.

FINISH

TOL. EX. AS NOTED
XX ± .010 FRAC. ± 1/64
XXX ± .005 ANG. ± 1°30' FINISH 125 V

SCALE FT | SHT | OF

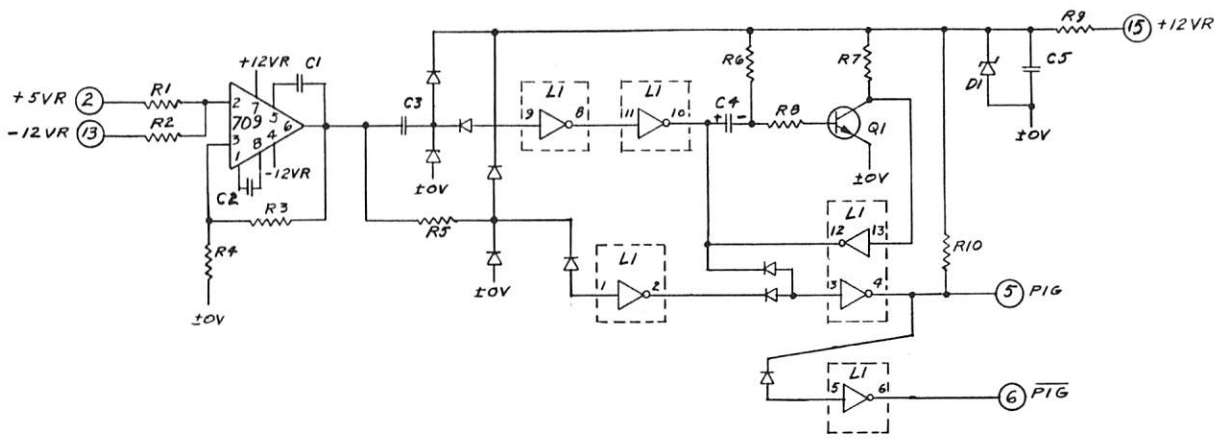
	BY	DATE	APPROVED BY	DATE
DWN	DS	5/3/70	E ENGR RJT	8-7-70
CHK			M ENGR	
E. C. CONTROL			MFG ENGR	

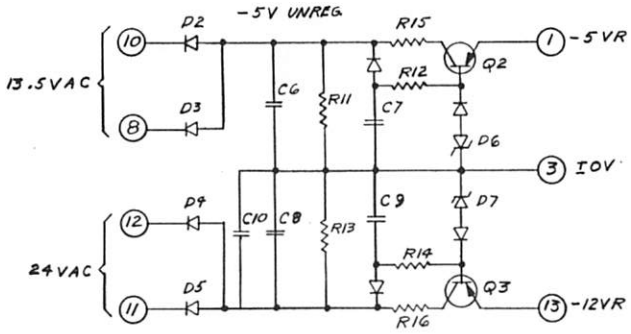
TITLE: CONNECTOR ASSEMBLY P3

WANG PART NUMBER	SIZE	DRAWING NUMBER	REV
7609	C	6092	2

NO	REVISION	DATE	BY
1	PER ECN #2453 WIRE AT PIN 2 WAS GRY/BLK APP. RJT	9/22/71	DS
2	PER ECN #3291 ADDED TAN WIRE TO PIN 2 APP'D SEH	10-9-72	DS

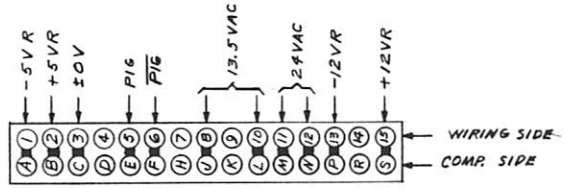
DO NOT SCALE





NOTE:
1. ALL DIODES ARE WL380-1001 UNLESS SPECIFIED OTHERWISE (QTY 14)
2. ALL RESISTORS ARE 1/4W

LOCATION	TYPE & MFR'R	TERM No. Vcc +5VR	TERM No. ±0V	QTY
L1	9935 DTAL FAIRCHILD	14	7	1



HOLE LEGEND

DRILLED OR PUNCHED HOLE TOLERANCES:	DIA.	TOL.
	.013 to .125	+.003 −.001
	.126 to .250	+.004 −.001
	.251 to .500	+.005 −.001

IDENT.	DESCRIPTION	QTY.
A		

COMP.	SIZE/TYPE	QTY
Q1	WL375-1006	1
Q2, 3	GT544	2
R1	10K	1
R10	2.2K	1
R2	27K	1
R3	1 MEG	1
R4, 11, 13	10K	3
R5, 7	4.7K	2
R6	33K	1
R8	220 Ω	1
R9	680 Ω	1
R12	560 Ω	1
C1, 2	220 MF	2
C3	.02 MF/25V	1
C4, 5	2.2 MF TANT.	2
C7	100 MF/16V	1
C6	200 MF/16V	1
D1, 6	IN751A	2
D2, 3, 4, 5	IN3253	4
709	WL376-0000	1
D7	IN758A	1
R14	330 Ω	1
C8, 9, 10	100 MF/25V	3
R15	27 Ω	1
R16	68 Ω	1

NO	REVISION	BY	DATE
1	PER ECN #'s 6 & 9 R1 WAS 1K R2 WAS 2.7K ADDED C10 100 MF 25V C9 WAS 100 MF 16V	JFD	3/30/70
2	PER ECN 3118 R14 WAS 1.5K ADDED R15 AND 16 APP'D SKH	JFD	8-8-72

WANG PART NO.	ITEM	QTY.	NAME
			WANG LABORATORIES, INC. TEWKSBURY, MASS. U.S.A.
			MATERIAL
			MODEL NO. SYSTEM 700
			SEE ENGRG SPECIFICATIONS No.
			FINISH
			TOL. EX. AS NOTED XX ± .010 FRAC ± 1/64 XXX ± .005 ANG ± 1°30' FINISH 125
			SCALE SHT OF

MATERIAL	DESCRIPTION			
BY	DATE	APPROVED BY	DATE	
DWN JFD	1/15/70	E ENGR SKH	3/30/70	
CHK RJT	3/13/70	M ENGR		
E. C. CONTROL		MFG ENGR		

TITLE SCHEMATIC LOGIBLOC #336 -5V, -12V REG. & POWER FAILURE PROTECTION CIRCUIT

WANG PART NUMBER	SIZE	DRAWING NUMBER	REV.
	C	6051	2